[12/24/2020 12:12] >[OPcs49h5AJ8]

[1] /

The Archive

[1.1] >

rish987.github.io

Introduction.

This is my personal archive, a place where I plan to record what I've learned in the fields of mathematics, computer science, and more. [1.2] >

A continual work in progress, the intent is for it to be a completely self-contained repository, with descriptions that are as concise as physically possible.

The archive is free and open-source, together with the archiver, the tool I use to build archives. You can find them below:

`github.com/rish987/archive`

`github.com/rish987/archiver`.

Navigation.

It's organized hierarchically into units called "references," which can be one of the following types: [1.3] >

- "root." The reference you're currently on! There's only one reference of this type. It's reserved for descriptions that apply to the entire archive. [1.4] >
- "topic" (denoted by T). These repeat a general format: [1.5] >
  - Describe a simple concept that relies on a simple physical intuition.
  - Correspond this intuition with some initial notation, and possibly define some more notation using this initial notation.
  - Use this notation to describe some facts that follow from it. These facts are merely stated, not proven.
- "definition" (denoted by D). These introduce notation that depend on a previously described physical intuition, previously defined notation, or a combination of both. [1.6] >
- "proof" (denoted by P). These state a fact and show why it is true, using previously defined notation. [1.7] >
- "note" (denoted by N). These may be one of the following: [1.8] >
  - A visual/verbal interpretation of a concept that isn't strictly necessary for understanding, but could be helpful.
  - A temporary stand-in for a reference that will be properly integrated into its own topic at a later time.

Each reference is delimited by a pair of horizontal rules. [1.9] >

At the upper-left corner of each reference is a "reference number" ("RN" for short), a unique ID that will never, ever change. When verbally referring to a particular reference, we say "RN" followed by the reference number, e.g. "RN53." The numbers themselves don't actually mean anything, other than roughly tracking the order in which I created the references. In fact, some reference numbers may not actually exist, if I decided to delete its former reference! [1.10] >

In the margins of a reference are labels for "reference parts" and their associated "reference part numbers." These are logical sections of the reference that should be verbally referred with "RNP" followed by the reference number, a period, and the reference part number, e.g. "RNP53.3." to indicate reference part 3 of reference 53. Similarly to reference numbers, the reference part numbers themselves don't mean anything other than roughly tracking the order in which I created them. [1.11] >

The symbol > denotes a link to a YouTube video for that reference. It should never be strictly necessary to watch the video to understand the reference, as all the details should be in the text. [1.12] >

At the upper-right corner of each reference is a "reference path," a unique path that identifies this reference's place in the hierarchy of references. It's written in UNIX-SHELL path style. [1.13] >

A reference is linked to either explicitly or implicitly. These links may either be "local" or "global." The difference is contextual – the first link to a reference in the hierarchy is a "local" link, in that following it will append to the current reference path. All other links are "global," in that the current reference path has no relation to that of the linked reference. [1.14] >

- An explicit link has the form $[\ldots]^{[\mathrm{RN}]}_{[\mathrm{type}]}$. [1.15] >
- An implicit link is present in the notation, and links to the reference where the notation was originally defined. Just as notation is nested, so too are the corresponding implicit links. [1.16] >

  We follow the convention that implicit links are always global; it follows that the first link to a reference is always explicit.

Annotations.

Equations may be annotated as, for example: [1.17] >

$$\overset{\texttt{i}}{a = b} \underset{\texttt{y}}{=} \overset{\texttt{j}}{c} \underset{\texttt{z}}{\geq} d,$$
$$\underset{\texttt{x}}{}$$

followed by

$$\overset{\texttt{(x-z)->}[\ldots]^{[\mathrm{RN}]}_{[\mathrm{type}]}}{g = h},$$

where blue underset boxes define "labels," and grey overset boxes provide "explanations" that explain why an (in)equality follows, using previously defined labels and other references. An explanation is formatted as follows:

- `x` indicates to consider the single (in)equality labeled by `x`. [1.18] >

  In the example above, `x` refers to $a = b$.

- `(x-y)` indicates that, when `x` and `y` appear in a chain (in)equality, to consider the equation that results from cropping out what appears in between the (in)equalities labeled by `x` and `y`. [1.19] >

  In the example above, `(x-z)` refers to $a \geq d$.

- `x->`$[\ldots]^{[\mathrm{RN}]}_{[\mathrm{type}]}$ indicates to use (in)equality `x` as the antecendent of the reference linked to by $[\ldots]^{[\mathrm{RN}]}_{[\mathrm{type}]}$. [1.20] >

  In the example above, `(x-z)->`$[\ldots]^{[\mathrm{RN}]}_{[\mathrm{type}]}$ refers to use $a \geq d$ as an antecedent of the reference with RN `[RN]`.

Where an explanation is not provided, the result should follow from simple algebra or arithmetic. [1.29] >

Note that reference paths and labels are simply meant to be unique, memorable identifiers, and by themselves don't actually matter. Though technically, they always are acronyms/abbreviations of *something* – see if you can figure out what they are in the YouTube video comments! [1.21] >

Formats.

The archive is available in the following formats: [1.22] >

- `tree_online`: Each reference has its own online link in a "reference tree" and corresponding PDF. [1.23] >

  Links are online hyperlinks.

- `tree`: Local download of the reference tree. [1.24] >

  Links are remote PDF links.

- `full`: All references are combined into a single PDF, ordered by depth (i.e., a linearized DAG), where each reference starts on a new page. [1.25] >

  Links are cross-referencing links within the same PDF.

- `full_compact`: Identical to `full`, except references may start on the same page. [1.26] >

  Links are cross-referencing links within the same PDF.

For the best experience, I highly recommend using `pdf.js` (native to the Firefox web browser, but also available on Chrome) as your PDF viewer, in "presentation mode" (`ctrl+alt+p`) on a vertical monitor. Note that with `pdf.js`, you can use your browser's history navigation functionality to traverse your link history (`alt+rightarrow` and `alt+leftarrow`). [1.27] >

For offline viewing, I highly recommend Atril Document Viewer. [1.28] >

Topics.

[Graph Theory]$_{\mathrm{T}}^{104}$

[12/25/2020 12:51] >[DJnqZUpY1wg]

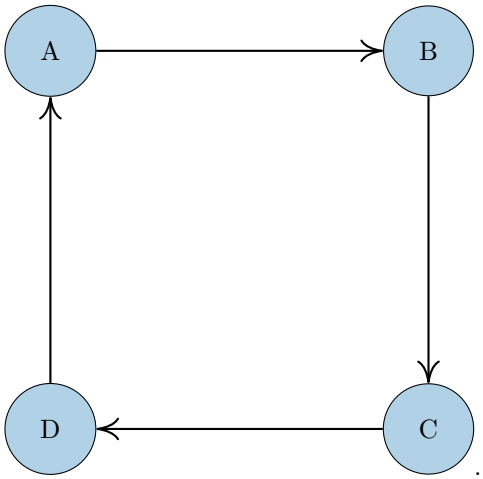
[104] /T/gr

Graph Theory

[104.1] >

Physical intuition.

Suppose that we have a "graph" consisting of a set of "nodes" and a set of "edges" between the nodes: [104.2] >



Here, our graph has the nodes $\{A, B, C, D\}$ and edges $\{[A, B], [B, C], [C, D], [D, A]\}$. [104.3] >

Notation.

[Define]$_{\text{D}}^{105}$

$$\mathrm{V} \coloneqq \text{a set of nodes.}$$

[Define]$_{\text{D}}^{106}$

$$\mathrm{E} \coloneqq \text{a set of edges, where } \mathrm{E} \subseteq \mathrm{V}^2.$$

[Define]$_{\text{D}}^{107}$

$$\mathrm{G} = [\mathrm{V}, \mathrm{E}], \text{ i.e., an arbitrary graph.}$$

[Define]$_{\text{D}}^{151}$

$$\mathrm{UD}\,(\mathrm{G}) \iff \forall [\mathrm{a}, \mathrm{b}] \in \mathrm{E}, [\mathrm{b}, \mathrm{a}] \in \mathrm{E}$$

or, for contextual G,

$$\mathrm{UD}.$$

[Define]$_{\text{D}}^{109}$ for all $\mathrm{V}' \subseteq \mathrm{V}$,

$$\mathrm{G} \underset{\rightarrow}{\setminus} \mathrm{V}' = \left[\mathrm{V} \setminus \mathrm{V}', \{[\mathrm{v}, \mathrm{v}'] \in \mathrm{E} \mid \mathrm{v}, \mathrm{v}' \notin \mathrm{V}'\}\right].$$

[Define]$_{\text{D}}^{142}$

$$\mathrm{PATHS}\,(\mathrm{G}) \coloneqq \{\mathrm{P} \in \mathrm{V}^* \mid \forall 0 \leq t \leq \|\mathrm{P}\| - 2, [\mathrm{P}\,[t], \mathrm{P}\,[t+1]] \in \mathrm{E}\},$$

or, for contextual G,

$$\mathrm{PATHS}.$$

[Define]$_{\text{D}}^{144}$ for all $\mathrm{P}, \mathrm{P}' \in \mathrm{PATHS}\,(\mathrm{G})$,

$$\left(\mathrm{G};\ \mathrm{P} \underset{\rightarrow}{=} \mathrm{P}'\right) \iff \mathrm{P}\,[0] = \mathrm{P}'\,[0], \mathrm{P}\,[-1] = \mathrm{P}'\,[-1],$$

or, for contextual G,

$$\mathrm{P} \underset{\rightarrow}{=} \mathrm{P}'.$$

[Define]$_{\text{D}}^{108}$ for all $\mathrm{v}, \mathrm{v}' \in \mathrm{V}^2$,

$$(\mathrm{G};\ \mathrm{v} \longrightarrow \mathrm{v}') \coloneqq \{\mathrm{P} \in \mathrm{PATHS}\,(\mathrm{G}) \mid \mathrm{P}\,[0] = \mathrm{v}, \mathrm{P}\,[-1] = \mathrm{v}'\},$$

or, for contextual G,

$$\mathrm{v} \longrightarrow \mathrm{v}'.$$

[Define]$_{\text{D}}^{110}$ for all $\mathrm{V}' \subseteq \mathrm{V}$ and $\mathrm{v}, \mathrm{v}' \in (\mathrm{V} \setminus \mathrm{V}')^2$,

$$(\mathrm{G};\ \mathrm{v} \longrightarrow \mathrm{v}' \setminus \mathrm{V}') \coloneqq \left(\mathrm{G} \underset{\rightarrow}{\setminus} \mathrm{V}';\ \mathrm{v} \longrightarrow \mathrm{v}'\right),$$

or, for contextual G,

$$\mathrm{v} \longrightarrow \mathrm{v}' \setminus \mathrm{V}'.$$

[Define]$_{\text{D}}^{111}$ for all $\mathrm{v}, \mathrm{v}' \in \mathrm{V}^2$,

$$(\mathrm{G};\ \mathrm{v} \longrightarrow \mathrm{v}') \iff \|(\mathrm{G};\ \mathrm{v} \longrightarrow \mathrm{v}')\| > 0$$

and

$$(\mathrm{G};\ \mathrm{v} \not\longrightarrow \mathrm{v}') \iff \|(\mathrm{G};\ \mathrm{v} \longrightarrow \mathrm{v}')\| = 0,$$

or, for contextual G,

$$\mathrm{v} \longrightarrow \mathrm{v}'$$

and

$$\mathrm{v} \not\longrightarrow \mathrm{v}',$$

respectively. [Define]$_{\text{D}}^{112}$ for all $\mathrm{V}' \subseteq \mathrm{V}$ and $\mathrm{v}, \mathrm{v}' \in (\mathrm{V} \setminus \mathrm{V}')^2$,

$$(\mathrm{G};\ \mathrm{v} \longrightarrow \mathrm{v}' \setminus \mathrm{V}') \iff \left(\mathrm{G} \underset{\rightarrow}{\setminus} \mathrm{V}';\ \mathrm{v} \longrightarrow \mathrm{v}'\right)$$

and

$$(\mathrm{G};\ \mathrm{v} \not\longrightarrow \mathrm{v}' \setminus \mathrm{V}') \iff \left(\mathrm{G} \underset{\rightarrow}{\setminus} \mathrm{V}';\ \mathrm{v} \not\longrightarrow \mathrm{v}'\right),$$

or, for contextual G,

$$\mathrm{v} \longrightarrow \mathrm{v}' \setminus \mathrm{V}'$$

and

$$\mathrm{v} \not\longrightarrow \mathrm{v}' \setminus \mathrm{V}',$$

respectively. [Define]$^{137}_{\mathrm{D}}$ for all $\mathrm{V}' \subseteq \mathrm{V}$ and $\mathrm{A}, \mathrm{B} \in \mathrm{V}'^2$,

$$\big(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'\big) \iff \big\|\big(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'\big)\big\| > 0$$

and

$$\big(\mathrm{G};\ \mathrm{A} \not\longrightarrow \mathrm{B} \setminus \mathrm{V}'\big) \iff \big\|\big(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'\big)\big\| = 0,$$

or, for contextual G,

$$\mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'$$

and

$$\mathrm{A} \not\longrightarrow \mathrm{B} \setminus \mathrm{V}',$$

respectively. [Define]$^{140}_{\mathrm{D}}$ for all $\mathrm{V}' \subseteq \mathrm{V}$ and $\mathrm{A}, \mathrm{B} \subseteq \mathrm{V}'$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B})$,

$$(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}') \coloneqq \left(\mathrm{G} \underset{\rightarrow}{\setminus} \mathrm{V}';\ \mathrm{A} \longrightarrow \mathrm{B}\right),$$

or, for contextual G,

$$\mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'.$$

[Define]$^{138}_{\mathrm{D}}$ for all $\mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B})$,

$$(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B}) \iff \|(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B})\| > 0$$

and

$$(\mathrm{G};\ \mathrm{A} \not\longrightarrow \mathrm{B}) \iff \|(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B})\| = 0,$$

or, for contextual G,

$$\mathrm{A} \longrightarrow \mathrm{B}$$

and

$$\mathrm{A} \not\longrightarrow \mathrm{B},$$

respectively. [Define]$^{139}_{\mathrm{D}}$ for all $\mathrm{V}' \subseteq \mathrm{V}$ and $\mathrm{A}, \mathrm{B} \in \mathrm{V}'^2$,

$$\big(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'\big) \iff \big\|\big(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'\big)\big\| > 0$$

and

$$\big(\mathrm{G};\ \mathrm{A} \not\longrightarrow \mathrm{B} \setminus \mathrm{V}'\big) \iff \big\|\big(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'\big)\big\| = 0,$$

or, for contextual G,

$$\mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'$$

and

$$\mathrm{A} \not\longrightarrow \mathrm{B} \setminus \mathrm{V}',$$

respectively.

Propositions.

[It can be shown]$_{\mathrm{P}}^{145}$ that, for contextual G and all $\mathrm{P} \in \mathrm{PATHS}$ and $0 \leq t_1 \leq t_2 \ \leq \|\mathrm{P}\|$, with $\mathrm{P}' \coloneqq \mathrm{P}\,[t_1 : t_2]$,

$$\mathrm{P}' \in \mathrm{PATHS}.$$

[It can be shown]$_{\mathrm{P}}^{146}$ that, for contextual G and all $\mathrm{P}_1, \mathrm{P}_2 \in \mathrm{PATHS}^2$, with $\mathrm{P} \coloneqq \mathrm{P}_1 \cup \mathrm{P}_2$,

$$[\mathrm{P}_1\,[-1], \mathrm{P}_2\,[0]] \in \mathrm{E} \implies \mathrm{P} \in \mathrm{PATHS}.$$

[It can be shown]$_{\mathrm{P}}^{143}$ that, for contextual G and all $\mathrm{P} \in \mathrm{PATHS}$, there exists $\mathrm{P}' \in \mathrm{PATHS}$ such that

$$\mathrm{P} \underset{\rightarrow}{=} \mathrm{P}' \text{ and } \forall \mathrm{v} \in \mathrm{P}', \|\mathrm{v};\ \mathrm{P}'\| = 1.$$

[It can be shown]$_{\mathrm{P}}^{147}$ that, for contextual G and all $\mathrm{a}, \mathrm{b}, \mathrm{c} \in \mathrm{V}^3$,

$$\mathrm{a} \longrightarrow \mathrm{c},\ \mathrm{c} \longrightarrow \mathrm{b} \implies \mathrm{a} \longrightarrow \mathrm{b}.$$

[It can be shown]$_{\mathrm{P}}^{153}$ that, for contextual G and all $\mathrm{a}, \mathrm{b} \in \mathrm{V}^2$, if UD,

$$\mathrm{a} \longrightarrow \mathrm{b} \implies \mathrm{b} \longrightarrow \mathrm{a}.$$

[It can be shown]$_{\mathrm{P}}^{150}$ that, for contextual G and all $\mathrm{a}, \mathrm{b} \in \mathrm{V}^2$, $\mathrm{V}' \subseteq \mathrm{V}$ such that $\mathrm{a}, \mathrm{b} \notin \mathrm{V}'$,

$$\mathrm{P} \in \mathrm{a} \longrightarrow \mathrm{b} \setminus \mathrm{V}' \implies \forall \mathrm{v} \in \mathrm{P}, \mathrm{v} \notin \mathrm{V}'.$$

[It can be shown]$_{\mathrm{P}}^{149}$ that, for contextual G and all $\mathrm{a}, \mathrm{b} \in \mathrm{V}^2$, $\mathrm{V}' \subseteq \mathrm{V}$ such that $\mathrm{a}, \mathrm{b} \notin \mathrm{V}'$, and all $\mathrm{P} \in \mathrm{a} \longrightarrow \mathrm{b}$,

$$\mathrm{P} \in \mathrm{a} \longrightarrow \mathrm{b} \setminus \mathrm{V}' \iff \forall \mathrm{v} \in \mathrm{P}, \mathrm{v} \notin \mathrm{V}'.$$

[It can be shown]$_{\mathrm{P}}^{148}$ that, for contextual G and all $\mathrm{a}, \mathrm{b} \in \mathrm{V}^2$, $\mathrm{V}' \subseteq \mathrm{V}$, and $\mathrm{C} \subseteq \mathrm{V}$ such that $\mathrm{a}, \mathrm{b} \notin \mathrm{V}' \cup \mathrm{C}$,

$$\mathrm{a} \longrightarrow \mathrm{b} \setminus \mathrm{V}' \cup \mathrm{C} \implies \mathrm{a} \longrightarrow \mathrm{b} \setminus \mathrm{V}'.$$

[It can be shown]$_{\mathrm{P}}^{141}$ that, for contextual G and all $\mathrm{a}, \mathrm{b}, \mathrm{c} \in \mathrm{V}^3$ such that $\mathrm{NE}\,(\mathrm{a}, \mathrm{b}, \mathrm{c})$, and $\mathrm{V}' \subseteq \mathrm{V}$ such that $\mathrm{a}, \mathrm{b}, \mathrm{c}, \notin \mathrm{V}'$,

$$\mathrm{a} \longrightarrow \mathrm{c} \setminus \mathrm{V}' \cup \{\mathrm{b}\},\ \mathrm{c} \longrightarrow \mathrm{b} \setminus \mathrm{V}' \cup \{\mathrm{a}\} \implies \mathrm{a} \longrightarrow \mathrm{b} \setminus \mathrm{V}'.$$

[It can be shown]$_{\mathrm{P}}^{152}$ that, for contextual G and all $\mathrm{a}, \mathrm{b}, \mathrm{c} \in \mathrm{V}^3$ such that $\mathrm{NE}\,(\mathrm{a}, \mathrm{b}, \mathrm{c})$, and $\mathrm{V}' \subseteq \mathrm{V}$ such that $\mathrm{a}, \mathrm{b}, \mathrm{c}, \notin \mathrm{V}'$, if UD,

$$\mathrm{a} \longrightarrow \mathrm{c} \setminus \mathrm{V}' \cup \{\mathrm{b}\},\ \mathrm{b} \longrightarrow \mathrm{c} \setminus \mathrm{V}' \cup \{\mathrm{a}\} \implies \mathrm{a} \longrightarrow \mathrm{b} \setminus \mathrm{V}'.$$

Topics.

[Probability Theory][35 T]

[12/25/2020 12:28]

[147] /T/gr/P/pathtrans

Proposition.

We have that for contextual G and all $a, b, c \in V^3$, [147.1]

$$a \underset{\texttt{pac}}{\longrightarrow} c,\ c \underset{\texttt{pcb}}{\longrightarrow} b \implies a \longrightarrow b.$$

Proof.

We have that $\exists P \in a \overset{\texttt{pac}}{\longrightarrow} c$ with $P[0] = a$ (pa) and $P[-1] = c$ (pc). [147.2]

Also, $\exists P' \in c \overset{\texttt{pcb}}{\longrightarrow} b$ with $P'[0] = c$ (ppc) and $P'[-1] = b$ (ppb).

There are two cases to consider: [147.3]

- $\|P\| = 1$. [147.4]

  Then, $a = P[0] = P[-1] = c$ (pa, pc), so

  $$a \overset{\texttt{pcb}}{\longrightarrow} b.$$

- $\|P\| > 1$. Then, $[P[-2], P'[0]] = [P[-2], P[-1]] \in E$ (pc,ppc; $[\ldots]^{142}_{D}$; ei) and $P[:-1] \in \text{PATHS}$ ($[\ldots]^{145}_{P}$; pi), so letting $P^+ := P[:-1] \cup P'$ (sg), we have [147.5]

  $$P^+ \in \text{PATHS},$$ ((ei,pi,pcb)-> $[\ldots]^{146}_{P}$)

  and

  $$P^+[0] = a \text{ and } P^+[-1] = b,$$ (sg,pa; ppb)

  so

  $$P^+ \in a \overset{[\ldots]^{108}_{D}}{\longrightarrow} b \overset{[\ldots]^{111}_{D}}{\implies} a \longrightarrow b.$$

[12/25/2020 12:51]

[152] /T/gr/P/pathswapud

Proposition.

We have that for contextual G and all $\mathrm{a}, \mathrm{b}, \mathrm{c} \in \mathrm{V}^3$ such that $\mathrm{NE}\,(\mathrm{a}, \mathrm{b}, \mathrm{c})$, and $\mathrm{V}' \subseteq \mathrm{V}$ such that $\mathrm{a}, \mathrm{b}, \mathrm{c}, \notin \mathrm{V}'$, if UD, [152.1]

ud

$$\mathrm{a} \longrightarrow \mathrm{c} \setminus \mathrm{V}' \cup \{\mathrm{b}\},\ \mathrm{b} \longrightarrow \mathrm{c} \setminus \mathrm{V}' \cup \{\mathrm{a}\} \implies \mathrm{a} \longrightarrow \mathrm{b} \setminus \mathrm{V}'.$$

pac pbc

Proof.

We have that [152.2]

$$\mathrm{c} \overset{\text{(pbc,ud)->}[\ldots]^{153}_{\mathrm{P}}}{\longrightarrow} \mathrm{b} \setminus \mathrm{V}' \cup \{\mathrm{a}\} \overset{\text{(pac,!)->}[\ldots]^{141}_{\mathrm{P}}}{\implies} \mathrm{a} \longrightarrow \mathrm{b} \setminus \mathrm{V}'.$$

[12/25/2020 12:47]

[148] /T/gr/P/pathsubless

Proposition.

We have that for contextual G and all $a, b \in V^2$, $V' \subseteq V$, and $C \subseteq V$ such that $a, b \notin V' \cup C$, [148.1]

$$a \longrightarrow b \setminus V' \cup C \implies a \longrightarrow b \setminus V'.$$

ptho

Proof.

We have that, for all $v \in P$, [148.2]

(ptho)->[...]$^{150}_{P}$

$$v \notin V' \cup C,$$

so

$$v \notin V'$$

[...]$^{149}_{P}$

$$\implies a \longrightarrow b \setminus V'.$$

[12/25/2020 12:02]

[146] /T/gr/P/subpathcomb

Proposition.

We have that for contextual G and all $\mathrm{P}_1, \mathrm{P}_2 \in \mathrm{PATHS}^2$, with $\mathrm{P} \coloneqq \mathrm{P}_1 \cup \mathrm{P}_2$, [146.1]

$$[\mathrm{P}_1[-1], \mathrm{P}_2[0]] \in \mathrm{E} \overset{\mathtt{ei}}{\implies} \mathrm{P} \in \mathrm{PATHS}.$$

Proof.

We have that [146.2]

$$\forall 0 \leq t \leq \|\mathrm{P}_1\| - 2, [\mathrm{P}[t], \mathrm{P}[t+1]] \in \mathrm{E},$$

$$\forall \|\mathrm{P}_1\| \leq t \leq \|\mathrm{P}_1\| + \|\mathrm{P}_2\| - 2, [\mathrm{P}[t], \mathrm{P}[t+1]] \in \mathrm{E},$$

and

$$[\mathrm{P}[\|\mathrm{P}_1\| - 1], \mathrm{P}[\|\mathrm{P}_1\|]] \overset{\mathtt{pd}}{=} [\mathrm{P}_1[-1], \mathrm{P}_2[0]] \overset{\mathtt{ei}}{\in} \mathrm{E}$$

$\overset{[\ldots]_{\mathrm{D}}^{142}}{\implies}$ $\mathrm{P} \in \mathrm{PATHS}$.

[12/25/2020 11:59]

[145] /T/gr/P/subpath

Proposition.

We have that for contextual G and all $\mathrm{P} \in \mathrm{PATHS}$ and $0 \leq t_1 \leq t_2 \ \leq \|\mathrm{P}\|$, with $\mathrm{P}' \coloneqq \mathrm{P}\left[t_1 : t_2\right]$, [145.1]

pp

$$\mathrm{P}' \in \mathrm{PATHS}.$$

Proof.

We have that [145.2]

[...]$^{142}_{\mathrm{D}}$

$$\forall 0 \leq t \leq \|\mathrm{P}\| - 2, [\mathrm{P}\left[t\right], \mathrm{P}\left[t+1\right]] \in \mathrm{E},$$

so

pp [...]$^{142}_{\mathrm{D}}$

$$\forall t_1 \leq t \leq t_2 - 2, [\mathrm{P}\left[t\right], \mathrm{P}\left[t+1\right]] \in \mathrm{E} \implies \forall 0 \leq t \leq \|\mathrm{P}'\| - 2, [\mathrm{P}'\left[t\right], \mathrm{P}'\left[t+1\right]] \in \mathrm{E} \implies \mathrm{P}' \in \mathrm{PATHS}.$$

[12/25/2020 12:23]

[143] /T/gr/P/pathuniq

Proposition.

We have that for contextual G and all $\mathrm{P} \in \mathrm{PATHS}$, there exists $\mathrm{P}' \in \mathrm{PATHS}$ such that [143.1]

$$\mathrm{P} \underset{\rightarrow}{=} \mathrm{P}' \text{ and } \forall \mathrm{v} \in \mathrm{P}', \|\mathrm{v};\ \mathrm{P}'\| = 1.$$

Proof.

Base case.

We have that, for all $\mathrm{P} \in \mathrm{PATHS}$ such that $\|\mathrm{P}\| \underset{\texttt{bc}}{=} 1$, letting $\mathrm{P}' \coloneqq \mathrm{P}$, we have [143.2]

$$\mathrm{P}'[0] = \mathrm{P}[0], \mathrm{P}'[-1] = \mathrm{P}[-1], \text{ so } \mathrm{P} \underset{\rightarrow}{\overset{[\ldots]^{144}_{\mathrm{D}}}{=}} \mathrm{P}',$$

and for all $\mathrm{v} \in \mathrm{P}'$ [143.3]

$$1 \leq \|\mathrm{v};\ \mathrm{P}'\| = \|\mathrm{v};\ \mathrm{P}\| \leq \|\mathrm{P}\| \overset{\texttt{bc}}{=} 1, \text{ so } \|\mathrm{v};\ \mathrm{P}'\| = 1.$$

Inductive hypothesis.

Suppose, for all $1 \leq n' \leq n$, with $n \geq 1$, and all $\mathrm{P} \in \mathrm{PATHS}$ such that $\|\mathrm{P}\| = n'$, there exists $\mathrm{P}' \in \mathrm{PATHS}$ such that [143.4]

$$\mathrm{P} \underset{\rightarrow}{\overset{\texttt{iha}}{=}} \mathrm{P}' \text{ and } \forall \mathrm{v} \in \mathrm{P}', \|\mathrm{v};\ \mathrm{P}'\| \overset{\texttt{ihb}}{=} 1.$$

Inductive step.

We have that, for all $\mathrm{P} \in \mathrm{PATHS}$ such that $\|\mathrm{P}\| \underset{\texttt{ps}}{=} n + 1$, there are two cases to consider: [143.5]

- $\forall \mathrm{v} \in \mathrm{P}, \|\mathrm{v};\ \mathrm{P}\| \underset{\texttt{mo}}{=} 1.$ [143.6]

  Then, letting $\mathrm{P}' \coloneqq \mathrm{P}$, we immediately have

  $$\mathrm{P} \underset{\rightarrow}{\overset{[\ldots]^{144}_{\mathrm{D}}}{=}} \mathrm{P}' \text{ and } \forall \mathrm{v} \in \mathrm{P}, \|\mathrm{v};\ \mathrm{P}'\| = \|\mathrm{v};\ \mathrm{P}\| \overset{\texttt{mo}}{=} 1.$$

- $\exists \mathrm{v} \in \mathrm{P} : \|\mathrm{v};\ \mathrm{P}\| \underset{\texttt{mg}}{>} 1.$ [143.7]

  Then, there are two cases to consider:

  - $\mathrm{v} \underset{\texttt{fe}}{=} \mathrm{P}[0]$. Then, letting $t \coloneqq \mathrm{P}\langle \mathrm{v} \rangle [1]$, we have that $\|\mathrm{P}\| \underset{\texttt{sgt}}{\overset{\texttt{mg}}{>}} t \underset{\texttt{tg}}{\overset{\texttt{mg,fe}}{>}} 0$, so letting [143.8]

$\mathrm{P}^- \coloneqq \mathrm{P}\,[t:]$, we have $\mathrm{P}^- \in \mathrm{PATHS}$, and

$$\mathrm{P}^-\,[0] = \mathrm{P}\,[0] \text{ and } \mathrm{P}^-\,[-1] = \mathrm{P}\,[-1] \implies \mathrm{P} \underset{\rightarrow}{=} \mathrm{P}^-,$$

with

$$\left\|\mathrm{P}^-\right\| = \|\mathrm{P}\| - t < \|\mathrm{P}\| = n + 1,$$

and so there exists $\mathrm{P}' \in \mathrm{PATHS}$ such that [143.9]

$$\mathrm{P}^- \underset{\rightarrow}{=} \mathrm{P}' \text{ and } \forall \mathrm{v} \in \mathrm{P}', \|\mathrm{v};\ \mathrm{P}'\| = 1,$$

and $\mathrm{P} \underset{\rightarrow}{=} \mathrm{P}'$.

– $\mathrm{v} \neq \mathrm{P}\,[0]$. Then, letting $t_1 \coloneqq \mathrm{P}\,\langle \mathrm{v} \rangle\,[0]$ and $t_2 \coloneqq \mathrm{P}\,\langle \mathrm{v} \rangle\,[1]$, we have that [143.10]

$$\|\mathrm{P}\| > t_2 > t_1 > 0,$$

so

$$[\mathrm{P}\,[t_1 - 1], \mathrm{P}\,[t_2]] = [\mathrm{P}\,[t_1 - 1], \mathrm{P}\,[t_1]] \in \mathrm{E},$$

and [143.11]

$$\mathrm{P}^- \coloneqq \mathrm{P}\,[: t_1] \cup \mathrm{P}\,[t_2 :] \in \mathrm{PATHS},$$

so we have

$$\mathrm{P}^-\,[0] = \mathrm{P}\,[0] \text{ and } \mathrm{P}^-\,[-1] = \mathrm{P}\,[-1] \implies \mathrm{P} \underset{\rightarrow}{=} \mathrm{P}^-,$$

with [143.12]

$$\left\|\mathrm{P}^-\right\| = t_1 + \|\mathrm{P}\| - t_2 = \|\mathrm{P}\| - (t_2 - t_1) < \|\mathrm{P}\| = n + 1,$$

and so there exists $\mathrm{P}' \in \mathrm{PATHS}$ such that

$$\mathrm{P}^- \underset{\rightarrow}{=} \mathrm{P}' \text{ and } \forall \mathrm{v} \in \mathrm{P}', \|\mathrm{v};\ \mathrm{P}'\| = 1,$$

and $\mathrm{P} \underset{\rightarrow}{=} \mathrm{P}'$.

[12/25/2020 12:56]

[149] /T/gr/P/pathsubcond

Proposition.

We have that for contextual G and all $\mathrm{a}, \mathrm{b} \in \mathrm{V}^2$, $\mathrm{V}' \subseteq \mathrm{V}$ such that $\mathrm{a}, \mathrm{b} \notin \mathrm{V}'$, and all $\mathrm{P} \in \mathrm{a} \longrightarrow \mathrm{b}$, [149.1]
pi

$$\mathrm{P} \in \mathrm{a} \longrightarrow \mathrm{b} \setminus \mathrm{V}' \iff \forall \mathrm{v} \in \mathrm{P}, \mathrm{v} \notin \mathrm{V}'.$$

Proof.

$\Longrightarrow$ suppose $\mathrm{P} \in \mathrm{a} \longrightarrow \mathrm{b} \setminus \mathrm{V}'$. [149.2]

Then,

$$\overset{[\ldots]^{150}_{\mathrm{P}}}{\forall \mathrm{v} \in \mathrm{P}, \mathrm{v} \notin \mathrm{V}'}.$$

$\Longleftarrow$ suppose $\forall \mathrm{v} \in \mathrm{P}, \mathrm{v} \notin \mathrm{V}'$. [149.3]
ni

We first note that [149.4]

$$\overset{\mathtt{pi},[\ldots]^{108}_{\mathrm{D}}}{\mathrm{P} \in \mathrm{PATHS}\,(\mathrm{G})},$$

so

$$\overset{[\ldots]^{142}_{\mathrm{D}}}{\forall \mathrm{v} \in \mathrm{P}, \mathrm{v} \in \mathrm{V}},$$

and

$$\overset{\mathtt{!,ni}}{\forall \mathrm{v} \in \mathrm{P}, \mathrm{v} \in \mathrm{V} \setminus \mathrm{V}'} \overset{[\ldots]^{109}_{\mathrm{D}}}{\Longrightarrow} \underset{\mathtt{gz}}{\mathrm{v} \in \mathrm{G} \underset{\rightarrow}{\setminus} \mathrm{V}'\,[0]},$$

and [149.5]

$$\forall 0 \leq t \leq \|\mathrm{P}\| - 2,\ \overset{\mathtt{pi},[\ldots]^{108}_{\mathrm{D}},[\ldots]^{142}_{\mathrm{D}}}{[\mathrm{P}\,[t], \mathrm{P}\,[t+1]] \in \mathrm{E},\ \mathrm{P}\,[t], \mathrm{P}\,[t+1] \notin \mathrm{V}'} \overset{\mathtt{ni}\ [\ldots]^{109}_{\mathrm{D}}}{\Longrightarrow} \underset{\mathtt{go}}{[\mathrm{P}\,[t], \mathrm{P}\,[t+1]] \in \mathrm{G} \underset{\rightarrow}{\setminus} \mathrm{V}'\,[1]},$$

so

$$\overset{\mathtt{gz,go}}{\mathrm{P} \in \mathrm{PATHS}\left(\mathrm{G} \underset{\rightarrow}{\setminus} \mathrm{V}'\right)},$$

and

$$\overset{\mathtt{!,pi}}{\mathrm{P} \in \mathrm{a} \longrightarrow \mathrm{b} \setminus \mathrm{V}'}.$$

[12/25/2020 12:34]

[153] /T/gr/P/pathud

Proposition.

We have that for contextual G and all $\mathrm{a}, \mathrm{b} \in \mathrm{V}^2$, if UD, [153.1]

$$\mathrm{a} \underset{\texttt{pab}}{\longrightarrow} \mathrm{b} \overset{\texttt{ud}}{\Longrightarrow} \mathrm{b} \longrightarrow \mathrm{a}.$$

Proof.

We have that $\exists \mathrm{P} \in \mathrm{a} \overset{\texttt{pab}}{\longrightarrow} \mathrm{b}$, where $\mathrm{P}[0] \underset{\texttt{pa}}{=} \mathrm{a}$ and $\mathrm{P}[-1] \underset{\texttt{pb}}{=} \mathrm{b}$, and [153.2]

$$\forall 0 \leq t \leq \|\mathrm{P}\| - 2, [\mathrm{P}[t], \mathrm{P}[t+1]] \in \mathrm{E} \overset{[\ldots]_{\mathrm{D}}^{108}, [\ldots]_{\mathrm{D}}^{142}}{\Longrightarrow} \overset{\texttt{ud}}{} \forall 0 \leq t \leq \|\mathrm{P}\| - 2, [\mathrm{P}[t+1], \mathrm{P}[t]] \in \mathrm{E}$$

so letting $\mathrm{P}' := \mathrm{P}[::-1]$, we have [153.2]

$$\forall 0 \leq t \leq \|\mathrm{P}'\| - 2, [\mathrm{P}'[t], \mathrm{P}'[t+1]] \in \mathrm{E}$$

$\overset{[\ldots]_{\mathrm{D}}^{142}}{\Longrightarrow} \mathrm{P}' \in \mathrm{PATHS}$, and where [153.3]

$$\mathrm{P}'[0] \underset{\texttt{pb}}{=} \mathrm{b} \text{ and } \mathrm{P}'[-1] \underset{\texttt{pa}}{=} \mathrm{a},$$

so

$$\mathrm{P}' \in \mathrm{b} \overset{\texttt{(ap,!)->}[\ldots]_{\mathrm{D}}^{108}}{\longrightarrow} \mathrm{a} \overset{[\ldots]_{\mathrm{D}}^{111}}{\Longrightarrow} \mathrm{b} \longrightarrow \mathrm{a}.$$

[12/25/2020 12:48]

[141] /T/gr/P/pathswap

Proposition.

We have that for contextual G and all $a, b, c \in V^3$ such that $\mathrm{NE}(a, b, c)$, and $V' \subseteq V$ such that $a, b, c, \notin V'$, [141.1]

$$\underbrace{a \longrightarrow c \setminus V' \cup \{b\}}_{\texttt{pac}},\ \underbrace{c \longrightarrow b \setminus V' \cup \{a\}}_{\texttt{pcb}} \implies a \longrightarrow b \setminus V'.$$

Proof.

We have that

$$\overset{\texttt{(pac)->}[\ldots]_{P}^{148}}{a \longrightarrow c \setminus V'} \text{ and } \overset{\texttt{(pcb)->}[\ldots]_{P}^{148}}{c \longrightarrow b \setminus V'}$$

$$\overset{[\ldots]_{P}^{147}}{\implies} a \longrightarrow b \setminus V'.$$

[12/25/2020 12:36]

[150] /T/gr/P/pathsubcondf

Proposition.

We have that for contextual G and all $a, b \in V^2$, $V' \subseteq V$ such that $a, b \notin V'$, [150.1]

$$\underset{\texttt{pi}}{P \in a \longrightarrow b \setminus V'} \implies \forall v \in P, v \notin V'.$$

Proof.

We have that [150.2]

$$\overset{\texttt{pi},[\ldots]^{108}_{D}}{P \in \mathrm{PATHS}\left(\underset{\rightarrow}{G \setminus V'}\right)},$$

so

$$\overset{[\ldots]^{142}_{D}}{\forall v \in P, v \in V \setminus V'},$$

and

$$\forall v \in P, v \notin V'.$$

[12/25/2020 14:21] >[NPpTHtxvpIg]

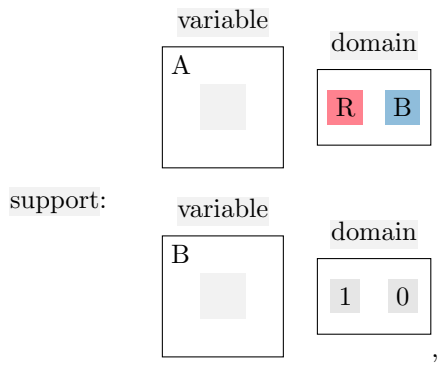
[35] /T/gr/T/pr

Probability Theory

[35.1] >

Physical intuition.

Suppose that we have a set of "variables," each with their own finite "domains", which we'll collectively refer to as the "support": [35.2] >



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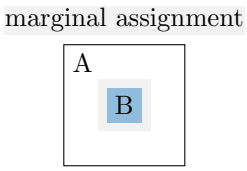
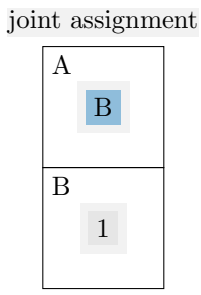
where a "marginal assignment" consists of associating some (but not all) of the variables of the support with something from their domains: [35.3] >



,

a "joint assignment" ("assignment" for short) consists of associating each variable of the support with something from its domain: [35.4] >



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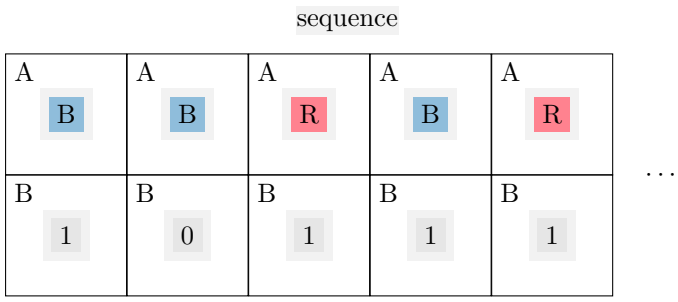
a "sequence" consists of an infinite, ordered list of such joint assignments: [35.5] >



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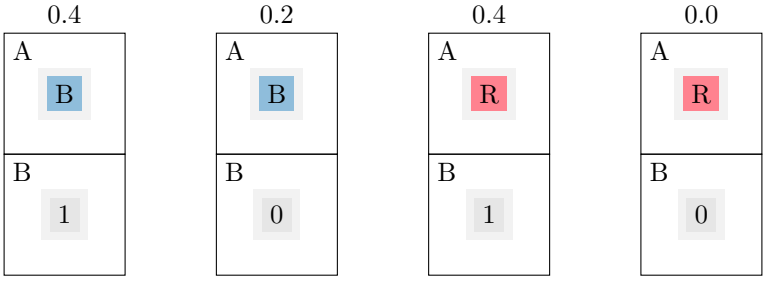
and a "trial" denotes a particular item (i.e. an indexed joint assignment) of this sequence.

Now, suppose something in particular about this sequence: if a joint assignment occurs at any trial, it will occur in infinitely many trials, and, additionally, as we the traverse sequence in order, the ratio of occurrences of the joint assignment to trials so far becomes arbitrarily close to some positive number. Otherwise, if an joint assignment never occurs, the ratio remains at zero. Let's call this ratio the "sequence probability" of the joint assignment. Let's refer to such a sequence as a probability sequence. [35.6] >

For example, if the sequence above repeated the first five trials, it would be a probability sequence defining the sequence probabilities: [35.7] >

sequence probabilities



Let a "distribution" be a set of probability sequences with the same sequence probability for every assignment, which, in the context of a distribution, we will simply refer to as the "joint probability" ("probability" for short) of that assignment. [35.8] >

Notation.

[Define]$^{36}_{\mathrm{D}}$

$$\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_{N_{\mathbf{X}}} \coloneqq \text{an arbitrary set of variable(s) with } N_{\mathbf{X}} \geq 1 \text{ and domains } \mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_{N_{\mathbf{X}}},$$

and we define $\mathrm{Y}_1, \mathrm{Y}_2, \ldots, \mathrm{Y}_{N_{\mathbf{Y}}}/\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_{N_{\mathbf{Y}}}/N_{\mathbf{Y}}$, $\mathrm{Z}_1, \mathrm{Z}_2, \ldots, \mathrm{Z}_{N_{\mathbf{Z}}}/\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_{N_{\mathbf{Z}}}/N_{\mathbf{Z}}$, and $\mathrm{W}_1, \mathrm{W}_2, \ldots, \mathrm{W}_{N_{\mathbf{W}}}/\mathbf{W}_1, \mathbf{W}_2, \ldots, \mathbf{W}_{N_{\mathbf{W}}}/N_{\mathbf{W}}$ similarly.

Define

$$?_1, ?_2, \ldots, ?_{N_{\mathbf{?}}} \coloneqq \text{an arbitrary set of variable(s) with } N_{\mathbf{?}} \geq 0 \text{ and domains } \mathbf{?}_1, \mathbf{?}_2, \ldots, \mathbf{?}_{N_{\mathbf{?}}}.$$

Define $?'_1, ?'_2, \ldots, ?'_{N_{\mathbf{?}'}}/\mathbf{?}'_1, \mathbf{?}'_2, \ldots, \mathbf{?}'_{N_{\mathbf{?}'}}/N_{\mathbf{?}'}$ similarly.

[Define]$^{102}_{\mathrm{D}}$

$$\mathrm{X} \coloneqq [\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_{N_{\mathbf{X}}}]$$

with

$$\mathbf{X} \coloneqq [\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_{N_{\mathbf{X}}}].$$

Define $\mathrm{Y}/\mathbf{Y}$, $\mathrm{Z}/\mathbf{Z}$, $\mathrm{W}/\mathbf{W}$, $?/\mathbf{?}$ and $?'/\mathbf{?}'$ similarly.

Note that $\|\mathrm{X}\| = N_{\mathbf{X}} \geq 1$, $\|\mathrm{Y}\| = N_{\mathbf{Y}} \geq 1$, $\|\mathrm{Z}\| = N_{\mathbf{Z}} \geq 1$, $\|\mathrm{W}\| = N_{\mathbf{W}} \geq 1$, $\|?\| = N_{\mathbf{?}} \geq 0$ and $\|?'\| = N_{\mathbf{?}'} \geq 0$.

[Let the notation]$^{103}_{\mathrm{D}}$

$$(\mathrm{X}, \mathbf{X}) \text{ indicate a “matching” between variables } \mathrm{X} \text{ and domains } \mathbf{X},$$

which expands as

$$(\mathrm{X}_1, \mathbf{X}_1), (\mathrm{X}_2, \mathbf{X}_2), \ldots, (\mathrm{X}_{N_{\mathbf{X}}}, \mathbf{X}_{N_{\mathbf{X}}}).$$

[Let the notation]$^{37}_{\mathrm{D}}$

$$\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\} \text{ establish a support consisting of variables } \mathrm{X} \text{ with domains } \mathbf{X}.$$

[Let]$^{57}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$ [or, $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$] and all $\mathrm{x} = [\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_{N_{\mathbf{X}}}] \in \mathbf{X}^{\times}$,

$$\mathrm{X} = \mathrm{x}$$

denote a joint assignment (assignment for short) [or, marginal assignment], which expands as

$$\mathrm{X}_1 = \mathrm{x}_1, \mathrm{X}_2 = \mathrm{x}_2, \ldots, \mathrm{X}_{N_{\mathbf{X}}} = \mathrm{x}_{N_{\mathbf{X}}}.$$

If a joint assignment [or, marginal assignment] was previously specified, we may later drop “X =”, for “implied assignment”:

$$\mathrm{x},$$

which expands as

$$\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_{N_{\mathbf{X}}}.$$

If a joint assignment [or, marginal assignment] was never specified, we assume an arbitrary $\mathrm{x} \in \mathbf{X}^{\times}$ for “implied generality”:

$$\mathrm{X},$$

which expands as

$$\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_{N_{\mathbf{X}}}.$$

[Let]$^{38}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{SQ} \text{ denote the set of all sequences.}$$

[Use]$^{63}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $1 \leq t < \infty$, $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x} \in \mathbf{X}^{\times}$,

$$I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) \coloneqq \begin{cases} 1 & \text{trial } t \text{ of sequence sq has assignment } \mathrm{X} = \mathrm{x} \\ 0 & \text{otherwise} \end{cases}$$

to describe a sequence $\mathrm{sq} \in \mathbf{SQ}$, where

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) = 1.$$

[Let]$^{64}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $1 \leq t < \infty$ and $\mathrm{sq} \in \mathbf{SQ}$,

$$I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) \coloneqq \sum_{? \in \mathbf{?}^{\times}} I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?\right).$$

[Define]$^{40}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, all $\mathrm{sq} \in \mathbf{SQ}$, and all $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x}\right) \coloneqq \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}\right)}{t} \right);$$

that is, the sequence probability of assignment $\mathrm{X} = \mathrm{x}$ in sequence sq.

[Define]$^{66}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x}\right) \coloneqq \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}\right)}{t} \right);$$

that is, the marginal sequence probability of marginal assignment $\mathrm{X} = \mathrm{x}$ in sequence sq.

[Define]$^{73}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right) \coloneqq \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right)}{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{Y} = \mathrm{y}\right)} \right);$$

that is, the conditional sequence probability of marginal assignment $\mathrm{X} = \mathrm{x}$ in sequence sq given marginal assignment $\mathrm{Y} = \mathrm{y}$.

[Let]$^{41}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{SQP} = \{\mathrm{sq} \in \mathbf{SQ} \mid \forall \mathrm{x} \in \mathbf{X}^{\times}, (\exists\, 1 \leq t < \infty : I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) = 1) \implies \Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x}\right) > 0\},$$

that is, the set of all probability sequences.

Propositions.

[It can be shown]$^{69}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$ and all $\mathrm{x} \in \mathbf{X}^{\times}$ and $\mathrm{sqp} \in \mathbf{SQP}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) \neq [\mathrm{null}].$$

[It can be shown]$^{71}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) = \sum_{? \in \mathbf{?}^{\times}} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, ? = ?).$$

[It can be shown]$^{90}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$, $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) = \sum_{\mathrm{y} \in \mathbf{Y}^{\times}} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

[It can be shown]$^{81}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $\mathrm{x} \in \mathbf{X}^{\times}$ and $\mathrm{sqp} \in \mathbf{SQP}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) \neq [\mathrm{null}].$$

[It can be shown]$^{91}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) = 0 \implies \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) = 0.$$

[It can be shown]$^{92}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$, $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) > 0 \implies \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) > 0.$$

[It can be shown]$^{98}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $1 \leq t < \infty$, $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x} \in \mathbf{X}^{\times}$,

$$I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = \begin{cases} 1 & \exists\, ? \in \mathbf{?}^{\times} : I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?) = 1 \\ 0 & \text{otherwise} \end{cases}$$

where

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = 1.$$

[It can be shown]$^{97}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $1 \leq t < \infty$ and $\mathrm{sqp} \in \mathbf{SQP}$,

$$\exists\, 1 \leq t < \infty : I(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}) = 1 \iff \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) > 0.$$

[It can be shown]$^{74}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})}.$$

[It can be shown]$^{89}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and for all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0 \iff \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}].$$

[It can be shown]$^{93}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and for all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) \neq [\mathrm{null}] \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}].$$

[It can be shown][94] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and for all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \times \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) = \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

[It can be shown][95] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, if $\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 1 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 1.$$

[It can be shown][96] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, if $\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 0.$$

[It can be shown][116] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Z}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}) = \sum_{\mathrm{y} \in \mathbf{Y}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}).$$

[It can be shown][118] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z})}.$$

[It can be shown][119] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}) = 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) = 0.$$

[It can be shown][120] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) > 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}) > 0.$$

[It can be shown][121] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and for all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$,

$$\begin{aligned}\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) > 0 \implies & \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) \times \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) \\ & = \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}).\end{aligned}$$

[It can be shown][128] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x})}{t} \right) \neq \text{[null]}$$

[It can be shown][127] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$,

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) = 1.$$

[It can be shown][129] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$,

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) = 1.$$

[It can be shown][130] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, and all $\mathrm{y} \in \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0 \implies \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 1.$$

[It can be shown][131] that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?}), (?', \mathbf{?}')\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, ? = ?) > 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) > 0.$$

Notation.

[Define]$^{42}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$,

$$\mathbf{D} = \{\emptyset \subset \mathrm{D} \subseteq \mathbf{SQP} \mid \forall\, \mathrm{d}, \mathrm{d}' \in \mathrm{D}^2, \forall\, \mathrm{x} \in \mathbf{X}^{\times}, \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}) = \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}\right)\}$$

that is, the set of all distributions.

[Define]$^{43}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, $\mathrm{D} \in \mathbf{D}$ and all $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\Pr(\mathrm{X} = \mathrm{x})$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \Pr(\mathrm{X} = \mathrm{x}) = \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}),$$

that is, a joint probability.

[Define]$^{46}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}^{\times}$,

$$\Pr(\mathrm{X} = \mathrm{x})$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \Pr(\mathrm{X} = \mathrm{x}) = \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}),$$

that is, a marginal probability.

[Define]$^{48}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$,

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y})$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y});$$

that is, a conditional probability.

Notation.

[Define]$^{60}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $f : \mathbf{X} \to \mathbb{R}$,

$$\mathbb{E}\left[\mathrm{sq}; f\right] = \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}\right) \times f(\mathrm{x})}{t} \right).$$

This is verbally referred to as the "sequence expectation" of f under sq.

[Define]$^{61}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$, $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$ and $\mathrm{y} \in \mathbf{Y}^{\times}$,

$$\mathbb{E}\left[\mathrm{sq}; f \mid \mathrm{Y} = \mathrm{y}\right] = \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) \times f(\mathrm{x}, \mathrm{y})}{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{Y} = \mathrm{y}\right)} \right).$$

This is verbally referred to as the "conditional sequence expectation" of f under sq given $\mathrm{Y} = \mathrm{y}$.

Propositions.

[It can be shown]$^{83}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $f : \mathbf{X} \to \mathbb{R}$,

$$\mathbb{E}\left[\mathrm{sq}; f\right] = \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x}\right) \times f(\mathrm{x}).$$

[It can be shown]$^{72}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$ and all $\mathrm{sqp} \in \mathbf{SQP}$, $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$, and $\mathrm{y} \in \mathbf{Y}^{\times}$,

$$\mathbb{E}\left[\mathrm{sqp}; f \mid \mathrm{Y} = \mathrm{y}\right] = \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right) \times f(\mathrm{x}, \mathrm{y}).$$

[It can be shown]$^{100}_{\mathrm{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$,

$$\mathbb{E}\left[\mathrm{sqp}; f\right] = \sum_{\mathrm{y} \in \mathbf{Y}^{\times} : \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0} \mathbb{E}\left[\mathrm{sqp}; f \mid \mathrm{Y} = \mathrm{y}\right] \times \Pr\left(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}\right).$$

Notation.

[Define]$^{58}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, $\mathrm{D} \in \mathbf{D}$, and all $f : \mathbf{X} \to \mathbb{R}$,

$$\mathbb{E}\left[f\right]$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \mathbb{E}\left[f\right] = \mathbb{E}\left[\mathrm{d}; f\right].$$

This is verbally referred to as the "expectation" of f under D.

[Define]$^{67}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, $\mathrm{D} \in \mathbf{D}$, and all $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$ and $\mathrm{y} \in \mathbf{Y}^{\times}$,

$$\mathbb{E}\left[f \mid \mathrm{Y} = \mathrm{y}\right]$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \mathbb{E}\left[f \mid \mathrm{Y} = \mathrm{y}\right] = \mathbb{E}\left[\mathrm{d}; f \mid \mathrm{Y} = \mathrm{y}\right].$$

This is verbally referred to as the "conditional expectation" of f under D given $\mathrm{Y} = \mathrm{y}$.

Notation.

[Define]$^{113}_{\mathrm{D}}$, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} \iff \forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}, \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}] \implies \Pr(\mathrm{x} \mid \mathrm{z}, \mathrm{y}) = \Pr(\mathrm{x} \mid \mathrm{z}),$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \iff \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \emptyset \iff \forall \mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}, \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}] \implies \Pr(\mathrm{x} \mid \mathrm{y}) = \Pr(\mathrm{x}).$$

Propositions.

[It can be shown]$^{115}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} \iff \forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) = \Pr(\mathrm{x} \mid \mathrm{z}) \times \Pr(\mathrm{y} \mid \mathrm{z}),$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \iff \forall \mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}, \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) = \Pr(\mathrm{x}) \times \Pr(\mathrm{y}).$$

[It can be shown]$^{122}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\begin{aligned} \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} \iff & \forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \\ & \Pr(\mathrm{Z} = \mathrm{z}) > 0 \implies \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = \Pr(\mathrm{x} \mid \mathrm{z}) \times \Pr(\mathrm{y}, \mathrm{z}). \end{aligned}$$

[It can be shown]$^{114}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} \iff \mathrm{Y} \perp\!\!\!\perp \mathrm{X} \mid \mathrm{Z},$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \iff \mathrm{Y} \perp\!\!\!\perp \mathrm{X}.$$

[It can be shown]$^{123}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (\mathrm{W}, \mathbf{W}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{W} \mid \mathrm{Z} \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z},\ \mathrm{X} \perp\!\!\!\perp \mathrm{W} \mid \mathrm{Z},$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{Z} \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y},\ \mathrm{X} \perp\!\!\!\perp \mathrm{Z}.$$

[It can be shown]$^{124}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (\mathrm{W}, \mathbf{W}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{W} \mid \mathrm{Z} \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z}, \mathrm{W},$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{Z} \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z}.$$

[It can be shown]$^{125}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (\mathrm{W}, \mathbf{W}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z},\ \ \mathrm{X} \perp\!\!\!\perp \mathrm{W} \mid \mathrm{Z}, \mathrm{Y}, \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{W} \mid \mathrm{Z},$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y},\ \ \mathrm{X} \perp\!\!\!\perp \mathrm{Z} \mid \mathrm{Y}, \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{Z}.$$

[It can be shown]$^{126}_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (\mathrm{W}, \mathbf{W}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z}, \mathrm{W},\ \ \mathrm{X} \perp\!\!\!\perp \mathrm{W} \mid \mathrm{Z}, \mathrm{Y},\ \ \forall \mathrm{y}, \mathrm{z}, \mathrm{w} \in \mathbf{Y}^{\times} \times \mathbf{Z}^{\times} \times \mathbf{W}^{\times}, \Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}, \mathrm{W} = \mathrm{w}) > 0$$

$\implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{W} \mid \mathrm{Z}$, and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z},\ \ \mathrm{X} \perp\!\!\!\perp \mathrm{Z} \mid \mathrm{Y},\ \ \forall \mathrm{y}, \mathrm{z} \in \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0 \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{Z}.$$

Notation.

[Define]$^{132}_{\mathtt{D}}$, for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, and all $\mathrm{E} \subseteq \mathrm{V}^2$, with $\mathrm{G} \coloneqq [\mathrm{V}, \mathrm{E}]$,

$$\mathrm{IMAP}\,(\mathrm{G}) \iff \forall\, \mathrm{C} \subseteq \mathrm{V},\ \emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V} : \mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}),\ \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \impliedby (\mathrm{G};\ \mathrm{A} \not\to \mathrm{B} \setminus \mathrm{C}).$$

[Define]$^{133}_{\mathtt{D}}$, for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, and all $\mathrm{E} \subseteq \mathrm{V}^2$, with $\mathrm{G} \coloneqq [\mathrm{V}, \mathrm{E}]$,

$$\mathrm{DMAP}\,(\mathrm{G}) \iff \forall\, \mathrm{C} \subseteq \mathrm{V},\ \emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V} : \mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}),\ \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \implies (\mathrm{G};\ \mathrm{A} \not\to \mathrm{B} \setminus \mathrm{C}).$$

[Define]$^{134}_{\mathtt{D}}$, for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, and all $\mathrm{E} \subseteq \mathrm{V}^2$, with $\mathrm{G} \coloneqq [\mathrm{V}, \mathrm{E}]$,

$$\mathrm{PMAP}\,(\mathrm{G}) \iff \mathrm{IMAP}\,(\mathrm{G}),\ \mathrm{DMAP}\,(\mathrm{G}).$$

Propositions.

[It can be shown][136]$_{\mathsf{P}}$ that, for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, if for all $\mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B}, \mathrm{S} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S},\ \mathrm{A} \perp\!\!\!\perp \mathrm{S} \mid \mathrm{C}, \mathrm{B} \implies \mathrm{A} \perp\!\!\!\perp \mathrm{B}, \mathrm{S} \mid \mathrm{C},$$

for all $\mathrm{C}, \mathrm{S} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \implies \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S},$$

and for all $\mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \implies \forall \mathrm{H} \in \mathrm{V} \setminus (\mathrm{A}, \mathrm{B}, \mathrm{C}),\ \mathrm{A} \perp\!\!\!\perp [\mathrm{H}] \mid \mathrm{C} \text{ or } [\mathrm{H}] \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}$$

then $\exists \mathrm{E} \subseteq \mathrm{V}^2 : \mathrm{PMAP}\,([\mathrm{V}, \mathrm{E}])$.

Topics.

[Reinforcement Learning Theory]$^{2}_{\mathrm{T}}$

[12/12/2020 16:47]

[106] /T/gr/D/edge

Definition.

We have by definition that [106.1]

$$\mathrm{E} \coloneqq \text{a set of edges, where } \mathrm{E} \subseteq \mathrm{V}^2.$$

[12/19/2020 19:22]

[138] /T/gr/D/spath

Definition.

We have by definition that for all $\mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B})$, [138.1]

$$(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B}) \iff \|(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B})\| > 0$$

and

$$(\mathrm{G};\ \mathrm{A} \not\longrightarrow \mathrm{B}) \iff \|(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B})\| = 0,$$

or, for contextual G,

$$\mathrm{A} \longrightarrow \mathrm{B}$$

and

$$\mathrm{A} \not\longrightarrow \mathrm{B},$$

respectively.

[12/19/2020 09:04] >[O7l8gMnpFI8]

[107] /T/gr/D/graph

Definition.

We let

$$G = [V, E], \text{ i.e., an arbitrary } \text{graph}.$$

[107.1] >

[12/22/2020 13:16] >[b3b3v3htxvw]

[112] /T/gr/D/pathsub

Definition.

We have by definition that for all $V' \subseteq V$ and $v, v' \in (V \setminus V')^2$, [112.1] >

$$(G;\ v \longrightarrow v' \setminus V') \iff \left(G \underset{\to}{\setminus} V';\ v \longrightarrow v'\right)$$

and

$$(G;\ v \not\to v' \setminus V') \iff \left(G \underset{\to}{\setminus} V';\ v \not\to v'\right),$$

or, for contextual G,

$$v \longrightarrow v' \setminus V'$$

and

$$v \not\to v' \setminus V',$$

respectively.

[12/19/2020 19:04]

[140] /T/gr/D/spathssub

Definition.

We have by definition that for all $\mathrm{V}' \subseteq \mathrm{V}$ and $\mathrm{A}, \mathrm{B} \subseteq \mathrm{V}'$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B})$, [140.1]

$$(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}') \coloneqq \left(\mathrm{G} \underset{\rightarrow}{\setminus} \mathrm{V}';\ \mathrm{A} \longrightarrow \mathrm{B}\right),$$

or, for contextual G,

$$\mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'.$$

[12/19/2020 11:07] >[PvbmXPxPXXg]

[111] /T/gr/D/path

Definition.

We have by definition that for all $\mathrm{v}, \mathrm{v}' \in \mathrm{V}^2$, [111.1] >

$$(\mathrm{G};\ \mathrm{v} \longrightarrow \mathrm{v}') \iff \|(\mathrm{G};\ \mathrm{v} \longrightarrow \mathrm{v}')\| > 0$$

and

$$(\mathrm{G};\ \mathrm{v} \not\longrightarrow \mathrm{v}') \iff \|(\mathrm{G};\ \mathrm{v} \longrightarrow \mathrm{v}')\| = 0,$$

or, for contextual G,

$$\mathrm{v} \longrightarrow \mathrm{v}'$$

and

$$\mathrm{v} \not\longrightarrow \mathrm{v}',$$

respectively.

[12/21/2020 13:46]

[144] /T/gr/D/allpathseq

Definition.

We have by definition that for all $\mathrm{P}, \mathrm{P}' \in \mathrm{PATHS}\,(\mathrm{G})$,

$$\left(\mathrm{G};\ \mathrm{P} \underset{\rightarrow}{=} \mathrm{P}'\right) \iff \mathrm{P}\,[0] = \mathrm{P}'\,[0], \mathrm{P}\,[-1] = \mathrm{P}'\,[-1],$$

or, for contextual G,

$$\mathrm{P} \underset{\rightarrow}{=} \mathrm{P}'.$$

[12/21/2020 11:28] >[n05ZAE_ei_Y]

[110] /T/gr/D/pathssub

Definition.

We have by definition that for all $V' \subseteq V$ and $v, v' \in (V \setminus V')^2$, [110.1] >

$$(G;\ v \longrightarrow v' \setminus V') := \left(G \underset{\rightarrow}{\setminus} V';\ v \longrightarrow v'\right),$$

or, for contextual G,

$$v \longrightarrow v' \setminus V'.$$

[12/23/2020 15:46]

[139] /T/gr/D/spathsub

Definition.

We have by definition that for all $\mathrm{V}' \subseteq \mathrm{V}$ and $\mathrm{A}, \mathrm{B} \in \mathrm{V}'^2$, [139.1]

$$\big(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'\big) \iff \big\|\big(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'\big)\big\| > 0$$

and

$$\big(\mathrm{G};\ \mathrm{A} \not\longrightarrow \mathrm{B} \setminus \mathrm{V}'\big) \iff \big\|\big(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'\big)\big\| = 0,$$

or, for contextual G,

$$\mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'$$

and

$$\mathrm{A} \not\longrightarrow \mathrm{B} \setminus \mathrm{V}',$$

respectively.

[12/23/2020 08:41]

[151] /T/gr/D/undir

Definition.

We have by definition that

$$\mathrm{UD}\,(\mathrm{G}) \iff \forall[\mathrm{a},\mathrm{b}] \in \mathrm{E}, [\mathrm{b},\mathrm{a}] \in \mathrm{E}$$

or, for contextual G,

$$\mathrm{UD}.$$

[12/19/2020 09:46] >[2ZwMcbGAgzw]

[109] /T/gr/D/graphsub

Definition.

We have by definition that for all $V' \subseteq V$, [109.1] >

$$G \underset{\rightarrow}{\setminus} V' = \big[V \setminus V', \{[v, v'] \in E \mid v, v' \notin V'\}\big].$$

[12/24/2020 20:01]

[137] /T/gr/D/spaths

Definition.

We have by definition that for all $\mathrm{V}' \subseteq \mathrm{V}$ and $\mathrm{A}, \mathrm{B} \in \mathrm{V}'^2$, [137.1]

$$\big(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'\big) \iff \big\|\big(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'\big)\big\| > 0$$

and

$$\big(\mathrm{G};\ \mathrm{A} \not\longrightarrow \mathrm{B} \setminus \mathrm{V}'\big) \iff \big\|\big(\mathrm{G};\ \mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'\big)\big\| = 0,$$

or, for contextual G,

$$\mathrm{A} \longrightarrow \mathrm{B} \setminus \mathrm{V}'$$

and

$$\mathrm{A} \not\longrightarrow \mathrm{B} \setminus \mathrm{V}',$$

respectively.

[12/21/2020 18:12]

[142] /T/gr/D/allpaths

Definition.

We have by definition that [142.1]

$$\mathrm{PATHS}\left(\mathrm{G}\right) \coloneqq \{\mathrm{P} \in \mathrm{V}^{*} \mid \forall 0 \leq t \leq \|\mathrm{P}\| - 2, [\mathrm{P}\left[t\right], \mathrm{P}\left[t+1\right]] \in \mathrm{E}\},$$

or, for contextual G,

$$\mathrm{PATHS}.$$

[12/12/2020 16:33] >[mN1wBvsg7mQ]

[105] /T/gr/D/node

Definition.

We have by definition that

$$V := \text{a set of nodes.}$$

[105.1] >

[12/21/2020 11:26] >[jOTSCixugsQ]

[108] /T/gr/D/paths

Definition.

We have by definition that for all $\mathrm{v}, \mathrm{v}' \in \mathrm{V}^2$, [108.1] >

$$(\mathrm{G};\ \mathrm{v} \longrightarrow \mathrm{v}') \coloneqq \{\mathrm{P} \in \mathrm{PATHS}\,(\mathrm{G}) \mid \mathrm{P}\,[0] = \mathrm{v}, \mathrm{P}\,[-1] = \mathrm{v}'\},$$

or, for contextual G,

$$\mathrm{v} \longrightarrow \mathrm{v}'.$$

[12/12/2020 15:43] >[EES1pxYDXyY]

[90] /T/gr/T/pr/P/sqmprobgexpr

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$, $\mathrm{x} \in \mathbf{X}^{\times}$, [90.1] >

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) = \sum_{\mathrm{y} \in \mathbf{Y}^{\times}} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

Proof.

We have by definition that for all $\mathrm{y} \in \mathbf{Y}^{\times}$, [90.2] >

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) \overset{[\ldots]^{71}_{\mathrm{P}}}{=} \sum_{?\in\mathbf{?}^{\times}} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, ? = ?),$$

mm

so [90.3] >

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) \overset{[\ldots]^{71}_{\mathrm{P}}}{=} \sum_{\mathrm{y} \in \mathbf{Y}^{\times}} \sum_{?\in\mathbf{?}^{\times}} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, ? = ?) \overset{\mathtt{mm}}{=} \sum_{\mathrm{y} \in \mathbf{Y}^{\times}} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

[12/25/2020 14:25] >[KLnx9hJxdzg]

[123] /T/gr/T/pr/P/inddecomp

Proposition.

We have that, for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y}), (Z, \mathbf{Z}), (W, \mathbf{W}), (?, \mathbf{?})\}$, $D \in \mathbf{D}$, [123.1] >

$$X \perp\!\!\!\perp Y, W \mid Z \overset{\text{ici}}{\implies} X \perp\!\!\!\perp Y \mid Z,\ X \perp\!\!\!\perp W \mid Z.$$

Proof.

We have that, for all $x, y, z \in \mathbf{X}^\times \times \mathbf{Y}^\times \times \mathbf{Z}^\times$, [123.2] >

$$\begin{aligned}
\Pr(X = x, Y = y \mid Z = z) &\overset{[\ldots]^{116}_{P}}{=} \sum_{w \in \mathbf{W}^\times} \Pr(x, y, W = w \mid z) \\
&\overset{\text{(ici)->}[\ldots]^{115}_{P}}{=} \sum_{w \in \mathbf{W}^\times} \Pr(x \mid z) \times \Pr(y, w \mid z) \\
&= \Pr(x \mid z) \times \sum_{w \in \mathbf{W}^\times} \Pr(y, w \mid z) \\
&\overset{[\ldots]^{116}_{P}}{=} \Pr(x \mid z) \times \Pr(y \mid z)
\end{aligned}$$

$\overset{[\ldots]^{115}_{P}}{\implies} X \perp\!\!\!\perp Y \mid Z.$

We can similarly show $X \perp\!\!\!\perp W \mid Z$. [123.3] >

Proposition.

We have that, for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y}), (Z, \mathbf{Z}), (?, \mathbf{?})\}$, $D \in \mathbf{D}$, [123.4] >

$$X \perp\!\!\!\perp Y, Z \overset{\text{ici}}{\implies} X \perp\!\!\!\perp Y,\ X \perp\!\!\!\perp Z.$$

Proof.

We have that, for all $x, y \in \mathbf{X}^\times \times \mathbf{Y}^\times$. [123.5] >

$$\begin{aligned}
\Pr(X = x, Y = y) &\overset{[\ldots]^{90}_{P}}{=} \sum_{z \in \mathbf{Z}^\times} \Pr(x, y, Z = z) \\
&\overset{\text{(ici)->}[\ldots]^{115}_{P}}{=} \sum_{z \in \mathbf{Z}^\times} \Pr(x) \times \Pr(y, z) \\
&= \Pr(x) \times \sum_{z \in \mathbf{Z}^\times} \Pr(y, z) \\
&\overset{[\ldots]^{90}_{P}}{=} \Pr(x) \times \Pr(y)
\end{aligned}$$

[...]$^{115}_{P}$

$\implies X \perp\!\!\!\perp Y.$

We can similarly show $X \perp\!\!\!\perp Z$. [123.6] >

[12/12/2020 15:43] >[htkQGdisynU]

[92] /T/gr/T/pr/P/sqmarggimpl

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$, $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, [92.1] >

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) > 0 \overset{\text{mg}}{\implies} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) > 0.$$

Proof.

We have that [92.2] >

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) \overset{[\ldots]^{90}_{\mathrm{P}}}{=} \sum_{\mathrm{y}' \in \mathbf{Y}^{\times}} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}') \geq \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) \overset{\text{mg}}{>} 0.$$

[12/25/2020 14:25] >[ZPz3PdS86Ak]

[126] /T/gr/T/pr/P/indinter

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (\mathrm{W}, \mathbf{W}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, [126.1] >

$$\underset{\texttt{icia}}{\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z}, \mathrm{W}}, \quad \underset{\texttt{icib}}{\mathrm{X} \perp\!\!\!\perp \mathrm{W} \mid \mathrm{Z}, \mathrm{Y}}, \quad \underset{\texttt{yzwg}}{\forall \mathrm{y}, \mathrm{z}, \mathrm{w} \in \mathbf{Y}^{\times} \times \mathbf{Z}^{\times} \times \mathbf{W}^{\times}, \Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}, \mathrm{W} = \mathrm{w}) > 0}$$

$$\implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{W} \mid \mathrm{Z}.$$

Proof.

We have that, for all $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, [126.2] >

$$\begin{aligned}
\Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) &\overset{\texttt{((yzwg)->}[\ldots]^{92}_{\mathrm{P}}\texttt{)->}[\ldots]^{130}_{\mathrm{P}}}{=} \Pr(\mathrm{x}, \mathrm{y} \mid \mathrm{z}) \times \sum_{\mathrm{w} \in \mathbf{W}^{\times}} \Pr(\mathrm{W} = \mathrm{w} \mid \mathrm{z}) \\
&= \sum_{\mathrm{w} \in \mathbf{W}^{\times}} \Pr(\mathrm{x}, \mathrm{y} \mid \mathrm{z}) \times \Pr(\mathrm{w} \mid \mathrm{z}) \\
&\overset{\texttt{((yzwg)->}[\ldots]^{131}_{\mathrm{P}}\texttt{)->}[\ldots]^{121}_{\mathrm{P}}}{=} \sum_{\mathrm{w} \in \mathbf{W}^{\times}} \Pr(\mathrm{x} \mid \mathrm{y}, \mathrm{z}) \times \Pr(\mathrm{y} \mid \mathrm{z}) \times \Pr(\mathrm{w} \mid \mathrm{z}) \\
&\overset{\texttt{((yzwg)->}[\ldots]^{89}_{\mathrm{P}}\texttt{)->icib}}{=} \sum_{\mathrm{w} \in \mathbf{W}^{\times}} \Pr(\mathrm{x} \mid \mathrm{y}, \mathrm{z}, \mathrm{w}) \times \Pr(\mathrm{y} \mid \mathrm{z}) \times \Pr(\mathrm{w} \mid \mathrm{z}) \\
&\overset{\texttt{((yzwg)->}[\ldots]^{89}_{\mathrm{P}}\texttt{)->icia}}{=} \sum_{\mathrm{w} \in \mathbf{W}^{\times}} \Pr(\mathrm{x} \mid \mathrm{z}, \mathrm{w}) \times \Pr(\mathrm{y} \mid \mathrm{z}) \times \Pr(\mathrm{w} \mid \mathrm{z}) \\
&\overset{[\ldots]^{118}_{\mathrm{P}}}{=} \sum_{\mathrm{w} \in \mathbf{W}^{\times}} \frac{\Pr(\mathrm{x}, \mathrm{w} \mid \mathrm{z})}{\Pr(\mathrm{w} \mid \mathrm{z})} \times \Pr(\mathrm{y} \mid \mathrm{z}) \times \Pr(\mathrm{w} \mid \mathrm{z}) \\
&\overset{\texttt{(yzwg)->}[\ldots]^{131}_{\mathrm{P}}}{=} \sum_{\mathrm{w} \in \mathbf{W}^{\times}} \Pr(\mathrm{x}, \mathrm{w} \mid \mathrm{z}) \times \Pr(\mathrm{y} \mid \mathrm{z}) \\
&\overset{[\ldots]^{116}_{\mathrm{P}}}{=} \Pr(\mathrm{x} \mid \mathrm{z}) \times \Pr(\mathrm{y} \mid \mathrm{z})
\end{aligned}$$

$$\overset{[\ldots]^{115}_{\mathrm{P}}}{\implies} \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} \overset{\texttt{(!,icib)->}[\ldots]^{125}_{\mathrm{P}}}{\implies} \mathrm{X} \perp\!\!\!\perp \mathrm{W}, \mathrm{Y} \mid \mathrm{Z}.$$ [126.3] >

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, [126.4] >

$$\underset{\texttt{icia}}{\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z}}, \quad \underset{\texttt{icib}}{\mathrm{X} \perp\!\!\!\perp \mathrm{Z} \mid \mathrm{Y}}, \quad \underset{\texttt{yzg}}{\forall \mathrm{y}, \mathrm{z} \in \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0} \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{Z}.$$

Proof.

We have that, for all $x, y \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, [126.5] >

$$
\begin{aligned}
\Pr(X = x, Y = y) &= \Pr(x, y) \times \sum_{w \in \mathbf{W}^{\times}} \Pr(Z = z) \quad [\ldots]_{P}^{129} \\
&= \sum_{w \in \mathbf{W}^{\times}} \Pr(x, y) \times \Pr(z) \\
&= \sum_{z \in \mathbf{Z}^{\times}} \Pr(x \mid y) \times \Pr(y) \times \Pr(z) \quad \texttt{((yzg)->}[\ldots]_{P}^{92}\texttt{)->}[\ldots]_{P}^{94} \\
&= \sum_{z \in \mathbf{Z}^{\times}} \Pr(x \mid y, z) \times \Pr(y) \times \Pr(z) \quad \texttt{((yzg)->}[\ldots]_{P}^{89}\texttt{)->icib} \\
&= \sum_{z \in \mathbf{Z}^{\times}} \Pr(x \mid z) \times \Pr(y) \times \Pr(z) \quad \texttt{((yzg)->}[\ldots]_{P}^{89}\texttt{)->icia} \\
&= \sum_{z \in \mathbf{Z}^{\times}} \frac{\Pr(x, z)}{\Pr(z)} \times \Pr(y) \times \Pr(z) \quad [\ldots]_{P}^{74} \\
&= \sum_{z \in \mathbf{Z}^{\times}} \Pr(x, z) \times \Pr(y) \quad \texttt{(yzg)->}[\ldots]_{P}^{92} \\
&= \Pr(x) \times \Pr(y). \quad [\ldots]_{P}^{90}
\end{aligned}
$$

$[\ldots]_{P}^{115}$ $\texttt{(!,icib)->}[\ldots]_{P}^{125}$

$\implies X \perp\!\!\!\perp Y \implies X \perp\!\!\!\perp Y, Z.$ [126.6] >

[12/13/2020 14:12] >[3rqsCIUeULk]

[94] /T/gr/T/pr/P/sqcprobtomprob

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and for all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, [94.1] >

$$\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \times \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) = \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

gz

Proof.

We have that [94.2] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \times \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) \overset{[\ldots]^{74}_{\mathrm{P}}}{=} \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})} \times \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})$$

gz

$$= \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

[12/13/2020 09:30] >[VU-2UUQsYJg]

[116] /T/gr/T/pr/P/sqcpmprobgexpr

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Z}^{\times}$, [116.1] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}) = \sum_{\mathrm{y} \in \mathbf{Y}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}).$$

Proof.

We have that [116.2] >

$$\begin{aligned}
\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}) &\overset{[\ldots]^{74}_{\mathrm{P}}}{=} \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Z} = \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{Z} = \mathrm{z})} \\
&\overset{[\ldots]^{90}_{\mathrm{P}}}{=} \frac{\sum_{\mathrm{y} \in \mathbf{Y}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{Z} = \mathrm{z})} \\
&= \sum_{\mathrm{y} \in \mathbf{Y}^{\times}} \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{Z} = \mathrm{z})} \\
&\overset{[\ldots]^{74}_{\mathrm{P}}}{=} \sum_{\mathrm{y} \in \mathbf{Y}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}).
\end{aligned}$$

[12/12/2020 15:43] >[qyuNtzbvOf0]

[96] /T/gr/T/pr/P/sqpzerocind

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, if $\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$, [96.1] >

mg

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 0.$$

cz

Proof.

We have that [96.2] >

[...]$^{74}_{\mathrm{P}}$

$$0 = \Pr(\mathrm{sqp}; \mathrm{x} \mid \mathrm{y}) = \frac{\Pr(\mathrm{sqp}; \mathrm{x}, \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{y})},$$

cz

so [96.3] >

$$\Pr(\mathrm{sqp}; \mathrm{x}, \mathrm{y}) = 0,$$

mz

and we have [96.4] >

(mz)->[...]$^{91}_{\mathrm{P}}$

$$\Pr(\mathrm{sqp}; \mathrm{x}, \mathrm{y}, \mathrm{z}) = 0,$$

fz

so [96.5] >

[...]$^{74}_{\mathrm{P}}$ fz,mg

$$\Pr(\mathrm{sqp}; \mathrm{x} \mid \mathrm{y}, \mathrm{z}) = \frac{\Pr(\mathrm{sqp}; \mathrm{x}, \mathrm{y}, \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{y}, \mathrm{z})} = 0.$$

[12/12/2020 15:43] >[sMCjQw2qy8E]

[93] /T/gr/T/pr/P/sqcprobdefimpl

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and for all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, [93.1] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) \neq [\mathrm{null}] \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}].$$

od

Proof.

We have that [93.2] >

(od)->$[\ldots]^{89}_{\mathrm{P}}$

$$\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0,$$

implying that

$[\ldots]^{92}_{\mathrm{P}}$

$$\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0,$$

and so

$[\ldots]^{89}_{\mathrm{P}}$

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}].$$

[12/14/2020 17:02] >[AVPM5hF2OLA]

[122] /T/gr/T/pr/P/indcptomp

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, [122.1] >

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} \iff \forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \\ \Pr(\mathrm{Z} = \mathrm{z}) > 0 \implies \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = \Pr(\mathrm{x} \mid \mathrm{z}) \times \Pr(\mathrm{y}, \mathrm{z}).$$

Proof.

$\implies$ suppose $\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z}$. [122.2] >

f

Then, for all $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, suppose $\Pr(\mathrm{z}) > 0$, and there are two cases to consider:

zg

- $\Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$. [122.3] >

  yzg

  Then,

$$\Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) \overset{[\ldots]^{94}_{\mathrm{P}}}{=} \Pr(\mathrm{x} \mid \mathrm{y}, \mathrm{z}) \times \Pr(\mathrm{y}, \mathrm{z}) \overset{((\mathrm{yzg})\text{->}[\ldots]^{89}_{\mathrm{P}})\text{->f}}{=} \Pr(\mathrm{x} \mid \mathrm{z}) \times \Pr(\mathrm{y}, \mathrm{z}).$$

- $\Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 0$. [122.4] >

  yze

  Then,

$$\Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) \overset{[\ldots]^{91}_{\mathrm{P}}}{=} 0 \overset{(\mathrm{zg})\text{->}[\ldots]^{89}_{\mathrm{P}},\mathrm{yze}}{=} \Pr(\mathrm{x} \mid \mathrm{z}) \times \Pr(\mathrm{y}, \mathrm{z}).$$

$\impliedby$ suppose [122.5] >

$$\forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \Pr(\mathrm{Z} = \mathrm{z}) > 0 \implies \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = \Pr(\mathrm{x} \mid \mathrm{z}) \times \Pr(\mathrm{y}, \mathrm{z}).$$

b

Then, for all $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, supposing $\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) \neq [\mathrm{null}]$, we have [122.6] >

xgyz

$$\Pr(\mathrm{y}, \mathrm{z}) \overset{(\mathrm{xgyz})\text{->}[\ldots]^{89}_{\mathrm{P}}}{>} 0,$$

yzgb

so

$$\Pr(\mathrm{z}) \overset{[\ldots]^{92}_{\mathrm{P}}}{>} 0,$$

zgb

and [122.7] >

$$
\begin{aligned}
\Pr(x \mid y, z) &= \frac{\Pr(x, y, z)}{\Pr(y, z)} \quad [\ldots]^{74}_{p} \\
&= \frac{\Pr(x \mid z) \times \Pr(y, z)}{\Pr(y, z)} \quad \texttt{(zgb)->b} \\
&= \Pr(x \mid z). \quad \texttt{yzgb}
\end{aligned}
$$

[12/16/2020 17:18] >[nt2nIFZL4Ck]

[131] /T/gr/T/pr/P/sqmarggimplcond

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?}), (?', \mathbf{?}')\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, [131.1] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, ? = ?) > 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) > 0.$$

xyqg

Proof.

We have that [131.2] >

[...]$^{74}_{\mathrm{P}}$ (xyqg)->[...]$^{92}_{\mathrm{P}}$

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})} > 0.$$

[12/16/2020 15:12] >[_44iRafvJRE]

[128] /T/gr/T/pr/P/sqpplimdef

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x} \in \mathbf{X}^{\times}$, [128.1] >

$$\lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}\right)}{t} \right) \neq [\mathrm{null}]$$

Proof.

There are two cases to consider: [128.2] >

– $\exists\, 1 \leq t < \infty : I\left(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}\right) = 1$. Then, $\Pr\left(\mathrm{sqp}; \mathrm{x}\right) > 0$, so in particular, [128.3] >

[...]$^{41}_{\mathrm{D}}$ xg

$$\lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}\right)}{t} \right) = \Pr\left(\mathrm{sqp}; \mathrm{x}\right) \neq [\mathrm{null}].$$

[...]$^{40}_{\mathrm{D}}$ xg

– $\forall\, 1 \leq t < \infty : I\left(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}\right) = 0$. Then [128.4] >

$$\lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}\right)}{t} \right) = \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} 0}{t} \right) = \lim_{t \to \infty} 0 = 0 \neq [\mathrm{null}].$$

[12/13/2020 09:40] >[vvsPtBkxdNo]

[118] /T/gr/T/pr/P/sqcpcprobexpr

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, [118.1] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z})}.$$

Proof.

There are two cases to consider: [118.2] >

- $\Pr(\mathrm{sqp}; \mathrm{Z} = \mathrm{z}) = 0$. [118.3] >

  co

  Then, [118.4] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) \overset{((\mathrm{co})\text{->}[\ldots]_{\mathrm{P}}^{91})\text{->}[\ldots]_{\mathrm{P}}^{89}}{=} [\mathrm{null}],$$

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) \overset{(\mathrm{co})\text{->}[\ldots]_{\mathrm{P}}^{89}}{=} [\mathrm{null}],$$

  and

$$\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) \overset{(\mathrm{co})\text{->}[\ldots]_{\mathrm{P}}^{89}}{=} [\mathrm{null}],$$

  so [118.6] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = [\mathrm{null}] = \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z})}.$$

- $\Pr(\mathrm{sqp}; \mathrm{Z} = \mathrm{z}) > 0$. [118.7] >

  ct

  Then, [118.8] >

$$\begin{aligned}
\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) &\overset{[\ldots]_{\mathrm{P}}^{74}}{=} \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z})} \\
&\overset{(\mathrm{ct})\text{->}[\ldots]_{\mathrm{P}}^{94}}{=} \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) \times \Pr(\mathrm{sqp}; \mathrm{Z} = \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) \times \Pr(\mathrm{sqp}; \mathrm{Z} = \mathrm{z})} \\
&= \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z})}.
\end{aligned}$$

[12/12/2020 15:43] >[jh3I3D0O99k]

[83] /T/gr/T/pr/P/sqevexpr

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $f : \mathbf{X} \to \mathbb{R}$, [83.1] >

$$\mathbb{E}\left[\mathrm{sq}; f\right] = \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x}\right) \times f(\mathrm{x}).$$

Proof.

We have that [83.2] >

$$\begin{aligned}
\mathbb{E}\left[\mathrm{sq}; f\right] &\overset{[\ldots]^{58}_{\mathrm{D}}}{=} \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}\right) \times f(\mathrm{x})}{t} \right) \\
&= \lim_{t \to \infty} \left(\sum_{\mathrm{x} \in \mathbf{X}^{\times}} \frac{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}\right) \times f(\mathrm{x})}{t} \right) \\
&= \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}\right) \times f(\mathrm{x})}{t} \right) \\
&= \sum_{\mathrm{x} \in \mathbf{X}^{\times}} f(\mathrm{x}) \times \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}\right)}{t} \right) \\
&\overset{[\ldots]^{40}_{\mathrm{D}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} f(\mathrm{x}) \times \Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x}\right).
\end{aligned}$$

[12/12/2020 15:43] >[SUHxluVewPM]

[97] /T/gr/T/pr/P/seqimimpl

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $1 \leq t < \infty$ and $\mathrm{sqp} \in \mathbf{SQP}$, [97.1] >

$$\exists\, 1 \leq t < \infty : I\left(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}\right) = 1 \iff \Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x}\right) > 0.$$

Proof.

$\implies$ suppose $\exists\, 1 \leq t < \infty : I\left(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}\right) = 1$. [97.2] >

We have that [97.3] >

$$1 = I\left(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}\right) \overset{[\ldots]^{64}_{\mathrm{D}}}{=} \sum_{?\in\mathbf{?}^{\times}} I\left(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}, ? = ?\right),$$

so we have

$$\exists ? \in \mathbf{?}^{\times}, \exists\, 1 \leq t < \infty : I\left(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}, ? = ?\right) = 1.$$

and therefore [97.4] >

$$\Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, ? = ?\right) \overset{[\ldots]^{41}_{\mathrm{D}}}{>} 0,$$

so

$$\Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x}\right) \overset{[\ldots]^{92}_{\mathrm{P}}}{>} 0.$$

$\impliedby$ suppose $\forall\, 1 \leq t < \infty, I\left(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}\right) \neq 1$. [97.5] >

Then, [97.6] >

$$\forall\, 1 \leq t < \infty, I\left(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}\right) \overset{[\ldots]^{98}_{\mathrm{P}}}{=} 0,$$

so

$$\Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x}\right) \overset{[\ldots]^{66}_{\mathrm{D}}}{=} \lim_{t\to\infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}\right)}{t}\right) = \lim_{t\to\infty} \frac{0}{t} = 0.$$

[12/12/2020 15:43] >[J_7WCWCs8Bk]

[98] /T/gr/T/pr/P/seqimzo

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $1 \leq t < \infty$, $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x} \in \mathbf{X}^{\times}$, [98.1] >

$$I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = \begin{cases} 1 & \exists\, ? \in \mathbf{?}^{\times} : I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?) = 1 \\ 0 & \text{otherwise} \end{cases}$$

where

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = 1.$$

Proof.

$\Longrightarrow$ suppose $\exists\, ? \in \mathbf{?}^{\times} : I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?) = 1$. [98.2] >

Then, [98.3] >

$$\forall\, ?' \in \mathbf{?}^{\times} : ?' \neq ?, I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?') = 0, \quad [\ldots]^{63}_{\mathrm{D}}$$

so [98.4] >

$$I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?) + \sum_{?' \in \mathbf{?}^{\times} : ?' \neq ?} I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?') = 1. \quad [\ldots]^{64}_{\mathrm{D}}$$

$\Longleftarrow$ suppose $\forall\, ? \in \mathbf{?}^{\times}, I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?) \neq 1$. [98.5] >

Then, [98.6] >

$$\forall\, ? \in \mathbf{?}^{\times}, I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?) = 0, \quad [\ldots]^{63}_{\mathrm{D}}$$

so [98.7] >

$$I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = \sum_{? \in \mathbf{?}^{\times}} I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?) = 0. \quad [\ldots]^{64}_{\mathrm{D}}$$

We have that [98.8] >

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \sum_{? \in \mathbf{?}^{\times}} I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?) = 1. \quad [\ldots]^{64}_{\mathrm{D}} \quad [\ldots]^{63}_{\mathrm{D}}$$

[12/12/2020 15:43] >[UuQzkRhewtk]

[69] /T/gr/T/pr/P/sqprobdef

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$ and all $\mathrm{x} \in \mathbf{X}^{\times}$ and $\mathrm{sqp} \in \mathbf{SQP}$, [69.1] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) \neq [\mathrm{null}].$$

Proof.

There are two cases to consider: [69.2] >

- Case 1: $\exists\, 1 \leq t < \infty : I(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}) = 1$. [69.3] > co

  Then, [69.4] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) \overset{\text{co->}[\ldots]^{41}_{\mathrm{D}}}{>} 0,$$

  and in particular

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) \neq [\mathrm{null}].$$

- Case 2: $\forall\, 1 \leq t < \infty, I(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}) \neq 1$. [69.6] >

  Then, [69.7] >

$$\forall\, 1 \leq t < \infty, I(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}) \overset{[\ldots]^{63}_{\mathrm{D}}}{=} 0,$$

  and [69.8] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) \overset{[\ldots]^{40}_{\mathrm{D}}}{=} \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x})}{t} \right) \overset{\text{iz}}{=} \lim_{t \to \infty} \left(\frac{0}{t} \right) = 0,$$

  therefore [69.9] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) = 0,$$

  and in particular

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) \neq [\mathrm{null}].$$

[12/13/2020 11:24] >[xVtNfgg4jNM]

[119] /T/gr/T/pr/P/sqcpmargzimpl

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, [119.1] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}) = 0 \overset{\text{mz}}{\implies} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) = 0.$$

Proof.

We have that [119.2] >

$$0 \overset{\text{mz}}{=} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}) \overset{[\ldots]^{116}_{\mathrm{P}}}{=} \sum_{\mathrm{y}' \in \mathbf{Y}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}' \mid \mathrm{Z} = \mathrm{z}),$$

so in particular [119.3] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) = 0.$$

[12/19/2020 11:17] >[RH82nTL9qbk]

[130] /T/gr/T/pr/P/sqcprobsumone

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, and all $\mathrm{y} \in \mathbf{Y}^{\times}$, [130.1] >

$$\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0 \overset{\text{yg}}{\implies} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = 1.$$

Proof.

We have that [130.2] >

$$\begin{aligned}
\sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) &\overset{[\ldots]^{74}_{\mathrm{P}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})} \\
&= \frac{\sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})} \\
&\overset{[\ldots]^{90}_{\mathrm{P}}}{=} \frac{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})} \overset{\text{yg}}{=} 1.
\end{aligned}$$

[12/12/2020 15:43] >[k60QFgPcs_k]

[91] /T/gr/T/pr/P/sqmargzimpl

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, [91.1] >

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) = 0 \overset{\text{mz}}{\implies} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) = 0.$$

Proof.

We have that [91.2] >

$$0 \overset{\text{mz}}{=} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) \overset{[\cdots]^{90}_{\mathrm{P}}}{=} \sum_{\mathrm{y}' \in \mathbf{Y}^{\times}} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}'),$$

so in particular [91.3] >

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) = 0.$$

[12/19/2020 11:17]

[129] /T/gr/T/pr/P/sqmprobsumone

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, [129.1]

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) = 1.$$

Proof.

We have that [129.2]

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) \overset{[\ldots]^{90}_{\mathrm{P}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \sum_{?\in \mathbf{?}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, ? = ?)$$
$$\overset{[\ldots]^{127}_{\mathrm{P}}}{=} 1.$$

[12/25/2020 10:04]

[136] /T/gr/T/pr/P/pmape

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, if for all $\mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B}, \mathrm{S} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$, [136.1]

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S},\ \mathrm{A} \perp\!\!\!\perp \mathrm{S} \mid \mathrm{C}, \mathrm{B} \underset{\texttt{int}}{\implies} \mathrm{A} \perp\!\!\!\perp \mathrm{B}, \mathrm{S} \mid \mathrm{C},$$

for all $\mathrm{C}, \mathrm{S} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \underset{\texttt{sun}}{\implies} \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S},$$

and for all $\mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \underset{\texttt{tran}}{\implies} \forall \mathrm{H} \in \mathrm{V} \setminus (\mathrm{A}, \mathrm{B}, \mathrm{C}),\ \mathrm{A} \perp\!\!\!\perp [\mathrm{H}] \mid \mathrm{C} \text{ or } [\mathrm{H}] \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}$$

then $\exists \mathrm{E} \subseteq \mathrm{V}^2 : \mathrm{PMAP}\,([\mathrm{V}, \mathrm{E}])$.
$\texttt{pmap}$

Proof.

Let $\mathrm{E} \coloneqq \{[\mathrm{H}, \mathrm{H}'] \in \mathrm{V}^2 \mid \mathrm{H} \neq \mathrm{H}',\ \forall \mathrm{V}' \subseteq \mathrm{V} \setminus \{\mathrm{H}, \mathrm{H}'\},\ [\mathrm{H}] \not\perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{V}'\}$ with contextual $\mathrm{G} \coloneqq [\mathrm{V}, \mathrm{E}]$. [136.2]
$\texttt{pmape}$

For all $\mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C})$, [136.3]

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \overset{\texttt{(int,sun)->}[\ldots]^{154}_{\mathrm{P}}}{\iff} \forall \mathrm{H}, \mathrm{H}' \in \mathrm{A} \times \mathrm{B}, [\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{C} \overset{\texttt{(int,sun,tran,pmape)->}[\ldots]^{156}_{\mathrm{P}}}{\iff} \forall \mathrm{H}, \mathrm{H}' \in \mathrm{A} \times \mathrm{B}, \mathrm{H} \not\to \mathrm{H}' \setminus \mathrm{C} \overset{[\ldots]^{138}_{\mathrm{D}}}{\iff} \mathrm{A} \not\to \mathrm{B} \setminus \mathrm{C}$$

$$\overset{[\ldots]^{134}_{\mathrm{D}}}{\implies} \mathrm{PMAP}\,(\mathrm{G}).$$ [136.4]

[12/12/2020 15:43]

[72] /T/gr/T/pr/P/sqevcexpr

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$ and all $\mathrm{sqp} \in \mathbf{SQP}$, $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$, and $\mathrm{y} \in \mathbf{Y}^{\times}$, [72.1]

$$\mathbb{E}\left[\mathrm{sqp}; f \mid \mathrm{Y} = \mathrm{y}\right] = \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right) \times f(\mathrm{x}, \mathrm{y}).$$

Proof.

We have that [72.2]

$$\begin{aligned}
\mathbb{E}\left[\mathrm{sqp}; f \mid \mathrm{Y} = \mathrm{y}\right] &\overset{[\ldots]^{61}_{\mathrm{D}}}{=} \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) \times f(\mathrm{x}, \mathrm{y})}{\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{Y} = \mathrm{y}\right)} \right) \\
&= \lim_{t \to \infty} \left(\sum_{\mathrm{x} \in \mathbf{X}^{\times}} \frac{\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) \times f(\mathrm{x}, \mathrm{y})}{\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{Y} = \mathrm{y}\right)} \right) \\
&= \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) \times f(\mathrm{x}, \mathrm{y})}{\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{Y} = \mathrm{y}\right)} \right) \\
&= \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \lim_{t \to \infty} \left(\frac{\left(\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right)\right)/t}{\left(\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{Y} = \mathrm{y}\right)\right)/t} \right) \times f(\mathrm{x}, \mathrm{y}) \\
&\overset{[\ldots]^{99}_{\mathrm{P}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \frac{\lim_{t \to \infty} \left(\left(\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right)\right)/t \right)}{\lim_{t \to \infty} \left(\left(\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{Y} = \mathrm{y}\right)\right)/t \right)} \times f(\mathrm{x}, \mathrm{y}) \\
&\overset{[\ldots]^{66}_{\mathrm{D}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \frac{\Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right)}{\Pr\left(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}\right)} \times f(\mathrm{x}, \mathrm{y}) \\
&\overset{[\ldots]^{74}_{\mathrm{P}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right) \times f(\mathrm{x}, \mathrm{y}).
\end{aligned}$$

[12/25/2020 14:40] >[OrnyeB_Uh0k]

[114] /T/gr/T/pr/P/indsymm

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, [114.1] >

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} \iff \mathrm{Y} \perp\!\!\!\perp \mathrm{X} \mid \mathrm{Z}.$$

Proof.

We have that [114.2] >

$$\begin{aligned} \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} &\overset{[\ldots]^{115}_{\mathrm{P}}}{\iff} \forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) = \Pr(\mathrm{x} \mid \mathrm{z}) \times \Pr(\mathrm{y} \mid \mathrm{z}) \\ &\overset{[\ldots]^{115}_{\mathrm{P}}}{\iff} \mathrm{Y} \perp\!\!\!\perp \mathrm{X} \mid \mathrm{Z}. \end{aligned}$$

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, [114.3] >

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \iff \mathrm{Y} \perp\!\!\!\perp \mathrm{X}.$$

Proof.

We have that [114.4] >

$$\begin{aligned} \mathrm{X} \perp\!\!\!\perp \mathrm{Y} &\overset{[\ldots]^{115}_{\mathrm{P}}}{\iff} \forall \mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}, \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) = \Pr(\mathrm{x}) \times \Pr(\mathrm{y}) \\ &\overset{[\ldots]^{115}_{\mathrm{P}}}{\iff} \mathrm{Y} \perp\!\!\!\perp \mathrm{X}. \end{aligned}$$

[12/12/2020 15:43] >[LORLTaSKQnA]

[71] /T/gr/T/pr/P/sqmprobexpr

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x} \in \mathbf{X}^{\times}$, [71.1] >

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) = \sum_{?\in\mathbf{?}^{\times}} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, ? = ?).$$

Proof.

We have that [71.2] >

$$\begin{aligned}
\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) &\overset{[\ldots]^{66}_{\mathrm{D}}}{=} \lim_{t\to\infty} \left(\frac{\sum_{t'=1}^{t} I(\mathrm{sq}, t', \mathrm{X} = \mathrm{x})}{t} \right) \\
&\overset{[\ldots]^{64}_{\mathrm{D}}}{=} \lim_{t\to\infty} \left(\frac{\sum_{t'=1}^{t} \sum_{?\in\mathbf{?}^{\times}} I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?)}{t} \right) \\
&= \lim_{t\to\infty} \left(\sum_{?\in\mathbf{?}^{\times}} \frac{\sum_{t'=1}^{t} I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?)}{t} \right) \\
&= \sum_{?\in\mathbf{?}^{\times}} \lim_{t\to\infty} \left(\frac{\sum_{t'=1}^{t} I(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?)}{t} \right) \\
&\overset{[\ldots]^{40}_{\mathrm{D}}}{=} \sum_{?\in\mathbf{?}^{\times}} \Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}, ? = ?)
\end{aligned}$$

[12/12/2020 15:43] >[OuSkDMvtB-8]

[95] /T/gr/T/pr/P/sqponecind

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, if $\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0$, [95.1] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \overset{\text{co}}{=} 1 \overset{\text{mg}}{\implies} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 1.$$

Proof.

We have that [95.2] >

$$1 \overset{\text{co}}{=} \Pr(\mathrm{sqp}; \mathrm{x} \mid \mathrm{y}) \overset{[\ldots]^{74}_{\mathrm{P}}}{=} \frac{\Pr(\mathrm{sqp}; \mathrm{x}, \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{y})},$$

so [95.3] >

$$\Pr(\mathrm{sqp}; \mathrm{x}, \mathrm{y}) \overset{\text{me}}{=} \Pr(\mathrm{sqp}; \mathrm{y}) \overset{[\ldots]^{90}_{\mathrm{P}}}{=} \Pr(\mathrm{sqp}; \mathrm{x}, \mathrm{y}) + \sum_{\mathrm{x}' \in \mathbf{X}^{\times}, \mathrm{x}' \neq \mathrm{x}} \Pr(\mathrm{sqp}; \mathrm{x}', \mathrm{y}),$$

and we have [95.4] >

$$\forall \mathrm{x}' \in \mathbf{X}^{\times}, \mathrm{x}' \neq \mathrm{x}, \Pr(\mathrm{sqp}; \mathrm{x}', \mathrm{y}) = 0,$$

so [95.5] >

$$\forall \mathrm{x}' \in \mathbf{X}^{\times}, \mathrm{x}' \neq \mathrm{x}, \Pr(\mathrm{sqp}; \mathrm{x}', \mathrm{y}, \mathrm{z}) \overset{[\ldots]^{91}_{\mathrm{P}}}{=} 0,$$

and therefore, [95.6] >

$$\Pr(\mathrm{sqp}; \mathrm{x}, \mathrm{y}, \mathrm{z}) \overset{\text{az}}{=} \Pr(\mathrm{sqp}; \mathrm{x}, \mathrm{y}, \mathrm{z}) \overset{\text{arm}}{+} \sum_{\mathrm{x}' \in \mathbf{X}^{\times}, \mathrm{x}' \neq \mathrm{x}} \Pr(\mathrm{sqp}; \mathrm{x}', \mathrm{y}, \mathrm{z}) \overset{[\ldots]^{90}_{\mathrm{P}}}{=} \Pr(\mathrm{sqp}; \mathrm{y}, \mathrm{z}) \overset{\text{rm}}{>} 0 \overset{\text{mg}}{\underset{\text{gf}}{}},$$

so [95.7] >

$$\Pr(\mathrm{sqp}; \mathrm{x} \mid \mathrm{y}, \mathrm{z}) \overset{[\ldots]^{74}_{\mathrm{P}}}{=} \frac{\Pr(\mathrm{sqp}; \mathrm{x}, \mathrm{y}, \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{y}, \mathrm{z})} \overset{\text{arm-rm,gf}}{=} 1.$$

[12/25/2020 14:24] >[LIlZNSoNlBQ]

[124] /T/gr/T/pr/P/indwun

Proposition.

We have that, for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y}), (Z, \mathbf{Z}), (W, \mathbf{W}), (?, \mathbf{?})\}$, $D \in \mathbf{D}$, [124.1] >

$$X \perp\!\!\!\perp Y, W \mid Z \implies X \perp\!\!\!\perp Y \mid Z, W.$$

ici

Proof.

For all $x, y, z, w \in \mathbf{X}^\times \times \mathbf{Y}^\times \times \mathbf{Z}^\times \times \mathbf{W}^\times$, there are two cases to consider: [124.2] >

- $\Pr(Y = y \mid Z = z, W = w) = [\text{null}]$. In this case, $\Pr(z, w) = 0$ [...]$^{89}_{P}$, so [124.3] >

zwz

$$\Pr(X = x, y \mid z, w) = \overset{\text{(zwz)->[...]}^{89}_{P}}{[\text{null}]} = \overset{\text{(zwz)->[...]}^{89}_{P}}{\Pr(x \mid z, w) \times \Pr(y \mid z, w)}.$$

- $\Pr(Y = y \mid Z = z, W = w) \neq [\text{null}]$. Then, $\Pr(z, w) > 0$ [...]$^{89}_{P}$, and there are two cases to consider: [124.4] >

zwg

  - $\Pr(y \mid z, w) = 0$. Then, we have: [124.5] >

ygzwz

$$\Pr(X = x, y \mid z, w) = \overset{[\ldots]^{119}_{P}}{0} = \overset{\text{(zwg)->[...]}^{89}_{P}\text{,ygzwz}}{\Pr(x \mid z, w) \times \Pr(y \mid z, w)}.$$

  - $\Pr(y \mid z, w) > 0$. Then, we have $\Pr(y, z, w) = \overset{\text{(zwg)->[...]}^{94}_{P}}{\Pr(y \mid z, w)} \times \overset{\text{ygzwg,zwg}}{\Pr(z, w)} > 0$. [124.6] >

ygzwg yzwg

    Therefore, noting that $X \perp\!\!\!\perp W \mid Z$, [124.7] >

ici(ii) -> [...]$^{123}_{P}$

icii

$$\begin{aligned}\Pr(X = x, y \mid z, w) &= \overset{\text{(ygzwg)->[...]}^{119}_{P}}{\Pr(x \mid y, z, w) \times \Pr(y \mid z, w)} \\ &= \overset{\text{((-yzwg)->[...]}^{89}_{P}\text{)->ici}}{\Pr(x \mid z) \times \Pr(y \mid z, w)} \\ &= \overset{\text{((zwg)->[...]}^{89}_{P}\text{)->icii}}{\Pr(x \mid z, w) \times \Pr(y \mid z, w)}.\end{aligned}$$

[...]$^{115}_{P}$

$\implies X \perp\!\!\!\perp Y \mid Z, W.$ [124.8] >

Proposition.

We have that, for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y}), (Z, \mathbf{Z}), (?, \mathbf{?})\}$, $D \in \mathbf{D}$, [124.9] >

$$X \perp\!\!\!\perp Y, Z \implies X \perp\!\!\!\perp Y \mid Z.$$

ici

Proof.

For all $x, y, z \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, there are two cases to consider: [124.10] >

- $\Pr(Y = y \mid Z = z) = [\text{null}]$. In this case, $\Pr(z) = 0$, so [124.11] >

  $[\ldots]_P^{89}$ zz

$$\Pr(X = x, y \mid z) = \overset{\texttt{(zz)->}[\ldots]_P^{89}}{[\text{null}]} \overset{\texttt{(zz)->}[\ldots]_P^{89}}{=} \Pr(x \mid z) \times \Pr(y \mid z).$$

- $\Pr(Y = y \mid Z = z) \neq [\text{null}]$. Then, $\Pr(z) > 0$, and there are two cases to consider: [124.12] >

  $[\ldots]_P^{89}$ zg

  - $\Pr(y \mid z) = 0$. Then, we have: [124.13] >

    ygzz

$$\Pr(X = x, y \mid z) = \overset{[\ldots]_P^{119}}{0} \overset{\texttt{(zg)->}[\ldots]_P^{89}\texttt{,ygzz}}{=} \Pr(x \mid z) \times \Pr(y \mid z).$$

  - $\Pr(y \mid z) > 0$. Then, we have $\Pr(y, z) \overset{\texttt{(zg)->}[\ldots]_P^{94}}{=} \Pr(y \mid z) \times \Pr(z) \overset{\texttt{ygzg,zg}}{>} 0$. [124.14] >

    ygzg yzg

    Therefore, noting that $X \overset{(\texttt{ici})\texttt{->}[\ldots]_P^{123}}{\perp\!\!\!\perp} Z$, [124.15] >

    icii

$$\begin{aligned}\Pr(X = x, y \mid z) &\overset{\texttt{(ygzg)->}[\ldots]_P^{119}}{=} \Pr(x \mid y, z) \times \Pr(y \mid z) \\ &\overset{\texttt{((-yzg)->}[\ldots]_P^{89}\texttt{)->ici}}{=} \Pr(x) \times \Pr(y \mid z) \\ &\overset{\texttt{((zg)->}[\ldots]_P^{89}\texttt{)->icii}}{=} \Pr(x \mid z) \times \Pr(y \mid z).\end{aligned}$$

$[\ldots]_P^{115}$

$\implies X \perp\!\!\!\perp Y \mid Z.$ [124.16] >

[12/19/2020 11:17] >[L94NBIoC7oI]

[127] /T/gr/T/pr/P/sqprobsumone

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, [127.1] >

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) = 1.$$

Proof.

We have that [127.2] >

$$\begin{aligned}
\sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) &\overset{[\ldots]^{40}_{\mathrm{D}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x})}{t} \right) \\
&\overset{[\ldots]^{128}_{\mathrm{P}}}{=} \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} I(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x})}{t} \right) \\
&\overset{[\ldots]^{63}_{\mathrm{D}}}{=} \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} 1}{t} \right) \\
&= \lim_{t \to \infty} 1 = 1.
\end{aligned}$$

[12/12/2020 15:43] >[xsxQq03wWZI]

[81] /T/gr/T/pr/P/sqmprobdef

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $\mathrm{x} \in \mathbf{X}^{\times}$ and $\mathrm{sqp} \in \mathbf{SQP}$, [81.1] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) \neq [\mathrm{null}].$$

Proof.

We have that [81.2] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}) \overset{[\cdots]^{71}_{\mathrm{P}}}{=} \sum_{?\in\mathbf{?}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, ? = ?) \overset{[\cdots]^{69}_{\mathrm{P}}}{\neq} [\mathrm{null}].$$

[12/13/2020 11:38] >[ieiYYQ19nsY]

[120] /T/gr/T/pr/P/sqcpmarggimpl

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, [120.1] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) > 0 \implies \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}) > 0.$$

mg

Proof.

We have that [120.2] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}) \overset{[\ldots]^{116}_{\mathrm{P}}}{=} \sum_{\mathrm{y}' \in \mathbf{Y}^{\times}} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}' \mid \mathrm{Z} = \mathrm{z}) \geq \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) > 0.$$

mg

[12/25/2020 14:41] >[Ql24lsFExU0]

[115] /T/gr/T/pr/P/indprod

Proposition.

We have that, for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y}), (Z, \mathbf{Z}), (?, ?)\}$, $D \in \mathbf{D}$, [115.1] >

$$X \perp\!\!\!\perp Y \mid Z \iff \forall x, y, z \in \mathbf{X}^\times \times \mathbf{Y}^\times \times \mathbf{Z}^\times, \Pr(X = x, Y = y \mid Z = z) = \Pr(x \mid z) \times \Pr(y \mid z).$$

Proof.

$\implies$ suppose $X \perp\!\!\!\perp Y \mid Z$. [115.2] >
f

Then, for all $x, y, z \in \mathbf{X}^\times \times \mathbf{Y}^\times \times \mathbf{Z}^\times$, there are two cases to consider:

- $\Pr(X = x, Y = y \mid Z = z) = [\text{null}]$. [115.4] >
  fco

  Then, $\Pr(z) = 0$ ($[\ldots]^{89}_{P}$), so $\Pr(x \mid z) = [\text{null}]$ ($[\ldots]^{89}_{P}$) and $\Pr(y \mid z) = [\text{null}]$ ((zz)->$[\ldots]^{89}_{P}$), so
  zz

  $$\Pr(x, y \mid z) = [\text{null}] = \Pr(x \mid z) \times \Pr(y \mid z).$$
  fco

- $\Pr(X = x, Y = y \mid Z = z) \neq [\text{null}]$. [115.5] >
  fct

  Then, we have $\Pr(z) > 0$ ($[\ldots]^{89}_{P}$), so $\Pr(y \mid z) \neq [\text{null}]$ ($[\ldots]^{89}_{P}$), and there are two cases to consider:
  zg

  - $\Pr(y \mid z) > 0$. [115.6] >
    fcto

    Then, [115.7] >

    $$\Pr(y, z) = \Pr(y \mid z) \times \Pr(z) > 0,$$
    (zg)->$[\ldots]^{94}_{P}$ fcto,zg yzg

    so [115.8] >

    $$\Pr(x, y \mid z) = \Pr(x \mid y, z) \times \Pr(y \mid z)$$
    (fcto)->$[\ldots]^{121}_{P}$

    $$= \Pr(x \mid z) \times \Pr(y \mid z).$$
    ((-yzg)->$[\ldots]^{89}_{P}$)->f

  - $\Pr(y \mid z) = 0$. [115.9] >
    fctt

Then,

$$\Pr(x, y \mid z) = \overset{[\ldots]^{120}_{P}}{} \overset{\texttt{fctt,(zg)->}[\ldots]^{89}_{P}}{0} = \Pr(x \mid z) \times \Pr(y \mid z).$$

$\Longleftarrow$ suppose [115.10] >

$$\forall x, y, z \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \Pr(X = x, Y = y \mid Z = z) \underset{\texttt{b}}{=} \Pr(X = x \mid Z = z) \times \Pr(Y = y \mid Z = z).$$

Then, for all $x, y, z \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, supposing $\Pr(X = x \mid Y = y, Z = z) \underset{\texttt{ba}}{\neq} [\text{null}]$, there are two cases to consider:

- $\Pr(z) > 0$. [115.11] >
  bco

  Then, we have that [115.12] >

$$\Pr(y \mid z) = \overset{[\ldots]^{74}_{P}}{} \overset{\texttt{bco,(ba)->}[\ldots]^{89}_{P}}{\frac{\Pr(y, z)}{\Pr(z)}} > 0.$$

  ygzg

  so [115.13] >

$$\begin{aligned}
\Pr(x \mid y, z) &= \overset{[\ldots]^{118}_{P}}{} \frac{\Pr(x, y \mid z)}{\Pr(y \mid z)} \\
&\overset{\texttt{b}}{=} \frac{\Pr(x \mid z) \times \Pr(y \mid z)}{\Pr(y \mid z)} \\
&\overset{\texttt{-ygzg}}{=} \Pr(x \mid z).
\end{aligned}$$

- $\Pr(z) = 0$. [115.14] >
  bct

  Then,

$$\Pr(x \mid y, z) = \overset{\texttt{((bct)->}[\ldots]^{91}_{P}\texttt{)->}[\ldots]^{89}_{P}}{[\text{null}]} \overset{\texttt{(bct)->}[\ldots]^{89}_{P}}{=} \Pr(x \mid z).$$

Proposition.

We have that, for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y}), (?, \mathbf{?})\}$, $D \in \mathbf{D}$, [115.20] >

$$X \perp\!\!\!\perp Y \iff \forall x, y \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}, \Pr(X = x, Y = y) = \Pr(x) \times \Pr(y).$$

Proof.

$\Longrightarrow$ suppose $X \underset{\texttt{f}}{\perp\!\!\!\perp} Y$. [115.15] >

Then, for all $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, there are two cases to consider:

– $\Pr(\mathrm{y}) > 0$. [115.16] >

fco

Then,

$$\overset{[\ldots]^{94}_{\mathrm{P}}}{\Pr(\mathrm{x}, \mathrm{y}) = \Pr(\mathrm{x} \mid \mathrm{y}) \times \Pr(\mathrm{y})}$$

$$\overset{\texttt{((fco)->}[\ldots]^{89}_{\mathrm{P}}\texttt{)->f}}{= \Pr(\mathrm{x}) \times \Pr(\mathrm{y})}.$$

– $\Pr(\mathrm{y}) = 0$. [115.17] >

fct

Then,

$$\Pr(\mathrm{x}, \mathrm{y}) = \overset{[\ldots]^{92}_{\mathrm{P}}}{0} \overset{\texttt{fct}}{=} \Pr(\mathrm{x}) \times \Pr(\mathrm{y}).$$

$\Longleftarrow$ suppose [115.18] >

$$\forall \mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}, \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) = \Pr(\mathrm{X} = \mathrm{x}) \times \Pr(\mathrm{Y} = \mathrm{y}).$$

b

Then, for all $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, supposing $\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}]$, we have that $\overset{[\ldots]^{89}_{\mathrm{P}}}{\Pr(\mathrm{y}) > 0}$, so [115.19] >

yg

$$\overset{[\ldots]^{74}_{\mathrm{P}}}{\Pr(\mathrm{x} \mid \mathrm{y}) = \frac{\Pr(\mathrm{x}, \mathrm{y})}{\Pr(\mathrm{y})}}$$

$$\overset{\texttt{b}}{= \frac{\Pr(\mathrm{x}) \times \Pr(\mathrm{y})}{\Pr(\mathrm{y})}}$$

$$\overset{\texttt{yg}}{= \Pr(\mathrm{x})}.$$

[12/12/2020 15:43] >[WiA2ngTJP-c]

[89] /T/gr/T/pr/P/sqcprobdef

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and for all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, [89.1] >

$$\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0 \iff \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}].$$

Proof.

$\implies$ suppose $\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0$. [89.2] >

gz

Then, [89.3] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) \overset{[\ldots]^{81}_{\mathrm{P}}}{\neq} [\mathrm{null}],$$

od

so,

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \overset{[\ldots]^{74}_{\mathrm{P}}}{=} \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})} \overset{\mathrm{od,gz}}{\neq} [\mathrm{null}].$$

$\impliedby$ suppose $\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) = 0$. [89.6] >

ez

We know that [89.7] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \overset{[\ldots]^{74}_{\mathrm{P}}}{=} \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})} \overset{\mathrm{ez}}{=} [\mathrm{null}].$$

[12/12/2020 15:43] >[6iPsvtiaLSQ]

[100] /T/gr/T/pr/P/sqevctoev

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$, $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$, [100.1] >

$$\mathbb{E}\left[\mathrm{sqp}; f\right] = \sum_{\mathrm{y} \in \mathbf{Y}^{\times} : \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0} \mathbb{E}\left[\mathrm{sqp}; f \mid \mathrm{Y} = \mathrm{y}\right] \times \Pr\left(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}\right).$$

Proof.

We have that [100.2] >

$$\begin{aligned}
\mathbb{E}\left[\mathrm{sqp}; f\right] &\overset{[\ldots]^{83}_{\mathrm{P}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \sum_{\mathrm{y} \in \mathbf{Y}^{\times}} f(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) \times \Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) \\
&\overset{[\ldots]^{91}_{\mathrm{P}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \sum_{\mathrm{y} \in \mathbf{Y}^{\times} : \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0} f(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) \times \Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) \\
&\overset{[\ldots]^{94}_{\mathrm{P}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \sum_{\mathrm{y} \in \mathbf{Y}^{\times} : \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0} f(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) \times \Pr\left(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}\right) \times \Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right) \\
&= \sum_{\mathrm{y} \in \mathbf{Y}^{\times} : \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0} \Pr\left(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}\right) \times \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right) f(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) \\
&\overset{[\ldots]^{72}_{\mathrm{P}}}{=} \sum_{\mathrm{y} \in \mathbf{Y}^{\times} : \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}) > 0} \Pr\left(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}\right) \times \mathbb{E}\left[\mathrm{sqp}; f \mid \mathrm{Y} = \mathrm{y}\right].
\end{aligned}$$

[12/25/2020 14:24] >[lZQ6g5Pya-A]

[125] /T/gr/T/pr/P/indcontr

Proposition.

We have that, for contextual $S \leftarrow \{(X, \mathbf{X}), (Y, \mathbf{Y}), (Z, \mathbf{Z}), (W, \mathbf{W}), (?, \mathbf{?})\}$, $D \in \mathbf{D}$, [125.1] >

$$\underset{\texttt{icia}}{X \perp\!\!\!\perp Y \mid Z}, \quad \underset{\texttt{icib}}{X \perp\!\!\!\perp W \mid Z, Y}, \implies X \perp\!\!\!\perp Y, W \mid Z.$$

Proof.

For all $x, y, z, w \in \mathbf{X}^\times \times \mathbf{Y}^\times \times \mathbf{Z}^\times \times \mathbf{W}^\times$, there are two cases to consider: [125.2] >

- $\Pr(Y = y, W = w \mid Z = z) = [\text{null}]$. In this case, $\Pr(z) = 0$ (zz: $[\ldots]^{89}_{P}$), so [125.3] >

$$\Pr(X = x, y, w \mid z) \overset{\texttt{(zz)->}[\ldots]^{89}_{P}}{=} [\text{null}] \overset{\texttt{(zz)->}[\ldots]^{89}_{P}}{=} \Pr(x \mid z) \times \Pr(y, w \mid z).$$

- $\Pr(Y = y, W = w \mid Z = z) \neq [\text{null}]$. Then, $\Pr(z) > 0$ (zg: $[\ldots]^{89}_{P}$), and there are two cases to consider: [125.4] >

  - $\Pr(y, w \mid z) = 0$ (ywgzz). Then, we have: [125.5] >

$$\Pr(X = x, y, w \mid z) \overset{[\ldots]^{119}_{P}}{=} 0 \overset{\texttt{(zg)->}[\ldots]^{89}_{P}\texttt{,ywgzz}}{=} \Pr(x \mid z) \times \Pr(y, w \mid z).$$

  - $\Pr(y, w \mid z) > 0$ (ywgzg). Then, we have $\Pr(y, z, w) \overset{\texttt{(zg)->}[\ldots]^{94}_{P}}{=} \Pr(y, w \mid z) \times \Pr(z) \overset{\texttt{ywgzg,zg}}{>} 0$ (yzwg). [125.6] >

    Therefore, [125.7] >

$$\begin{aligned}\Pr(X = x, y, w \mid z) &\overset{\texttt{(ywgzg)->}[\ldots]^{121}_{P}}{=} \Pr(x \mid y, z, w) \times \Pr(y, w \mid z) \\ &\overset{\texttt{((-yzwg)->}[\ldots]^{89}_{P}\texttt{)->icib}}{=} \Pr(x \mid y, z) \times \Pr(y, w \mid z) \\ &\overset{\texttt{(((-yzwg)->}[\ldots]^{92}_{P}\texttt{)->}[\ldots]^{89}_{P}\texttt{)->icia}}{=} \Pr(x \mid z) \times \Pr(y, w \mid z)\end{aligned}$$

$$\overset{[\ldots]^{115}_{P}}{\implies} X \perp\!\!\!\perp Y, W \mid Z.$$ [125.8] >

Proposition.

We have that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, [125.9] >

$$\underbrace{\mathrm{X} \perp\!\!\!\perp \mathrm{Y}}_{\texttt{icia}}, \quad \underbrace{\mathrm{X} \perp\!\!\!\perp \mathrm{Z} \mid \mathrm{Y}}_{\texttt{icib}}, \implies \mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{Z}.$$

Proof.

For all $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, there are two cases to consider: [125.10] >

- $\overbrace{\Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) = 0}^{\texttt{yzz}}$. Then, [125.11] >

$$\Pr(\mathrm{X} = \mathrm{x}, \mathrm{y}, \mathrm{z}) = \overbrace{0}^{\texttt{(yzz)->}[\ldots]_{\mathrm{P}}^{89}} \overbrace{= \Pr(\mathrm{x}) \times \Pr(\mathrm{y}, \mathrm{z})}^{\texttt{yzz}}.$$

- $\overbrace{\Pr(\mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) > 0}^{\texttt{yzg}}$. Then, [125.12] >

$$\begin{aligned}
\Pr(\mathrm{X} = \mathrm{x}, \mathrm{y}, \mathrm{z}) &\overset{\texttt{(yzg)->}[\ldots]_{\mathrm{P}}^{119}}{=} \Pr(\mathrm{x} \mid \mathrm{y}, \mathrm{z}) \times \Pr(\mathrm{y}, \mathrm{z}) \\
&\overset{\texttt{((yzg)->}[\ldots]_{\mathrm{P}}^{89}\texttt{)->icib}}{=} \Pr(\mathrm{x} \mid \mathrm{y}) \times \Pr(\mathrm{y}, \mathrm{z}) \\
&\overset{\texttt{(((yzg)->}[\ldots]_{\mathrm{P}}^{92}\texttt{)->}[\ldots]_{\mathrm{P}}^{89}\texttt{)->icia}}{=} \Pr(\mathrm{x}) \times \Pr(\mathrm{y}, \mathrm{z})
\end{aligned}$$

$\overset{[\ldots]_{\mathrm{P}}^{115}}{\implies} \mathrm{X} \perp\!\!\!\perp \mathrm{Y}, \mathrm{Z}.$ [125.13] >

[12/12/2020 15:43] >[sIaGXp-pwiU]

[74] /T/gr/T/pr/P/sqcprobexpr

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, [74.1] >

$$\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})}.$$

Proof.

We have that [74.2] >

$$\begin{aligned}
\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) &\overset{[\ldots]^{73}_{\mathrm{D}}}{=} \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\sum_{t'=1}^{t} I(\mathrm{sqp}, t', \mathrm{Y} = \mathrm{y})} \right) \\
&= \lim_{t \to \infty} \left(\frac{\left(\sum_{t'=1}^{t} I(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})\right)/t}{\left(\sum_{t'=1}^{t} I(\mathrm{sqp}, t', \mathrm{Y} = \mathrm{y})\right)/t} \right) \\
&\overset{[\ldots]^{99}_{\mathrm{P}}}{=} \frac{\lim_{t \to \infty} \left(\left(\sum_{t'=1}^{t} I(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})\right)/t \right)}{\lim_{t \to \infty} \left(\left(\sum_{t'=1}^{t} I(\mathrm{sqp}, t', \mathrm{Y} = \mathrm{y})\right)/t \right)} \\
&\overset{[\ldots]^{66}_{\mathrm{D}}}{=} \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y})}.
\end{aligned}$$

[12/13/2020 15:06] >[fk-jmNJ5CX0]

[121] /T/gr/T/pr/P/sqcpcprobtomprob

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, and for all $\mathrm{sqp} \in \mathbf{SQP}$, $\mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}$, [121.1] >

$$\begin{aligned} \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) > 0 \overset{\text{gz}}{\implies} & \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) \times \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) \\ & = \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}). \end{aligned}$$

Proof.

We have that [121.2] >

$$\begin{aligned} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}, \mathrm{Z} = \mathrm{z}) \times \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) &= \frac{\Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z})}{\Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z})} \times \\ & \Pr(\mathrm{sqp}; \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}) \\ & \overset{\text{gz}}{=} \Pr(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y} \mid \mathrm{Z} = \mathrm{z}). \end{aligned}$$

[12/12/2020 15:43] >[DpI2Wd37gV8]

[2] /T/gr/T/pr/T/rl

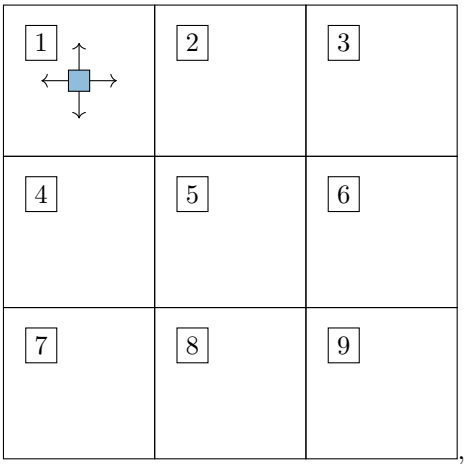
Reinforcement Learning Theory

[2.1] >

Adapted from *Neuro-Dynamic Programming* by
Dmitri P. Bertsekas and John N. Tsitsiklis.

Physical intuition.

Consider the following "environment": [2.2] >



,

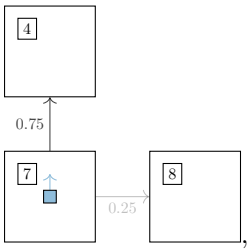
where there are a total of 9 "states" (□), an "agent" (■), and a set of "actions" (→) that the agent can take at any state.

At every "step" in time, the agent is in one of the states, and must take one of the actions. [2.3] >

At any step, if the agent takes an action in a state, it will be in a state at the next step according to a "transition distribution" (that only depends on the current state and action, not the current step). [2.4] >

For example, the agent could

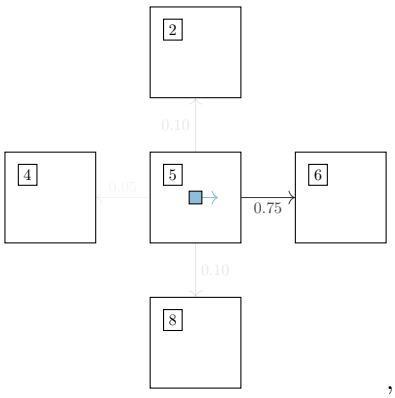
– take action ↑ at state 7:



,

and

– take action → at state 5:



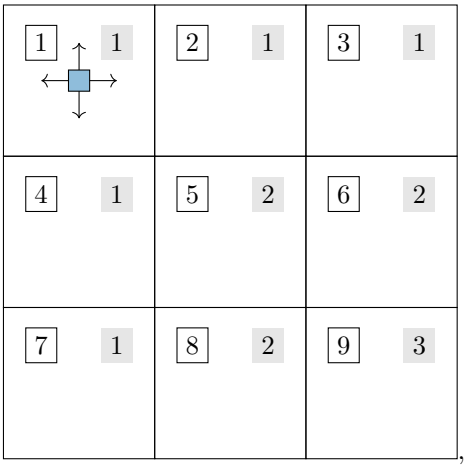
,

showing the conditional probability of being in another state at the next step according to the transition distribution.

However, this begs the question, what state is the agent at initially (step 1)? To decide this, there is also an initial distribution over the states. [2.5] >

Now, let's make things a bit more interesting. [2.6] >

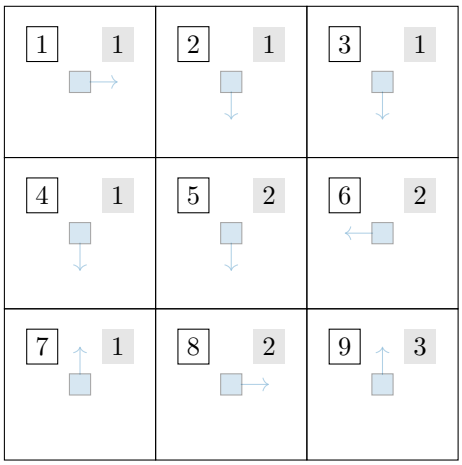
Let every state also have a return that the agent recieves from it, leaving our environment looking something like:

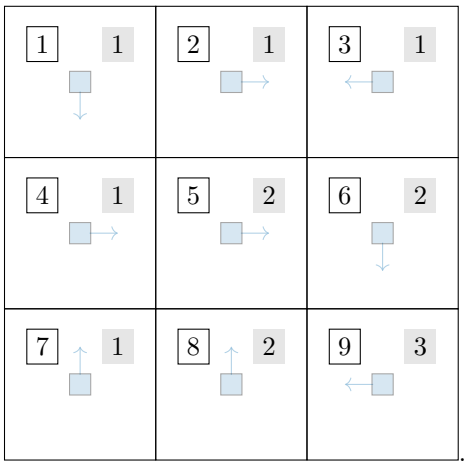


with the returns (·) shown.

Also, let there be some fixed "step limit" such that, when the current step reaches it, everything ends. [2.7] >

Now, suppose you can "program" the agent as follows: for every state and step, you can fix the action that the agent takes. For example, here's what you might program the agent to do at every state for two different steps: [2.8] >





We'll call each such programming a "strategy".

We know that the agent starts in some state according to the initial distribution. After this, we follow the strategy, transitioning states according to the transition distribution, recieving returns at every state until we reach the step limit. Let's refer to this process as a "run", and the sum of the returns over all a run's states as the "total return". [2.9] >

Because there is some initial distribution and transition distribution that are fixed over runs, we know there must be some average total return for every strategy. [2.10] >

Now, consider the following question: what strategy has the greatest average total return? [2.11] >

Notation.

[Define]$^{26}_{\mathrm{D}}$

$$\mathbf{S} \coloneqq \text{the set of states},$$

where $\mathbf{S}$ is finite.

[Define]$^{30}_{\mathrm{D}}$

$$\mathbf{A} \coloneqq \text{the set of actions},$$

where $\mathbf{A}$ is finite.

[Define]$^{19}_{\mathrm{D}}$

$$R(\cdot) : \mathbf{S} \to \mathbb{R} \coloneqq \text{mapping from states to returns}.$$

[Define]$^{21}_{\mathrm{D}}$

$$T \coloneqq \text{the step limit, where } T \geq 1,$$

where t denotes a step index, with $1 \leq t \leq T$.

[Define]$^{34}_{\mathrm{D}}$

$$\mathbf{\Pi} \coloneqq \text{the set of all strategies},$$

where, for each strategy $\pi \in \mathbf{\Pi}$ and each $1 \leq t \leq T$,

$$\pi(t; \cdot) : \mathbf{S} \to \mathbf{A} \coloneqq \text{a step-strategy}.$$

[Note that]$^{4}_{\mathrm{P}}$ the total number of strategies can be written as

$$|\mathbf{\Pi}| = (T-1) \times |\mathbf{S}| \times |\mathbf{A}|.$$

[Define]$^{23}_{\mathrm{D}}$ for any $\mathrm{s} \in \mathbf{S}$,

$$\mathrm{P_I}(\mathrm{s}) \coloneqq \text{the initial distribution probability of starting in state s}.$$

[Define]$^{28}_{\mathrm{D}}$ for any $\mathrm{s}, \mathrm{a} \in \mathbf{S} \times \mathbf{A}$, $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T-1$,

$$\mathrm{P_A}(\pi, t : \mathrm{a} \mid \mathrm{s}) \coloneqq \begin{cases} 1 & \pi(t; \mathrm{s}) = \mathrm{a} \\ 0 & \pi(t; \mathrm{s}) \neq \mathrm{a}. \end{cases}$$

[Define]$^{31}_{\mathrm{D}}$ for any $\mathrm{s}, \mathrm{s}', \mathrm{a} \in \mathbf{S}^2 \times \mathbf{A}$,

$$\mathrm{P_T}(\mathrm{s} \mid \mathrm{a}, \mathrm{s}') \coloneqq \text{the transition distribution probability of going to state s after action a in state s}'.$$

[Define]$^{29}_{\mathrm{D}}$ for all $1 \leq t \leq T$,

$$\mathrm{S}_t \coloneqq \text{variable for the state at step } t,$$

with domain $\mathbf{S}$.

[Define]$^{27}_{\mathrm{D}}$ for all $1 \leq t \leq T-1$,

$$\mathrm{A}_t \coloneqq \text{variable for the action at step } t,$$

with domain $\mathbf{A}$.

[Define]$^{88}_{\mathrm{D}}$ for all $1 \leq t \leq T$,

$$R(t; \cdot) : (\mathbf{S} \times \mathbf{A})^{T-1} \times \mathbf{S} \to \mathbb{R}$$

such that

$$R(t; \mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T) \coloneqq \sum_{t'=t}^{T} R(\mathrm{s}'_t).$$

[Let]$^{49}_{\mathrm{D}}$

$$\mathrm{S} \leftarrow \{(\mathrm{S}_1, \mathbf{S}), (\mathrm{A}_1, \mathbf{A}), (\mathrm{S}_2, \mathbf{S}), \ldots, (\mathrm{A}_{T-1}, \mathbf{A}), (\mathrm{S}_T, \mathbf{S})\},$$

and, for each $\pi \in \mathbf{\Pi}$, let $\mathrm{D}_\pi \in \mathbf{D}$ be a distribution.

[We know that]$^{25}_{\mathrm{D}}$, for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and for all $\mathrm{s}, \mathrm{a} \in \mathbf{S} \times \mathbf{A}$, $1 \leq t \leq T - 1$, if $\Pr(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) \neq$ [null],

$$\Pr(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) = \mathrm{P}_\mathrm{A}(\pi, t : \mathrm{a} \mid \mathrm{s}),$$

[from which it follows]$^{3}_{\mathrm{P}}$ that A_t is independent of all previous states and actions given S_t. Notationally, for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and for all $1 \leq t \leq T - 1$, when $\Pr(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) \neq$ [null],

$$\Pr(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) = \Pr(\mathrm{A}_t \mid \mathrm{S}_t).$$

[We know that]$^{18}_{\mathrm{D}}$ for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and for all $\mathrm{s} \in \mathbf{S}$,

$$\Pr(\mathrm{S}_1 = \mathrm{s}) = \mathrm{P}_\mathrm{I}(\mathrm{s}).$$

[We know that]$^{33}_{\mathrm{D}}$ for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and all $\mathrm{s}, \mathrm{s}', \mathrm{a} \in \mathbf{S}^2 \times \mathbf{A}$, $2 \leq t \leq T$, if $\Pr(\mathrm{S}_t = \mathrm{s} \mid \mathrm{A}_{t-1} = \mathrm{a}, \mathrm{S}_{t-1} = \mathrm{s}') \neq$ [null],

$$\Pr(\mathrm{S}_t = \mathrm{s} \mid \mathrm{A}_{t-1} = \mathrm{a}, \mathrm{S}_{t-1} = \mathrm{s}') = \mathrm{P}_\mathrm{T}(\mathrm{s} \mid \mathrm{a}, \mathrm{s}'),$$

[where]$^{24}_{\mathrm{D}}$ S_t is independent of all previous states and actions given A_{t-1} and S_{t-1}. Notationally, for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and for all $2 \leq t \leq T$, when $\Pr(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) \neq$ [null],

$$\Pr(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) = \Pr(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}).$$

Propositions.

[It can be shown]$^{5}_{\mathrm{P}}$ that, for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_{\pi}$ and for any $\mathrm{s}_1, \ldots, \mathrm{s}_T, \mathrm{a}_1, \ldots, \mathrm{a}_{T-1} \in \mathbf{S}^T \times \mathbf{A}^{T-1}$, we can write

$$\Pr\left(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_T = \mathrm{s}_T\right)$$

as

$$\Pr\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T\right) = \mathrm{P}_{\mathrm{I}}\left(\mathrm{s}_1\right) \times \prod_{t=1}^{T-1} \mathrm{P}_{\mathrm{A}}\left(\pi, t : \mathrm{a}_t \mid \mathrm{s}_t\right) \mathrm{P}_{\mathrm{T}}\left(\mathrm{s}_{t+1} \mid \mathrm{a}_t, \mathrm{s}_t\right).$$

[It can be shown]$^{101}_{\mathrm{P}}$ that, for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_{\pi}$ and for all $1 \leq t \leq T-1$ and $\mathrm{s}_t \in \mathbf{S}$, if $\Pr\left(\mathrm{S}_t = \mathrm{s}_t\right) > 0$,

$$\mathbb{E}\left[R\left(t; \cdot\right) \mid \mathrm{S}_t = \mathrm{s}_t\right] = R\left(\mathrm{s}_t\right) + \sum_{\mathrm{s}_{t+1} \in \mathbf{S} : \mathrm{P}_{\mathrm{T}}\left(\mathrm{s}_{t+1} \mid \pi\left(t; \mathrm{s}_t\right), \mathrm{s}_t\right) > 0} \mathrm{P}_{\mathrm{T}}\left(\mathrm{s}_{t+1} \mid \pi\left(t; \mathrm{s}_t\right), \mathrm{s}_t\right) \times \mathbb{E}\left[R\left(t+1; \cdot\right) \mid \mathrm{S}_{t+1} = \mathrm{s}_{t+1}\right].$$

[12/23/2020 18:01] >[Ltj4hRy5dYM]

[133] /T/gr/T/pr/D/dmap

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, and all $\mathrm{E} \subseteq \mathrm{V}^2$, with $\mathrm{G} \coloneqq [\mathrm{V}, \mathrm{E}]$, [133.1] >

$$\mathrm{DMAP}\,(\mathrm{G}) \iff \forall\, \mathrm{C} \subseteq \mathrm{V},\ \emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V} : \mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}),\ \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \implies (\mathrm{G};\ \mathrm{A} \not\to \mathrm{B} \setminus \mathrm{C}).$$

[12/13/2020 09:57]

[113] /T/gr/T/pr/D/ind

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (\mathrm{Z}, \mathbf{Z}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, [113.1]

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \mathrm{Z} \iff \forall \mathrm{x}, \mathrm{y}, \mathrm{z} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \times \mathbf{Z}^{\times}, \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Z} = \mathrm{z}, \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}] \implies \Pr(\mathrm{x} \mid \mathrm{z}, \mathrm{y}) = \Pr(\mathrm{x} \mid \mathrm{z}),$$

and, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$,

$$\mathrm{X} \perp\!\!\!\perp \mathrm{Y} \iff \mathrm{X} \perp\!\!\!\perp \mathrm{Y} \mid \emptyset \iff \forall \mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}, \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \neq [\mathrm{null}] \implies \Pr(\mathrm{x} \mid \mathrm{y}) = \Pr(\mathrm{x}).$$

[12/13/2020 09:19] >[1LNwOOFVXb0]

[103] /T/gr/T/pr/D/match

Definition.

We let the notation [103.1] >

$$(\mathrm{X}, \mathbf{X}) \text{ indicate a “matching” between variables } \mathrm{X} \text{ and domains } \mathbf{X},$$

which expands as

$$(\mathrm{X}_1, \mathbf{X}_1), (\mathrm{X}_2, \mathbf{X}_2), \ldots, (\mathrm{X}_{N_{\mathbf{X}}}, \mathbf{X}_{N_{\mathbf{X}}}).$$

[12/12/2020 15:43] >[Ti4HlgXBtpo]

[64] /T/gr/T/pr/D/seqim

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $1 \leq t < \infty$ and $\mathrm{sq} \in \mathbf{SQ}$, [64.1] >

$$I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) \coloneqq \sum_{?\in\mathbf{?}^{\times}} I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}, ? = ?\right).$$

[12/12/2020 15:43] >[qJ3nE38NCWI]

[73] /T/gr/T/pr/D/sqcprob

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, [73.1] >

$$\Pr\left(\mathrm{sq}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right) \coloneqq \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right)}{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{Y} = \mathrm{y}\right)} \right);$$

that is, the conditional sequence probability of marginal assignment $\mathrm{X} = \mathrm{x}$ in sequence sq given marginal assignment $\mathrm{Y} = \mathrm{y}$.

[12/12/2020 15:43] >[pO8PFJXbKd4]

[60] /T/gr/T/pr/D/sqev

Definition.

We have by definition that, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $f : \mathbf{X} \to \mathbb{R}$, [60.1] >

$$\mathbb{E}\left[\mathrm{sq}; f\right] = \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}\right) \times f(\mathrm{x})}{t} \right).$$

This is verbally referred to as the "sequence expectation" of f under sq.

[12/23/2020 18:01] >[ujtPeggYykQ]

[132] /T/gr/T/pr/D/imap

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, and all $\mathrm{E} \subseteq \mathrm{V}^2$, with $\mathrm{G} \coloneqq [\mathrm{V}, \mathrm{E}]$, [132.1] >

$$\mathrm{IMAP}\,(\mathrm{G}) \iff \forall\, \mathrm{C} \subseteq \mathrm{V},\ \emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V} : \mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}),\ \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \impliedby (\mathrm{G};\ \mathrm{A} \not\rightarrow \mathrm{B} \setminus \mathrm{C}).$$

[12/12/2020 15:43] >[eRtR1ki8Jwc]

[58] /T/gr/T/pr/D/ev

Definition.

Define, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, $\mathrm{D} \in \mathbf{D}$, and all $f : \mathbf{X} \to \mathbb{R}$, [58.1] >

$$\mathbb{E}\left[f\right]$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \mathbb{E}\left[f\right] = \mathbb{E}\left[\mathrm{d}; f\right].$$

This is verbally referred to as the "expectation" of f under D.

Justification.

This definition is enabled by [the fact that]$^{82}_{\mathrm{P}}$ for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$ and all $f : \mathbf{X} \to \mathbb{R}$, $\mathrm{D} \in \mathbf{D}$, and $\mathrm{d}, \mathrm{d}' \in \mathrm{D}^2$, [58.2] >

$$\mathbb{E}\left[\mathrm{d}; f\right] = \mathbb{E}\left[\mathrm{d}'; f\right].$$

[12/12/2020 15:43] >[ix7nKFxJj_Y]

[41] /T/gr/T/pr/D/seqp

Definition.

We let for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, [41.1] >

$$\mathbf{SQP} = \{\mathrm{sq} \in \mathbf{SQ} \mid \forall \mathrm{x} \in \mathbf{X}^{\times}, (\exists\, 1 \leq t < \infty : I\,(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}) = 1) \implies \Pr\,(\mathrm{sq}; \mathrm{X} = \mathrm{x}) > 0\},$$

that is, the set of all probability sequences.

[12/12/2020 15:43] >[bRTvqkt75zM]

[46] /T/gr/T/pr/D/mprob

Definition.

We define, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x} \in \mathbf{X}^{\times}$, [46.1] >

$$\Pr(\mathrm{X} = \mathrm{x})$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \Pr(\mathrm{X} = \mathrm{x}) = \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}),$$

that is, a marginal probability.

Justification.

This definition is enabled by [the fact that]$^{76}_{\mathrm{P}}$ for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$ and all $\mathrm{D} \in \mathbf{D}$, $\mathrm{d}, \mathrm{d}' \in \mathrm{D}^2$ and $\mathrm{x} \in \mathbf{X}^{\times}$, [46.2] >

$$\Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}) = \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}\right).$$

[12/12/2020 15:43] >[vH0hacP_Wvo]

[42] /T/gr/T/pr/D/dist

Definition.

We define, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, [42.1] >

$$\mathbf{D} = \{\emptyset \subset \mathrm{D} \subseteq \mathbf{SQP} \mid \forall\, \mathrm{d}, \mathrm{d}' \in \mathrm{D}^2, \forall\, \mathrm{x} \in \mathbf{X}^{\times}, \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}) = \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}\right)\}$$

that is, the set of all distributions.

[12/12/2020 15:43] >[XT9yqjNkGqw]

[57] /T/gr/T/pr/D/asn

Definition.

We let, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$ [or, $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$] and all $\mathrm{x} = [\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_{N_{\mathbf{X}}}] \in \mathbf{X}^{\times}$, [57.1] >

$$\mathrm{X} = \mathrm{x}$$

denote a joint assignment (assignment for short) [or, marginal assignment], which expands as

$$\mathrm{X}_1 = \mathrm{x}_1, \mathrm{X}_2 = \mathrm{x}_2, \ldots, \mathrm{X}_{N_{\mathbf{X}}} = \mathrm{x}_{N_{\mathbf{X}}}.$$

If a joint assignment [or, marginal assignment] was previously specified, we may later drop "X =", for "implied assignment":

$$\mathrm{x},$$

which expands as

$$\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_{N_{\mathbf{X}}}.$$

If a joint assignment [or, marginal assignment] was never specified, we assume an arbitrary $\mathrm{x} \in \mathbf{X}^{\times}$ for "implied generality":

$$\mathrm{X},$$

which expands as

$$\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_{N_{\mathbf{X}}}.$$

[12/12/2020 15:43] >[JFxZjVEPk2o]

[43] /T/gr/T/pr/D/prob

Definition.

We define, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, $\mathrm{D} \in \mathbf{D}$ and all $\mathrm{x} \in \mathbf{X}^{\times}$, [43.1] >

$$\Pr(\mathrm{X} = \mathrm{x})$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \Pr(\mathrm{X} = \mathrm{x}) = \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}),$$

that is, a joint probability.

Justification.

This definition is enabled by [the fact that][42]$_{\mathrm{D}}$ [43.2] >

$$\forall\ \mathrm{d}, \mathrm{d}' \in \mathrm{D}^2, \forall\ \mathrm{x} \in \mathbf{X}^{\times}, \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}) = \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}\right).$$

[12/16/2020 17:11] >[ZiLjG8olMQc]

[102] /T/gr/T/pr/D/xyzv

Definition.

We define [102.1] >

$$\mathrm{X} \coloneqq [\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_{N_{\mathbf{X}}}]$$

with

$$\mathbf{X} \coloneqq [\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_{N_{\mathbf{X}}}].$$

Define $\mathrm{Y}/\mathbf{Y}$, $\mathrm{Z}/\mathbf{Z}$, $\mathrm{W}/\mathbf{W}$, $?/\mathbf{?}$ and $?'/\mathbf{?}'$ similarly.

Note that $\|\mathrm{X}\| = N_{\mathbf{X}} \geq 1$, $\|\mathrm{Y}\| = N_{\mathbf{Y}} \geq 1$, $\|\mathrm{Z}\| = N_{\mathbf{Z}} \geq 1$, $\|\mathrm{W}\| = N_{\mathbf{W}} \geq 1$, $\|?\| = N_{\mathbf{?}} \geq 0$ and $\|?'\| = N_{\mathbf{?}'} \geq 0$.

[12/12/2020 15:43] >[7Icbh4Nj4mk]

[61] /T/gr/T/pr/D/sqevc

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$, $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$ and $\mathrm{y} \in \mathbf{Y}^{\times}$, [61.1] >

$$\mathbb{E}\left[\mathrm{sq}; f \mid \mathrm{Y} = \mathrm{y}\right] = \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} I\left(\mathrm{sq}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) \times f(\mathrm{x}, \mathrm{y})}{\sum_{t'=1}^{t} I\left(\mathrm{sq}, t', \mathrm{Y} = \mathrm{y}\right)} \right).$$

This is verbally referred to as the "conditional sequence expectation" of f under sq given $\mathrm{Y} = \mathrm{y}$.

[12/12/2020 15:43] >[iuk_qr7jFwk]

[38] /T/gr/T/pr/D/seq

Definition.

We let, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, [38.1] >

$$\mathbf{SQ} \text{ denote the set of all sequences.}$$

[12/12/2020 15:43] >[0kQRxOmB7EM]

[67] /T/gr/T/pr/D/evc

Definition.

Define, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, $\mathrm{D} \in \mathbf{D}$, and all $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$ and $\mathrm{y} \in \mathbf{Y}^{\times}$, [67.1] >

$$\mathbb{E}\left[f \mid \mathrm{Y} = \mathrm{y}\right]$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \mathbb{E}\left[f \mid \mathrm{Y} = \mathrm{y}\right] = \mathbb{E}\left[\mathrm{d}; f \mid \mathrm{Y} = \mathrm{y}\right].$$

This is verbally referred to as the "conditional expectation" of f under D given $\mathrm{Y} = \mathrm{y}$.

Justification.

This definition is enabled by [the fact that]$^{84}_{\mathrm{P}}$ for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$ and all $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$, $\mathrm{y} \in \mathbf{Y}^{\times}$, $\mathrm{D} \in \mathbf{D}$ and $\mathrm{d}, \mathrm{d}' \in \mathrm{D}^2$, [67.2] >

$$\mathbb{E}\left[\mathrm{d}; f \mid \mathrm{Y} = \mathrm{y}\right] = \mathbb{E}\left[\mathrm{d}'; f \mid \mathrm{Y} = \mathrm{y}\right].$$

[12/12/2020 15:43] >[XbUw6Y4O2Bs]

[63] /T/gr/T/pr/D/seqi

Definition.

We use, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, and all $1 \leq t < \infty$, $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x} \in \mathbf{X}^{\times}$, [63.1] >

$$I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) := \begin{cases} 1 & \text{trial } t \text{ of sequence sq has assignment } \mathrm{X} = \mathrm{x} \\ 0 & \text{otherwise} \end{cases}$$

to describe a sequence $\mathrm{sq} \in \mathbf{SQ}$, where

$$\sum_{\mathrm{x} \in \mathbf{X}^{\times}} I\left(\mathrm{sq}, t, \mathrm{X} = \mathrm{x}\right) = 1.$$

[12/12/2020 15:43] >[emhaZGBGG3Y]

[66] /T/gr/T/pr/D/sqmprob

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$, and all $\mathrm{sq} \in \mathbf{SQ}$ and $\mathrm{x} \in \mathbf{X}^{\times}$, [66.1] >

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) \coloneqq \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I(\mathrm{sq}, t', \mathrm{X} = \mathrm{x})}{t} \right);$$

that is, the marginal sequence probability of marginal assignment $\mathrm{X} = \mathrm{x}$ in sequence sq.

[12/12/2020 15:43] >[G1J5iiU-oR0]

[40] /T/gr/T/pr/D/sqprob

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$, all $\mathrm{sq} \in \mathbf{SQ}$, and all $\mathrm{x} \in \mathbf{X}^{\times}$, [40.1] >

$$\Pr(\mathrm{sq}; \mathrm{X} = \mathrm{x}) \coloneqq \lim_{t \to \infty} \left(\frac{\sum_{t'=1}^{t} I(\mathrm{sq}, t', \mathrm{X} = \mathrm{x})}{t} \right);$$

that is, the sequence probability of assignment $\mathrm{X} = \mathrm{x}$ in sequence sq.

[12/23/2020 18:02] >[h6UfcZ06hSM]

[134] /T/gr/T/pr/D/pmap

Definition.

We have by definition that for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, and all $\mathrm{E} \subseteq \mathrm{V}^2$, with $\mathrm{G} \coloneqq [\mathrm{V}, \mathrm{E}]$, [134.1] >

$$\mathrm{PMAP}\,(\mathrm{G}) \iff \mathrm{IMAP}\,(\mathrm{G}),\ \mathrm{DMAP}\,(\mathrm{G}).$$

[12/16/2020 17:11] >[AbWIOhwABf0]

[36] /T/gr/T/pr/D/xyz

Definition.

We have by definition that [36.1] >

$$\mathrm{X}_1, \mathrm{X}_2, \ldots, \mathrm{X}_{N_{\mathbf{X}}} \coloneqq \text{an arbitrary set of variable(s) with } N_{\mathbf{X}} \geq 1 \text{ and domains } \mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_{N_{\mathbf{X}}},$$

and we define $\mathrm{Y}_1, \mathrm{Y}_2, \ldots, \mathrm{Y}_{N_{\mathbf{Y}}}/\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_{N_{\mathbf{Y}}}/N_{\mathbf{Y}}$, $\mathrm{Z}_1, \mathrm{Z}_2, \ldots, \mathrm{Z}_{N_{\mathbf{Z}}}/\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_{N_{\mathbf{Z}}}/N_{\mathbf{Z}}$, and $\mathrm{W}_1, \mathrm{W}_2, \ldots, \mathrm{W}_{N_{\mathbf{W}}}/\mathbf{W}_1, \mathbf{W}_2, \ldots, \mathbf{W}_{N_{\mathbf{W}}}/N_{\mathbf{W}}$ similarly.

Define

$$?_1, ?_2, \ldots, ?_{N_?} \coloneqq \text{an arbitrary set of variable(s) with } N_? \geq 0 \text{ and domains } \mathbf{?}_1, \mathbf{?}_2, \ldots, \mathbf{?}_{N_?}.$$

Define $?'_1, ?'_2, \ldots, ?'_{N_{?'}}/\mathbf{?}'_1, \mathbf{?}'_2, \ldots, \mathbf{?}'_{N_{?'}}/N_{?'}$ similarly.

[12/12/2020 15:43] >[nKeW82L3LrY]

[37] /T/gr/T/pr/D/spt

Definition.

We let the notation [37.1] >

$$S \leftarrow \{(X, \mathbf{X})\} \text{ establish a support consisting of variables } X \text{ with domains } \mathbf{X}.$$

[12/12/2020 15:43] >[LJXYeyVhits]

[48] /T/gr/T/pr/D/cprob

Definition.

We define, for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, and all $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, [48.1] >

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y})$$

such that

$$\forall \mathrm{d} \in \mathrm{D}, \Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y});$$

that is, a conditional probability.

Justification.

This definition is enabled by [the fact that]$^{77}_{\mathrm{P}}$ for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$ and all $\mathrm{D} \in \mathbf{D}$, $\mathrm{d}, \mathrm{d}' \in \mathrm{D}^2$ and $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, [48.2] >

$$\Pr(\mathrm{d}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right).$$

[12/25/2020 10:33]

[154] /T/gr/T/pr/P/pmape/P/conddecomp

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} := \mathrm{X} \cup \mathrm{Y} \cup ?$, if for all $\mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B}, \mathrm{S} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$, [154.1]

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S},\ \mathrm{A} \perp\!\!\!\perp \mathrm{S} \mid \mathrm{C}, \mathrm{B} \underset{\texttt{int}}{\implies} \mathrm{A} \perp\!\!\!\perp \mathrm{B}, \mathrm{S} \mid \mathrm{C}$$

and for all $\mathrm{C}, \mathrm{S} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \underset{\texttt{sun}}{\implies} \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S},$$

then, for all $\mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \iff \forall \mathrm{H}, \mathrm{H}' \in \mathrm{A} \times \mathrm{B}, [\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{C}.$$

Proof.

$\implies$. [154.2]

We have that

$$\begin{aligned}
\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} &\overset{[\ldots]^{123}_{\mathrm{P}}}{\implies} \forall \mathrm{H}' \in \mathrm{B},\ \mathrm{A} \perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{C} \\
&\overset{[\ldots]^{114}_{\mathrm{P}}}{\implies} \forall \mathrm{H}' \in \mathrm{B},\ [\mathrm{H}'] \perp\!\!\!\perp \mathrm{A} \mid \mathrm{C} \\
&\overset{[\ldots]^{123}_{\mathrm{P}}}{\implies} \forall \mathrm{H},\ \mathrm{H}' \in \mathrm{A} \times \mathrm{B}, [\mathrm{H}'] \perp\!\!\!\perp [\mathrm{H}] \mid \mathrm{C} \\
&\overset{[\ldots]^{114}_{\mathrm{P}}}{\implies} \forall \mathrm{H},\ \mathrm{H}' \in \mathrm{A} \times \mathrm{B}, [\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{C}.
\end{aligned}$$

$\impliedby$. [154.3]

We have that

$$\begin{aligned}
\forall \mathrm{H}, \mathrm{H}' \in \mathrm{A} \times \mathrm{B},\ [\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{C}. &\overset{\texttt{(int,sun)->}[\ldots]^{155}_{\mathrm{P}}}{\implies} \forall \mathrm{H} \in \mathrm{A},\ [\mathrm{H}] \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \\
&\overset{[\ldots]^{114}_{\mathrm{P}}}{\implies} \forall \mathrm{H} \in \mathrm{A},\ \mathrm{B} \perp\!\!\!\perp [\mathrm{H}] \mid \mathrm{C} \\
&\overset{\texttt{(int,sun)->}[\ldots]^{155}_{\mathrm{P}}}{\implies} \forall \mathrm{H} \in \mathrm{A},\ \mathrm{B} \perp\!\!\!\perp \mathrm{A} \mid \mathrm{C} \\
&\overset{[\ldots]^{114}_{\mathrm{P}}}{\implies} \forall \mathrm{H} \in \mathrm{A},\ \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}.
\end{aligned}$$

[12/25/2020 11:49]

[156] /T/gr/T/pr/P/pmape/P/indgrpatheq

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} \coloneqq ?$, if [156.1]
for all $\mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B}, \mathrm{S} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S},\ \mathrm{A} \perp\!\!\!\perp \mathrm{S} \mid \mathrm{C}, \mathrm{B} \underset{\texttt{int}}{\implies} \mathrm{A} \perp\!\!\!\perp \mathrm{B}, \mathrm{S} \mid \mathrm{C},$$

for all $\mathrm{C}, \mathrm{S} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \underset{\texttt{sun}}{\implies} \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S},$$

and for all $\mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \underset{\texttt{tran}}{\implies} \forall \mathrm{H} \in \mathrm{V} \setminus (\mathrm{A}, \mathrm{B}, \mathrm{C}),\ \mathrm{A} \perp\!\!\!\perp [\mathrm{H}] \mid \mathrm{C} \text{ or } [\mathrm{H}] \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C},$$

then, with $\mathrm{E} \coloneqq \{[\mathrm{H}, \mathrm{H}'] \in \mathrm{V}^2 \mid \mathrm{H} \neq \mathrm{H}',\ \forall \mathrm{V}' \subseteq \mathrm{V} \setminus \{\mathrm{H}, \mathrm{H}'\},\ [\mathrm{H}] \not\perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{V}'\}$ and contextual
$\mathrm{G} \coloneqq [\mathrm{V}, \mathrm{E}]$, for all $\mathrm{H}, \mathrm{H}' \in \mathrm{V}^2$ such that $\mathrm{H} \neq \mathrm{H}'$ and $\mathrm{C} \subseteq (\mathrm{V} \setminus \{\mathrm{H}, \mathrm{H}'\})$, pmape

$$[\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{C} \underset{\texttt{css}}{\iff} \mathrm{H} \not\longrightarrow \mathrm{H}' \setminus \mathrm{C}.$$

Proof.

$\implies$ suppose $\mathrm{H} \longrightarrow \mathrm{H}' \setminus \mathrm{C}$. [156.2]

Then, [156.3]

$$\exists \mathrm{P} \in \mathrm{H} \overset{[\ldots]^{143}_{\mathrm{P}}}{\longrightarrow} \mathrm{H}' \setminus \mathrm{C} : \forall \mathrm{v} \in \mathrm{P}, \|\mathrm{v};\ \mathrm{P}\| = 1,$$

so, for all $0 \leq t, t', t'' \leq \|\mathrm{P}\| - 1$ such that $\mathrm{NE}\,(t, t', t'')$,

$$\mathrm{NE}\,(\mathrm{P}\,[t], \mathrm{P}\,[t'], \mathrm{P}\,[t'']),$$

therefore

$$\mathrm{DJ}\,([\mathrm{P}\,[t]], [\mathrm{P}\,[t'']], \mathrm{C}) \underset{\texttt{adj}}{\text{ and }} \mathrm{P}\,[t'] \in (\mathrm{V} \setminus \{\mathrm{P}\,[t]\}, \{\mathrm{P}\,[t'']\}, \mathrm{C}),$$

so we have, for all $1 \leq t \leq \|\mathrm{P}\| - 2$, [156.5]

$$[\mathrm{P}\,[0]] \not\perp\!\!\!\perp [\mathrm{P}\,[t]] \mid \mathrm{C} \text{ and } [\mathrm{P}\,[t]] \not\perp\!\!\!\perp [\mathrm{P}\,[t+1]] \mid \mathrm{C} \underset{\texttt{atran}}{\overset{\texttt{(adj,ani)->¬tran}}{\implies}} [\mathrm{P}\,[0]] \not\perp\!\!\!\perp [\mathrm{P}\,[t+1]] \mid \mathrm{C},$$

and, for all $0 \leq t \leq \|\mathrm{P}\| - 2$, [156.6]

$$[[\mathrm{P}\,[t]], [\mathrm{P}\,[t+1]]] \in \mathrm{E} \overset{[\ldots]^{142}_{\mathrm{D}}}{\underset{\texttt{pmape}}{\implies}} [\mathrm{P}\,[t]] \not\perp\!\!\!\perp [\mathrm{P}\,[t+1]] \mid \mathrm{C}.$$

Therefore, [156.7]

$$[\mathrm{P}\,[0]] \not\perp\!\!\!\perp [\mathrm{P}\,[\|\mathrm{P}\| - 1]] \mid \mathrm{C}, \quad \texttt{atran,!}$$

$[\ldots]_{\mathrm{D}}^{108}$

so $[\mathrm{H}] \not\perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{C}$.

$\Longleftarrow$.

Firstly, note that [156.8]

$$\forall [\mathrm{H}, \mathrm{H}'] \in \mathrm{E}, [\mathrm{H}', \mathrm{H}] \in \mathrm{E} \quad \overset{\texttt{pmape},\neg[\ldots]_{\mathrm{P}}^{114}\ [\ldots]_{\mathrm{D}}^{151}}{\Longrightarrow} \quad \underset{\texttt{ud}}{\mathrm{UD}}.$$

Base case.

Suppose that $\|\mathrm{C}\| \underset{\texttt{smt}}{=} \|\mathrm{V}\| - 2$ and $\mathrm{H} \underset{\texttt{npath}}{\not\rightarrow} \mathrm{H}' \setminus \mathrm{C}$. [156.9]

Then, [156.10]

$$\mathrm{C} \underset{\texttt{cd}}{\overset{\texttt{smt}}{=}} \mathrm{V} \setminus \{\mathrm{H}, \mathrm{H}'\},$$

so [156.11]

$$[\mathrm{H}, \mathrm{H}'] \notin \mathrm{E} \overset{\texttt{npath}}{\Longrightarrow} \overset{\texttt{pmape}}{\exists} \mathrm{V}' \subseteq \mathrm{V} \setminus \{\mathrm{H}, \mathrm{H}'\} : [\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{V}'$$
$$\overset{\texttt{cd,sun}}{\Longrightarrow} [\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{C}.$$

Inductive hypothesis.

Let $n \leq \|\mathrm{V}\| - 2$. Suppose, for all n' such that $n \leq n' \leq \|\mathrm{V}\| - 2$, that for all $\mathrm{H}, \mathrm{H}' \in \mathrm{V}^2$ such that $\mathrm{H} \overset{\texttt{nd}}{\neq} \mathrm{H}'$ and $\mathrm{C} \subseteq (\mathrm{V} \setminus \{\mathrm{H}, \mathrm{H}'\})$ with $\|\mathrm{C}\| = n'$, [156.12]

$$\mathrm{H} \not\rightarrow \mathrm{H}' \setminus \mathrm{C} \underset{\texttt{ih}}{\Longrightarrow} [\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{C}.$$

Inductive step.

Let $\|\mathrm{C}\| \underset{\texttt{cs}}{=} n - 1$, and suppose $\mathrm{H} \underset{\texttt{npath}}{\not\rightarrow} \mathrm{H}' \setminus \mathrm{C}$. [156.13]

Then, [156.14]

$$\|\mathrm{V} \setminus \{\mathrm{H}, \mathrm{H}'\} \cup \mathrm{C}\| > 0, \text{ so } \overset{\texttt{cs,nd,css}}{\exists} \mathrm{H}'' \in \mathrm{V} \setminus \{\mathrm{H}, \mathrm{H}'\} \cup \mathrm{C},$$

with [156.15]

$$\|\mathrm{C}, \mathrm{H}\|, \|\mathrm{C}, \mathrm{H}'\|, \|\mathrm{C}, \mathrm{H}''\| = \|\mathrm{C}\| + 1 \underset{\texttt{eqn}}{\overset{\texttt{cs}}{=}} n \underset{\texttt{lts}}{\overset{\texttt{nd}}{\leq}} \|\mathrm{V}\| - 2$$

and [156.16]

$$\mathrm{H} \underset{\texttt{snpath}}{\overset{\texttt{(npath)->}\neg[\ldots]_{\mathrm{P}}^{148}}{\not\rightarrow}} \mathrm{H}' \setminus \mathrm{C}, \mathrm{H}'' \overset{\texttt{(-eqn,-lts)->ih}}{\Longrightarrow} \underset{\texttt{inda}}{[\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{C} \cup [\mathrm{H}'']}$$

and [156.17]

$$\overset{\texttt{(npath,ud)->}\neg[\ldots]^{152}_{\mathrm{P}}}{\mathrm{H} \not\to \mathrm{H}'' \setminus \mathrm{C}, \mathrm{H}' \text{ or } \mathrm{H}' \not\to \mathrm{H}'' \setminus \mathrm{C}, \mathrm{H}},$$

so there are two cases to consider:

– $\mathrm{H} \not\to \mathrm{H}'' \setminus \mathrm{C}, \mathrm{H}'$. [156.18]

co

Then,

$$\begin{aligned}
\overset{\texttt{(co,-eqn,-lts)->ih}}{[\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}''] \mid \mathrm{C} \cup [\mathrm{H}']} &\overset{\texttt{(inda,!)->int}}{\implies} [\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}', \mathrm{H}''] \mid \mathrm{C} \\
&\overset{[\ldots]^{123}_{\mathrm{P}}}{\implies} [\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{C}.
\end{aligned}$$

– $\mathrm{H}' \not\to \mathrm{H}'' \setminus \mathrm{C}, \mathrm{H}$. [156.19]

ct

Then,

$$\begin{aligned}
\overset{\texttt{(ct,-eqn,-lts)->ih}}{[\mathrm{H}'] \perp\!\!\!\perp [\mathrm{H}''] \mid \mathrm{C} \cup [\mathrm{H}]} \text{ and } \overset{\texttt{(inda)->}[\ldots]^{114}_{\mathrm{P}}}{[\mathrm{H}'] \perp\!\!\!\perp [\mathrm{H}] \mid \mathrm{C} \cup [\mathrm{H}'']} &\overset{\texttt{int}}{\implies} [\mathrm{H}'] \perp\!\!\!\perp [\mathrm{H}, \mathrm{H}''] \mid \mathrm{C} \\
&\overset{[\ldots]^{123}_{\mathrm{P}}}{\implies} [\mathrm{H}'] \perp\!\!\!\perp [\mathrm{H}] \mid \mathrm{C} \\
&\overset{[\ldots]^{114}_{\mathrm{P}}}{\implies} [\mathrm{H}] \perp\!\!\!\perp [\mathrm{H}'] \mid \mathrm{C}.
\end{aligned}$$

[12/12/2020 15:43] >[YeD7n3V99ig]

[99] /T/gr/T/pr/P/sqcprobexpr/P/seqimqlim

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$, and all $\mathrm{sqp} \in \mathbf{SQP}$ and $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, [99.1] >

$$\lim_{t \to \infty} \left(\frac{\left(\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) \right) / t}{\left(\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{Y} = \mathrm{y}\right) \right) / t} \right) = \frac{\lim_{t \to \infty} \left(\left(\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) \right) / t \right)}{\lim_{t \to \infty} \left(\left(\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{Y} = \mathrm{y}\right) \right) / t \right)}.$$

Proof.

There are two cases to consider: [99.2] >

– Case 1: $\Pr\left(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}\right) = 0$. [99.3] >

mz

Then, we have that [99.4] >

(mz)->$[\ldots]^{97}_{\mathrm{P}}$

$$\forall\ 1 \leq t < \infty, I\left(\mathrm{sqp}, t, \mathrm{Y} = \mathrm{y}\right) = 0,$$

mzi

and

(mz)->$[\ldots]^{91}_{\mathrm{P}}$

$$\Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) = 0,$$

jz

so

(mz)->$[\ldots]^{41}_{\mathrm{D}}$

$$\forall\ 1 \leq t < \infty, I\left(\mathrm{sqp}, t, \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) = 0,$$

jzi

and we have [99.5] >

mzi,jzi

$$\begin{aligned}
\lim_{t \to \infty} \left(\frac{\left(\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) \right) / t}{\left(\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{Y} = \mathrm{y}\right) \right) / t} \right) &= \lim_{t \to \infty} \frac{0}{0} \\
&= [\mathrm{null}] \\
&= \frac{0}{0} \\
&\overset{\mathrm{mz,jz}}{=} \frac{\Pr\left(\mathrm{sqp}; \mathrm{Y} = \mathrm{y}\right)}{\Pr\left(\mathrm{sqp}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right)} \\
&\overset{[\ldots]^{40}_{\mathrm{D}}, [\ldots]^{66}_{\mathrm{D}}}{=} \frac{\lim_{t \to \infty} \left(\left(\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right) \right) / t \right)}{\lim_{t \to \infty} \left(\left(\sum_{t'=1}^{t} I\left(\mathrm{sqp}, t', \mathrm{Y} = \mathrm{y}\right) \right) / t \right)}.
\end{aligned}$$

– Case 2: $\Pr(\text{sqp}; \text{Y} = \text{y}) > 0$. [99.6] >

gz

Then, we have that [99.7] >

[...]66D,[...]81P,gz

$$\lim_{t\to\infty}\left(\frac{\left(\sum_{t'=1}^{t} I\left(\text{sqp}, t', \text{X} = \text{x}, \text{Y} = \text{y}\right)\right)/t}{\left(\sum_{t'=1}^{t} I\left(\text{sqp}, t', \text{Y} = \text{y}\right)\right)/t}\right) = \frac{\lim_{t\to\infty}\left(\left(\sum_{t'=1}^{t} I\left(\text{sqp}, t', \text{X} = \text{x}, \text{Y} = \text{y}\right)\right)/t\right)}{\lim_{t\to\infty}\left(\left(\sum_{t'=1}^{t} I\left(\text{sqp}, t', \text{Y} = \text{y}\right)\right)/t\right)}.$$

[12/12/2020 15:43] >[WRi2dpYp-Dc]

[3] /T/gr/T/pr/T/rl/P/actcind

Proposition.

We have that A_t is independent of all previous states and actions given S_t. Notationally, for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and for all $1 \leq t \leq T - 1$, when [3.1] >
$\Pr(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) \neq [\text{null}]$,

$$\Pr(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) \overset{\texttt{def}}{=} \Pr(\mathrm{A}_t \mid \mathrm{S}_t).$$

Proof.

We have that [3.3] >

$$\Pr(\mathrm{A}_t \mid \mathrm{S}_t) \overset{\texttt{(def)->}[\ldots]^{93}_{\mathrm{P}}}{\neq} [\text{null}],$$

Suppose $\mathrm{A}_t = \mathrm{a} \in \mathbf{A}$ and $\mathrm{S}_t = \mathrm{s} \in \mathbf{S}$. We have [3.4] >

$$\Pr(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) \overset{[\ldots]^{25}_{\mathrm{D}}}{=} \mathrm{P}_{\mathrm{A}}(\pi, t : \mathrm{a} \mid \mathrm{s}),$$

so [there are two cases]$^{28}_{\mathrm{D}}$ to consider: [3.5] >

- Case 1: $\mathrm{P}_{\mathrm{A}}(\pi, t : \mathrm{a} \mid \mathrm{s}) \overset{\texttt{po}}{=} 1$. [3.6] >

  Then, [3.7] >

$$\Pr(\mathrm{A}_t \mid \mathrm{S}_t) \overset{\texttt{po,ap}}{=} 1,$$

  so [3.8] >

$$\Pr(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) \overset{\texttt{(mg,poi)->}[\ldots]^{95}_{\mathrm{P}}}{=} 1 \overset{\texttt{poi}}{=} \Pr(\mathrm{A}_t \mid \mathrm{S}_t).$$

- Case 2: $\mathrm{P}_{\mathrm{A}}(\pi, t : \mathrm{a} \mid \mathrm{s}) \overset{\texttt{pz}}{=} 0$. [3.9] >

  Then, [3.10] >

$$\Pr(\mathrm{A}_t \mid \mathrm{S}_t) \overset{\texttt{pz,ap}}{=} 0,$$

  so [3.11] >

$$\Pr(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) \overset{\texttt{(pzi)->}[\ldots]^{96}_{\mathrm{P}}}{=} 0 \overset{\texttt{pzi}}{=} \Pr(\mathrm{A}_t \mid \mathrm{S}_t).$$

[12/12/2020 15:43] >[k3QCBLKPbSE]

[4] /T/gr/T/pr/T/rl/P/npol

Proposition.

We have that the total number of strategies can be written as [4.1] >

$$|\mathbf{\Pi}| = (T-1) \times |\mathbf{S}| \times |\mathbf{A}|.$$

Proof.

This follows from simple combinatorics. Consider an arbitrary strategy [4.2] >

$$\pi = (\pi(1;\cdot), \ldots, \pi(T-1;\cdot)).$$

Because each step-strategy $\pi(t;\cdot) : \mathbf{S} \to \mathbf{A}$, there are a total of $|\mathbf{S}| \times |\mathbf{A}|$ distinct $\pi(t;\cdot)$ possible.

Because each strategy $\pi \in \mathbf{\Pi}$ consists of $T-1$ such step-strategies $\pi(t;\cdot)$, there are a total of [4.3] >

$$|\mathbf{\Pi}| = (T-1) \times |\mathbf{S}| \times |\mathbf{A}|$$

distinct π possible.

[12/12/2020 15:43] >[Q3Pxk5LPHwQ]

[5] /T/gr/T/pr/T/rl/P/pexpr

Proposition.

We have that for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and for any $\mathrm{s}_1, \ldots, \mathrm{s}_T, \mathrm{a}_1, \ldots, \mathrm{a}_{T-1} \in \mathbf{S}^T \times \mathbf{A}^{T-1}$, we can write [5.1] >

$$\Pr\left(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_T = \mathrm{s}_T\right)$$

as

$$\Pr\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t=1}^{T-1} \mathrm{P}_\mathrm{A}\left(\pi, t : \mathrm{a}_t \mid \mathrm{s}_t\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t+1} \mid \mathrm{a}_t, \mathrm{s}_t\right).$$

Proof.

[It can be shown inductively]$^{11}_{\mathrm{P}}$ that for any $1 \leq t \leq T$, for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and for any $\mathrm{s}_1, \ldots, \mathrm{s}_t, \mathrm{a}_1, \ldots, \mathrm{a}_{t-1} \in \mathbf{S}^t \times \mathbf{A}^{t-1}$ we can write [5.2] >

$$\Pr\left(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_t = \mathrm{s}_t\right)$$

as

$$\Pr\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-1} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right),$$

for which the result follows in particular when $t = T$. [5.3] >

[101] /T/gr/T/pr/T/rl/P/evexpr

Proposition.

We have that for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and for all $1 \leq t \leq T-1$ and $\mathrm{s}_t \in \mathbf{S}$, if $\Pr(\mathrm{S}_t = \mathrm{s}_t) > 0$,

$$\mathbb{E}\left[R(t;\cdot) \mid \mathrm{S}_t = \mathrm{s}_t\right] = R(\mathrm{s}_t) + \sum_{\mathrm{s}_{t+1} \in \mathbf{S}: \mathrm{P_T}(\mathrm{s}_{t+1} \mid \pi(t;\mathrm{s}_t), \mathrm{s}_t) > 0} \mathrm{P_T}(\mathrm{s}_{t+1} \mid \pi(t;\mathrm{s}_t), \mathrm{s}_t) \times \mathbb{E}\left[R(t+1;\cdot) \mid \mathrm{S}_{t+1} = \mathrm{s}_{t+1}\right].$$

Proof.

We have that

$$\begin{aligned}
\mathbb{E}\left[R(t;\cdot) \mid \mathrm{S}_t = \mathrm{s}_t\right] &= \sum_{\mathrm{s}_1,\mathrm{a}_1,\ldots,\mathrm{a}_{t-1},\mathrm{a}_t,\mathrm{s}_{t+1},\ldots,\mathrm{s}_T} \Pr(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \ldots, \mathrm{A}_{t-1} = \mathrm{a}_{t-1}, \mathrm{A}_t = \mathrm{a}_t, \mathrm{S}_{t+1} = \mathrm{s}_{t+1}, \ldots, \mathrm{S}_T = \mathrm{s}_T \mid \mathrm{S}_t = \mathrm{s}_t) \\
&\qquad \times \sum_{t'=t}^{T} R(\mathrm{s}'_t) \\
&= \sum_{\mathrm{s}_1,\mathrm{a}_1,\ldots,\mathrm{a}_{t-1},\mathrm{a}_t,\mathrm{s}_{t+1},\ldots,\mathrm{s}_T} \Pr(\mathrm{s}_1, \mathrm{a}_1, \ldots, \mathrm{a}_{t-1}, \mathrm{a}_t, \mathrm{s}_{t+1}, \ldots, \mathrm{s}_T \mid \mathrm{s}_t) \times R(\mathrm{s}_t) + \\
&\qquad \sum_{\mathrm{s}_1,\mathrm{a}_1,\ldots,\mathrm{a}_{t-1},\mathrm{a}_t,\mathrm{s}_{t+1},\ldots,\mathrm{s}_T} \Pr(\mathrm{s}_1, \mathrm{a}_1, \ldots, \mathrm{a}_{t-1}, \mathrm{a}_t, \mathrm{s}_{t+1}, \ldots, \mathrm{s}_T \mid \mathrm{s}_t) \times \sum_{t'=t+1}^{T} R(\mathrm{s}'_t) \\
&= R(\mathrm{s}_t) \sum_{\mathrm{s}_1,\mathrm{a}_1,\ldots,\mathrm{a}_{t-1},\mathrm{a}_t,\mathrm{s}_{t+1},\ldots,\mathrm{s}_T} \Pr(\mathrm{s}_1, \mathrm{a}_1, \ldots, \mathrm{a}_{t-1}, \mathrm{a}_t, \mathrm{s}_{t+1}, \ldots, \mathrm{s}_T \mid \mathrm{s}_t) + \\
&\qquad \sum_{\mathrm{a}_t,\mathrm{s}_{t+1},\ldots,\mathrm{s}_T} \sum_{t'=t+1}^{T} R(\mathrm{s}'_t) \times \sum_{\mathrm{s}_1,\mathrm{a}_1,\ldots,\mathrm{a}_{t-1}} \Pr(\mathrm{s}_1, \mathrm{a}_1, \ldots, \mathrm{a}_{t-1}, \mathrm{a}_t, \mathrm{s}_{t+1}, \ldots, \mathrm{s}_T \mid \mathrm{s}_t) \\
&= R(\mathrm{s}_t) + \sum_{\mathrm{a}_t,\mathrm{s}_{t+1},\ldots,\mathrm{s}_T} \Pr(\mathrm{a}_t, \mathrm{s}_{t+1}, \ldots, \mathrm{s}_T \mid \mathrm{s}_t) \times \sum_{t'=t+1}^{T} R(\mathrm{s}'_t) \\
&= R(\mathrm{s}_t) + \\
&\qquad \sum_{\mathrm{a}_t,\mathrm{s}_{t+1},\ldots,\mathrm{s}_T} \Pr(\mathrm{s}_{t+1} \mid \mathrm{a}_t, \mathrm{s}_t) \times \Pr(\mathrm{a}_t \mid \mathrm{s}_t) \times \Pr(\mathrm{a}_{t+1}, \mathrm{s}_{t+2}, \ldots, \mathrm{s}_T \mid \mathrm{a}_t, \mathrm{s}_{t+1}, \mathrm{s}_t) \times \sum_{t'=t+1}^{T} R(\mathrm{s}'_t) \\
&= R(\mathrm{s}_t) + \\
&\qquad \sum_{\mathrm{a}_t,\mathrm{s}_{t+1},\ldots,\mathrm{s}_T} \Pr(\mathrm{s}_{t+1} \mid \mathrm{a}_t, \mathrm{s}_t) \times \Pr(\mathrm{a}_t \mid \mathrm{s}_t) \times \Pr(\mathrm{a}_{t+1}, \mathrm{s}_{t+2}, \ldots, \mathrm{s}_T \mid \mathrm{s}_{t+1}) \times \sum_{t'=t+1}^{T} R(\mathrm{s}'_t) \\
&= R(\mathrm{s}_t) + \\
&\qquad \sum_{\mathrm{a}_t,\mathrm{s}_{t+1}} \Pr(\mathrm{s}_{t+1} \mid \mathrm{a}_t, \mathrm{s}_t) \times \Pr(\mathrm{a}_t \mid \mathrm{s}_t) \times \sum_{\mathrm{a}_{t+1},\mathrm{s}_{t+2},\ldots,\mathrm{s}_T} \Pr(\mathrm{a}_{t+1}, \mathrm{s}_{t+2}, \ldots, \mathrm{s}_T \mid \mathrm{s}_{t+1}) \times \sum_{t'=t+1}^{T} R(\mathrm{s}'_t)
\end{aligned}$$

[12/12/2020 15:43] >[uv1voiyTLLE]

[18] /T/gr/T/pr/T/rl/D/ind

Definition.

We have by definition that for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and for all $\mathrm{s} \in \mathbf{S}$, [18.1] >

$$\Pr\left(\mathrm{S}_1 = \mathrm{s}\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}\right).$$

[12/12/2020 15:43] >[2RUCE-Q1HaM]

[19] /T/gr/T/pr/T/rl/D/rewards

Definition.

We have by definition that [19.1] >

$$R(\cdot) : \mathbf{S} \to \mathbb{R} := \text{mapping from states to returns.}$$

[12/12/2020 15:43] >[eHxIWesY4Tg]

[21] /T/gr/T/pr/T/rl/D/steplim

Definition.

We have by definition that [21.1] >

$$T := \text{the step limit, where } T \geq 1,$$

where t denotes a step index, with $1 \leq t \leq T$.

[12/12/2020 15:43] >[rubDKv_sK94]

[23] /T/gr/T/pr/T/rl/D/inddef

Definition.

We have by definition that for any $s \in \mathbf{S}$, [23.1] >

$$P_I(s) := \text{the initial distribution probability of starting in state } s.$$

[12/12/2020 15:43] >[SN4128e751A]

[24] /T/gr/T/pr/T/rl/D/trdcind

Definition.

We have by definition that S_t is independent of all previous states and actions given A_{t-1} and S_{t-1}. Notationally, for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and for all $2 \leq t \leq T$, when $\Pr(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) \neq [\text{null}]$, [24.1] >

$$\Pr(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) = \Pr(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}).$$

[12/12/2020 15:43] >[xiP0jaq43Vk]

[25] /T/gr/T/pr/T/rl/D/pact

Definition.

We have by definition that, for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and for all $\mathrm{s}, \mathrm{a} \in \mathbf{S} \times \mathbf{A}$, $1 \leq t \leq T - 1$, if $\Pr(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) \neq$ [null], [25.1] >

$$\Pr(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) = \mathrm{P}_\mathrm{A}(\pi, t : \mathrm{a} \mid \mathrm{s}).$$

[12/12/2020 15:43] >[g7RBaFHBekc]

[26] /T/gr/T/pr/T/rl/D/states

Definition.

We have by definition that [26.1] >

$$\mathbf{S} := \text{the set of states},$$

where $\mathbf{S}$ is finite.

[12/12/2020 15:43]

[88] /T/gr/T/pr/T/rl/D/remret

Definition.

We define for all $1 \leq t \leq T$,

$$R\left(t;\cdot\right) : \left(\mathbf{S} \times \mathbf{A}\right)^{T-1} \times \mathbf{S} \to \mathbb{R}$$

such that

$$R\left(t; \mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T\right) \coloneqq \sum_{t'=t}^{T} R\left(\mathrm{s}'_t\right).$$

[12/12/2020 15:43] >[u38FCwwJ87s]

[27] /T/gr/T/pr/T/rl/D/rvat

Definition.

We have by definition that for all $1 \leq t \leq T - 1$, [27.1] >

$$\mathrm{A}_t \coloneqq \text{variable for the action at step } t,$$

with domain $\mathbf{A}$.

[12/12/2020 15:43] >[w1IWPp3I0Sk]

[28] /T/gr/T/pr/T/rl/D/pactdef

Definition.

We have by definition that for any $\mathrm{s}, \mathrm{a} \in \mathbf{S} \times \mathbf{A}$, $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T-1$, [28.1] >

$$\mathrm{P}_{\mathrm{A}}\left(\pi, t : \mathrm{a} \mid \mathrm{s}\right) \coloneqq \begin{cases} 1 & \pi\left(t; \mathrm{s}\right) = \mathrm{a} \\ 0 & \pi\left(t; \mathrm{s}\right) \neq \mathrm{a}. \end{cases}$$

[12/12/2020 15:43] >[Ph2hDAturbM]

[29] /T/gr/T/pr/T/rl/D/rvst

Definition.

We have by definition that for all $1 \leq t \leq T$, [29.1] >

$$\mathrm{S}_t \coloneqq \text{variable for the state at step } t,$$

with domain $\mathbf{S}$.

[12/12/2020 15:43] >[pP2D-trk9DI]

[30] /T/gr/T/pr/T/rl/D/actions

Definition.

We have by definition that [30.1] >

$$\mathbf{A} := \text{the set of actions},$$

where $\mathbf{A}$ is finite.

[12/12/2020 15:43] >[H0VdYaJ43WI]

[49] /T/gr/T/pr/T/rl/D/psp

Definition.

We have by definition that [49.1] >

$$\mathrm{S} \leftarrow \{(\mathrm{S}_1, \mathbf{S}), (\mathrm{A}_1, \mathbf{A}), (\mathrm{S}_2, \mathbf{S}), \ldots, (\mathrm{A}_{T-1}, \mathbf{A}), (\mathrm{S}_T, \mathbf{S})\},$$

and, for each $\pi \in \mathbf{\Pi}$, let $\mathrm{D}_\pi \in \mathbf{D}$ be a distribution.

[12/12/2020 15:43] >[U75FxZHgu3c]

[31] /T/gr/T/pr/T/rl/D/trddef

Definition.

We have by definition that for any $\mathrm{s}, \mathrm{s}', \mathrm{a} \in \mathbf{S}^2 \times \mathbf{A}$, [31.1] >

$\mathrm{P}_{\mathrm{T}}(\mathrm{s} \mid \mathrm{a}, \mathrm{s}') \coloneqq$ the transition distribution probability of going to state s after action a in state s'.

[12/12/2020 15:43] >[Zz_J2SSzJ9Y]

[33] /T/gr/T/pr/T/rl/D/trd

Definition.

We have by definition that for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and all $\mathrm{s}, \mathrm{s}', \mathrm{a} \in \mathbf{S}^2 \times \mathbf{A}$, $2 \leq t \leq T$, if $\Pr(\mathrm{S}_t = \mathrm{s} \mid \mathrm{A}_{t-1} = \mathrm{a}, \mathrm{S}_{t-1} = \mathrm{s}') \neq$ [null], [33.1] >

$$\Pr(\mathrm{S}_t = \mathrm{s} \mid \mathrm{A}_{t-1} = \mathrm{a}, \mathrm{S}_{t-1} = \mathrm{s}') = \mathrm{P}_\mathrm{T}(\mathrm{s} \mid \mathrm{a}, \mathrm{s}').$$

[12/12/2020 15:43] >[QIU9zIwNiak]

[34] /T/gr/T/pr/T/rl/D/strs

Definition.

We have by definition that [34.1] >

$$\mathbf{\Pi} \coloneqq \text{the set of all strategies},$$

where, for each strategy $\pi \in \mathbf{\Pi}$ and each $1 \leq t \leq T$,

$$\pi\left(t; \cdot\right) : \mathbf{S} \to \mathbf{A} \coloneqq \text{a step-strategy}.$$

[12/12/2020 15:43] >[wikNQg8Yscw]

[82] /T/gr/T/pr/D/ev/P/evdeq

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X})\}$ and all $f : \mathbf{X} \to \mathbb{R}$, $\mathrm{D} \in \mathbf{D}$, and $\mathrm{d}, \mathrm{d}' \in \mathrm{D}^2$, [82.1] >

$$\mathbb{E}\left[\mathrm{d}; f\right] = \mathbb{E}\left[\mathrm{d}'; f\right].$$

Proof.

We have that [82.2] >

$$\mathbb{E}\left[\mathrm{d}; f\right] \overset{[\ldots]^{83}_{\mathrm{P}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr\left(\mathrm{d}; \mathrm{X} = \mathrm{x}\right) \times f(\mathrm{x}) \overset{[\ldots]^{42}_{\mathrm{D}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}\right) \times f(\mathrm{x}) \overset{[\ldots]^{83}_{\mathrm{P}}}{=} \mathbb{E}\left[\mathrm{d}'; f\right].$$

[12/12/2020 15:43] >[laWqGMhX4Bs]

[76] /T/gr/T/pr/D/mprob/P/mprobdeq

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (?, \mathbf{?})\}$ and all $\mathrm{D} \in \mathbf{D}$, $\mathrm{d}, \mathrm{d}' \in \mathrm{D}^2$ and $\mathrm{x} \in \mathbf{X}^\times$, [76.1] >

$$\Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}) = \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}\right).$$

Proof.

We have that [76.2] >

$$\Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}) \overset{[\ldots]^{71}_{\mathrm{P}}}{=} \sum_{?\in\mathbf{?}^\times} \Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}, ? = ?) \overset{[\ldots]^{42}_{\mathrm{D}}}{=} \sum_{?\in\mathbf{?}^\times} \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}, ? = ?\right) \overset{[\ldots]^{71}_{\mathrm{P}}}{=} \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}\right).$$

[12/12/2020 15:43] >[17vPrN7TQkE]

[84] /T/gr/T/pr/D/evc/P/evcdeq

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y})\}$ and all $f : \mathbf{X}^{\times} \times \mathbf{Y}^{\times} \to \mathbb{R}$, $\mathrm{y} \in \mathbf{Y}^{\times}$, $\mathrm{D} \in \mathbf{D}$ and $\mathrm{d}, \mathrm{d}' \in \mathrm{D}^2$, [84.1] >

$$\mathbb{E}\left[\mathrm{d}; f \mid \mathrm{Y} = \mathrm{y}\right] = \mathbb{E}\left[\mathrm{d}'; f \mid \mathrm{Y} = \mathrm{y}\right].$$

Proof.

We have that [84.2] >

$$\begin{aligned}
\mathbb{E}\left[\mathrm{d}; f \mid \mathrm{Y} = \mathrm{y}\right] &\overset{[\ldots]^{72}_{\mathrm{P}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr\left(\mathrm{d}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right) \times f(\mathrm{x}, \mathrm{y}) \\
&\overset{[\ldots]^{77}_{\mathrm{P}}}{=} \sum_{\mathrm{x} \in \mathbf{X}^{\times}} \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right) \times f(\mathrm{x}, \mathrm{y}) \\
&\overset{[\ldots]^{72}_{\mathrm{P}}}{=} \mathbb{E}\left[\mathrm{d}'; f \mid \mathrm{Y} = \mathrm{y}\right].
\end{aligned}$$

[12/12/2020 15:43] >[1ICsc3iKpKg]

[77] /T/gr/T/pr/D/cprob/P/cprobdeq

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$ and all $\mathrm{D} \in \mathbf{D}$, $\mathrm{d}, \mathrm{d}' \in \mathrm{D}^2$ and $\mathrm{x}, \mathrm{y} \in \mathbf{X}^{\times} \times \mathbf{Y}^{\times}$, [77.1] >

$$\Pr(\mathrm{d}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right).$$

Proof.

We have that [77.2] >

$$\Pr(\mathrm{d}; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) \overset{[\ldots]^{74}_{\mathrm{P}}}{=} \frac{\Pr(\mathrm{d}; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{d}; \mathrm{Y} = \mathrm{y})} \overset{[\ldots]^{76}_{\mathrm{P}}}{=} \frac{\Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}\right)}{\Pr\left(\mathrm{d}'; \mathrm{Y} = \mathrm{y}\right)} \overset{[\ldots]^{74}_{\mathrm{P}}}{=} \Pr\left(\mathrm{d}'; \mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}\right)$$

[12/25/2020 10:39]

[155] /T/gr/T/pr/P/pmape/P/conddecomp/P/inddecomprev

Proposition.

We have that for contextual $\mathrm{S} \leftarrow \{(\mathrm{X}, \mathbf{X}), (\mathrm{Y}, \mathbf{Y}), (?, \mathbf{?})\}$, $\mathrm{D} \in \mathbf{D}$, with $\mathrm{V} := \mathrm{X} \cup \mathrm{Y} \cup ?$, [155.1]
if
for all $\mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B}, \mathrm{S} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S},\ \mathrm{A} \perp\!\!\!\perp \mathrm{S} \mid \mathrm{C}, \mathrm{B} \underset{\texttt{int}}{\implies} \mathrm{A} \perp\!\!\!\perp \mathrm{B}, \mathrm{S} \mid \mathrm{C}$$

and for all $\mathrm{C}, \mathrm{S} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$,

$$\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C} \underset{\texttt{sun}}{\implies} \mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S},$$

then, for all $\emptyset \subseteq \mathrm{C} \subseteq \mathrm{V}$ and $\emptyset \subset \mathrm{A}, \mathrm{B}, \mathrm{S} \subseteq \mathrm{V}$ such that $\mathrm{DJ}\,(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{S})$,

$$\underset{\texttt{io}}{\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}},\ \underset{\texttt{it}}{\mathrm{A} \perp\!\!\!\perp \mathrm{S} \mid \mathrm{C}} \implies \mathrm{A} \perp\!\!\!\perp \mathrm{B}, \mathrm{S} \mid \mathrm{C}.$$

Proof.

We have that [155.2]

$$\overset{\texttt{(io)->sun}}{\mathrm{A} \perp\!\!\!\perp \mathrm{B} \mid \mathrm{C}, \mathrm{S}},\ \overset{\texttt{(it)->sun}}{\mathrm{A} \perp\!\!\!\perp \mathrm{S} \mid \mathrm{C}, \mathrm{B}} \overset{\texttt{int}}{\implies} \mathrm{A} \perp\!\!\!\perp \mathrm{B}, \mathrm{S} \mid \mathrm{C}.$$

[12/12/2020 15:43] >[JRkW63ijRw8]

[11] /T/gr/T/pr/T/rl/P/pexpr/P/pexprt

Proposition.

We have that for any $1 \leq t \leq T$, for contextual $\pi \in \mathbf{\Pi}$, $\mathrm{D} = \mathrm{D}_\pi$ and for any $\mathrm{s}_1, \ldots, \mathrm{s}_t, \mathrm{a}_1, \ldots, \mathrm{a}_{t-1} \in \mathbf{S}^t \times \mathbf{A}^{t-1}$ we can write [11.1] >

$$\Pr\left(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_t = \mathrm{s}_t\right)$$

as

$$\Pr\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-1} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right).$$

Proof.

Base case. When $t = 1$, we have [by definition]$^{18}_{\mathrm{D}}$ that $\Pr\left(\mathrm{s}_1\right) = \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right)$. [11.2] >

Inductive hypothesis. Assume that for $t-1$ such that $1 \leq t-1 \leq T-1$, the proposition holds. That is, [11.3] >

$$\Pr\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right) \underset{\texttt{ih}}{=} \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-2} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right).$$

Inductive step. We want to show that the proposition holds for t. [11.4] >

There are two cases to consider: [11.5] >

- Case 1: $\Pr\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right) = 0$. [11.6] >
  co

  Then, [11.7] >

  $$\Pr\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right) \underset{\texttt{coz}}{\overset{\texttt{(co)->}[\ldots]^{91}_{\mathrm{P}}}{=}} 0,$$

  and [11.8] >

  $$\begin{aligned}
  &\mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-1} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right) \\
  &= \mathrm{P}_\mathrm{I}\left(\mathrm{s}_1\right) \times \prod_{t'=1}^{t-2} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right) \times \\
  &\quad \mathrm{P}_\mathrm{A}\left(\pi, t-1 : \mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}\right) \times \mathrm{P}_\mathrm{T}\left(\mathrm{s}_t \mid \mathrm{a}_{t-1}, \mathrm{s}_{t-1}\right) \\
  &\underset{\texttt{ih}}{=} \Pr\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right) \times \mathrm{P}_\mathrm{A}\left(\pi, t-1 : \mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}\right) \times \mathrm{P}_\mathrm{T}\left(\mathrm{s}_t \mid \mathrm{a}_{t-1}, \mathrm{s}_{t-1}\right) \\
  &\underset{\texttt{co}\ \texttt{coz}}{=} 0 = \Pr\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t\right).
  \end{aligned}$$

- Case 2: $\Pr\left(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}\right) > 0$. [11.9] >
  ct

Then, [11.10] >

$$\overset{\texttt{(ct)->}[\ldots]^{89}_{\mathrm{P}}}{\Pr\left(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}\right) \neq [\text{null}]}, \overset{\texttt{(daf)->}[\ldots]^{93}_{\mathrm{P}}}{\Pr\left(a_{t-1} \mid s_{t-1}\right) \neq [\text{null}]},$$

daf da

and we have [11.11] >

$$\overset{\texttt{(daf)->}[\ldots]^{3}_{\mathrm{P}}}{\Pr\left(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}\right) = \Pr\left(a_{t-1} \mid s_{t-1}\right)} \overset{\texttt{(da)->}[\ldots]^{25}_{\mathrm{D}}}{= \mathrm{P_A}\left(\pi, t-1 : a_{t-1} \mid s_{t-1}\right)}.$$

cia ad

[There are two cases]$^{28}_{\mathrm{D}}$ to consider: [11.12] >

– Case 2.1: $\mathrm{P_A}\left(\pi, t-1 : a_{t-1} \mid s_{t-1}\right) = 1.$ [11.13] >

cto

Then, [11.14] >

$$\begin{aligned}
\Pr\left(s_1, a_1, s_2, \ldots, s_{t-1}, a_{t-1}\right) &\overset{\texttt{(ct)->}[\ldots]^{94}_{\mathrm{P}}}{=} \Pr\left(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}\right) \times \Pr\left(s_1, a_1, s_2, \ldots, s_{t-1}\right) \\
&\overset{\texttt{cia-ad}}{=} \mathrm{P_A}\left(\pi, t-1 : a_{t-1} \mid s_{t-1}\right) \times \Pr\left(s_1, a_1, s_2, \ldots, s_{t-1}\right) \\
&\overset{\texttt{cto}}{=} \Pr\left(s_1, a_1, s_2, \ldots, s_{t-1}\right) \\
&\overset{\texttt{ct}}{>} 0.
\end{aligned}$$

md mg

Therefore, [11.15] >

$$\overset{\texttt{(md-mg)->}[\ldots]^{89}_{\mathrm{P}}}{\Pr\left(s_t \mid a_{t-1}, s_1, a_1, s_2, \ldots, s_{t-1}\right) \neq [\text{null}]}, \overset{\texttt{(dsf)->}[\ldots]^{93}_{\mathrm{P}}}{\Pr\left(s_t \mid a_{t-1}, s_{t-1}\right) \neq [\text{null}]}$$

dsf ds

and we have [11.16] >

$$\overset{\texttt{(dsf)->}[\ldots]^{24}_{\mathrm{D}}}{\Pr\left(s_t \mid a_{t-1}, s_1, a_1, s_2, \ldots, s_{t-1}\right) = \Pr\left(s_t \mid a_{t-1}, s_{t-1}\right)} \overset{\texttt{(ds)->}[\ldots]^{33}_{\mathrm{D}}}{= \mathrm{P_T}\left(s_t \mid a_{t-1}, s_{t-1}\right)}.$$

cis sd

Therefore, [11.17] >

$$\begin{aligned}
\Pr\left(s_1, a_1, s_2, \ldots, s_t\right) &\overset{\texttt{(md-mg)->}[\ldots]^{94}_{\mathrm{P}}\texttt{,(ct)->}[\ldots]^{94}_{\mathrm{P}}}{=} \Pr\left(s_1, a_1, s_2, \ldots, s_{t-1}\right) \times \Pr\left(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}\right) \times \\
&\qquad \Pr\left(s_t \mid a_{t-1}, s_1, a_1, s_2, \ldots, s_{t-1}\right) \\
&\overset{\texttt{cia,cis}}{=} \Pr\left(s_1, a_1, s_2, \ldots, s_{t-1}\right) \times \Pr\left(a_{t-1} \mid s_{t-1}\right) \times \Pr\left(s_t \mid a_{t-1}, s_{t-1}\right) \\
&\overset{\texttt{ih,ad,sd}}{=} \mathrm{P_I}\left(s_1\right) \times \prod_{t'=1}^{t-1} \mathrm{P_A}\left(\pi, t' : a_{t'} \mid s_{t'}\right) \mathrm{P_T}\left(s_{t'+1} \mid a_{t'}, s_{t'}\right).
\end{aligned}$$

– Case 2.2: $\mathrm{P_A}\left(\pi, t-1 : a_{t-1} \mid s_{t-1}\right) = 0.$ [11.18] >

ctt

Then, [11.19] >

$$\overset{\texttt{(ct)->}[\ldots]^{94}_{\mathrm{P}}}{\Pr(s_1, a_1, s_2, \ldots, s_{t-1}, a_{t-1})} = \underset{\texttt{afd}}{\Pr(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1})} \times \Pr(s_1, a_1, s_2, \ldots, s_{t-1})$$

$$\overset{\texttt{cia-ad}}{= P_A(\pi, t-1 : a_{t-1} \mid s_{t-1})} \times \Pr(s_1, a_1, s_2, \ldots, s_{t-1})$$

$$\overset{\texttt{ctt}}{= 0.}$$

$$\texttt{afz}$$

Therefore, [11.20] >

$$\overset{\texttt{(afd-afz)->}[\ldots]^{91}_{\mathrm{P}}}{\Pr(s_1, a_1, s_2, \ldots, s_t) = 0,}$$

$$\texttt{cttz}$$

so [11.21] >

$$\Pr(s_1, a_1, s_2, \ldots, s_t) = \overset{\texttt{cttz}}{0} = \overset{\texttt{ctt}}{P_I(s_1)} \times \prod_{t'=1}^{t-1} P_A(\pi, t' : a_{t'} \mid s_{t'}) P_T(s_{t'+1} \mid a_{t'}, s_{t'}).$$